ploidyspec

Reading polyploidy, its origin and its return to diploidy
from haplotype-resolved genome assemblies

The method, the tool, every output and diagnostic plot,
worked examples, and a summary of every species run so far

What this is
One place to understand the whole project: why it exists (Chapters 1–3), how to run it (4), how each number is computed (5–6), what every output file and plot means (7–8), four walkthroughs on real species (9), how it was validated and corrected (10–11), how to interpret a result (12), what it says about each of the 59 assemblies (13), and how it stands against the literature (14).

Status
Reflects the code and results of 12 October 2026, after the panel was re-run with every correction in Chapter 11. Generated parts (Chapter 13) are rebuilt from the result files by `docs/guide/make_guide_species.py`.

Contents

1 At a glance 4
2 Background: why this question, why now 5
2.1 Polyploidy is common, and its categories are blurry 5
2.2 Meiosis decides what the sequence records 5
2.3 Rediploidization: the time axis of the continuum 6
2.4 Homoeologous exchange blurs things from the other side 6
2.5 Repeats and centromeres turn over fastest 7
2.6 Divergence records isolation, not duplication 7
2.7 Other ways to tell auto from allo, and how ploidyspec differs 7
2.8 Haplotype-resolved assemblies make this possible 8
2.9 What we are trying to achieve 8
3 Auto, allo and rediploidization: what the data can show 10
3.1 Two axes, not one label 10
3.2 What each process leaves in the sequence 10
3.3 What one individual cannot tell you 11
3.4 How the four answers are built 11
4 Running ploidyspec 12
4.1 Input 12
4.2 Installing and running 12
4.3 The pipeline 13
4.4 Vocabulary used throughout 13
5 Measurement primitives 14
5.1 k-mer tables 14
5.2 Whole-chromosome distance: Mash with a noise floor 14
5.3 Window distance: containment 14
5.4 Position-free windows 14
5.5 Statistical tools used repeatedly 14
6 The stages, algorithm by algorithm 16
6.1 prepare: from a manifest to chromosome copies 16
6.2 kmers 16
6.3 matrix: distances, renumbering, extra sets 16
6.4 homeologs: whole-chromosome pairs 17
6.5 windowed: along each chromosome, same number 18
6.6 windowed-homeologs: the homeology map, then pair tracks 18
6.7 structure: partitions and synchrony 19
6.8 te-markers (optional) 19
6.9 rediploidization: fusions and per-chromosome states 20
6.10 Residual tetrasomy and homeologous exchange 21
6.11 summary: the decision logic 22
6.12 report, panel, simulate, cleanup 22
7 What every output means 23
8 Reading the diagnostic plots 25
8.1 Conventions shared by every heatmap 25
8.2 Whole-chromosome distance heatmap (`matrix/whole_chrom_distance_heatmap.png`) 25
8.3 Contrast heatmap (`whole_chrom_distance_heatmap_contrast.png`) 26
8.4 k-resolution diagnostic (`matrix/k_resolution_diagnostic.png`) 26
8.5 Homeolog candidates (`homeologs/homeolog_pairs.png`) 27
8.6 Reordered chromosome-number heatmap (`homeolog_pairs_reordered_heatmap.png`) 28
8.7 Homeology map (`homeologs/homeology_map.png`) 28
8.8 Windowed tracks along a chromosome (`windowed/windowed_chrNN.png`, `_heatmap.png`) 29

8.9 Genome overview (`windowed/windowed_genome_overview*.png`) ... 31
8.10 Tracks between homeologs (`homeologs/windowed_chrAAxBB*.png`, overview) ... 32
8.11 TE-marker painting (`subgenomes/te_markers_windowed_*.png`) ... 32
8.12 Fusion profile (`rediploidization/fusion_profile_<scaffold>.png`) ... 33
8.13 Panel PCA (`meta/supplementary/poly_space_pca.png`) ... 33
9 Walkthroughs ... 35
9.1 Glechoma hederacea: a two-haplotype allotetraploid ... 35
9.2 Schizothorax curvilabiatus: rediploidization by fusion ... 36
9.3 Strophosoma melanogrammum: an AAB triploid in two files ... 37
9.4 Oncorhynchus masou: when a guard is missing ... 37
10 Validation ... 39
10.1 Simulated genomes ... 39
10.2 Real anchors ... 39
10.3 Fusion profiles: divergence spreads from the fusion site ... 40
11 What we got wrong, and fixed ... 41
12 Interpreting results, and the limits ... 43
12.1 A reading order for any species ... 43
12.2 Patterns and what they mean ... 43
12.3 Limits ... 43
12.4 Where ploidyspec sits in poly-space ... 44
12.5 Open questions ... 44
13 Every species so far ... 45
13.1 Two-haplotype genomes with duplicated sets (allo-like anchors and relatives) ... 48
13.2 Cryptic or partial duplication in two-haplotype genomes ... 59
13.3 Genomes assembled with three or more copies ... 62
13.4 Salmonids ... 68
13.5 Diploid-like and negative controls ... 71
13.6 One usable haplotype ... 76
14 The literature: where ploidyspec stands ... 79
14.1 The methods landscape ... 79
14.2 Where ploidyspec is new, and where it is not ... 80
14.3 Our conclusions checked against the literature ... 81
14.4 The panel in the light of the literature ... 82
14.5 What the literature changed in ploidyspec ... 83
14.6 Open questions the literature raises ... 84
14.7 The literature by theme ... 84
15 Appendix A: parameters ... 85
16 Appendix B: glossary ... 86
17 Appendix C: references ... 87

1 At a glance

Give ploidyspec the haplotype-resolved assembly of one individual (two or more haplotype FASTA files, or one file with haplotypes named in the headers) and it answers four questions, each with the evidence behind it and a confidence:

Question	How ploidyspec answers it	Where
1. What is the ploidy?	Counts the haplotype copies of each chromosome (an extra chromosome set held in one file is split out), then looks for chromosome numbers that are themselves duplicated: homeolog pairs, genome partitions, and blocks of segmental homeology.	Sections 6.3–6.7, 12
2. Auto-like or allo-like?	Whether the copies of each chromosome are interchangeable or fall into lineages that do not mix, consistently across chromosomes; for two-haplotype assemblies, through homeolog pairs and residual tetrasomy.	Sections 6.9–6.11
3. What do the TE markers say?	Whether repeat (transposable element) k-mers private to one copy track the lineages: a within-genome contrast, not an absolute score.	Sections 6.8, 8.11
4. Is there evidence of rediploidization?	Fusions between copies and the divergence spreading from them, chromosomes split into lineages along all or part of their length, residual tetrasomy, homeologous exchange and lineage switches, chromosomes rearranged since the duplication, and pairs or blocks diverged to different depths.	Sections 6.9–6.11

It does this from DNA sequence alone: no gene annotation, no reference genome, no read data, no alignment. The core measurement is k-mer distance between chromosome copies, whole and in windows. Chapter 14 compares this with published methods and checks the panel's conclusions against the literature.

> **The one thing to remember**
> Sequence from a single individual shows how its chromosome copies relate to each other today. It shows inheritance-like structure, not inheritance itself (that needs crosses or populations), and not origin. A long rediploidized autopolyploid can look exactly like an allopolyploid. The outputs are worded accordingly: "tetrasomic-like", "resolved lineages", "allo-like", never "is an autopolyploid".

2 Background: why this question, why now

This chapter summarizes the literature the method rests on. Chapter 14 goes further: it sets ploidyspec beside every comparable method, checks our conclusions against the published record, and lists what the literature changed in the tool. All works are listed in Appendix C.

2.1 Polyploidy is common, and its categories are blurry

Whole-genome duplication (WGD) has happened again and again in the history of plants and animals. Most lineages have been through several cycles of duplication and diploidization, and all seed plants descend from polyploid ancestors (Doyle & Sherman-Broyles, 2017; Peer et al., 2009). Phylogenomic surveys keep finding new WGDs; duplicate genes from them are lost roughly exponentially, with a half-life of about 21.6 million years in angiosperms (Qiao et al., 2019; Ren et al., 2018). Established WGDs may be rare successes among many failed polyploids, but they coincide with radiations and novelty (Peer et al., 2009), and much basic biology of polyploids is still open (Soltis et al., 2010).

Polyploids are traditionally sorted into two kinds. **Autopolyploids** double a genome within one species; their chromosome copies are near-identical and pair freely at meiosis (**polysomic**, or **tetrasomic** in a tetraploid, inheritance). **Allopolyploids** combine the genomes of two diverged species; each chromosome has a preferred partner from its own parent (**disomic** inheritance), and the two parental sets remain as **subgenomes**. Stebbins (1947) already added a third class, the **segmental allopolyploid**, whose parents were only partly diverged, so that some chromosomes pair disomically and some polysomically.

The two kinds are defined in two incompatible ways. **Taxonomic** (mode-of-origin) definitions ask whether one species or two contributed the genomes; **genetic** or **cytological** definitions ask whether inheritance is polysomic or disomic (Doyle & Sherman-Broyles, 2017). The two often disagree. Lv et al. (2024) found pure or predominantly polysomic inheritance in 39 of 43 putative autopolyploids (91%), but also in seven species that are taxonomically allopolyploid, with a further five partly polysomic; inheritance can relabel species. Complex polyploids with more than two contributing lineages, and introgression between cytotypes, blur the picture further (Leal et al., 2024).

Twyford et al. (2025) argue that the binary is better seen as a **continuum**, or a multidimensional "poly-space", because the definitions disagree and because genomes move through that space over time. Their Box 2 makes the point that matters most here: after duplication a genome undergoes **diploidization** (or **rediploidization**), and a diploidized autopolyploid can have diverged duplicate chromosomes "indistinguishable from those in a classic allopolyploid". They suggest measurable axes to place genomes in poly-space: heterozygosity, sequence divergence between duplicated sets, gene-level divergence, and structural variation, ideally from haplotype-resolved genomes. Phased genomes are what now make these questions answerable: which part of the genome came from which lineage, and whether the lineages have mixed (Thomas, 2026). ploidyspec measures several of these axes directly (Section 12.4).

2.2 Meiosis decides what the sequence records

What a polyploid genome looks like after some time depends on how its chromosomes pair. In a new polyploid, related chromosomes can form multivalents or pair with any partner; stable polyploids usually evolve diploid-like, bivalent pairing, under genetic control that is now partly known (Grandont et al., 2013; Soares et al., 2021). The transition from polysomic to disomic inheritance need not wait for dedicated pairing genes: in simulations, drift together with a threshold for pairing fidelity is enough, inversions and reduced chiasma numbers help, and sub- or neofunctionalization speeds it up (Le Comber et al., 2010).

The transition is rarely genome-wide at once. In tetraploid rose, pairing is **partially preferential**, differs between chromosomes, varies along chromosome arms, and depends on the genotype (Bourke et al., 2017). In apomictic *Brachiaria decumbens*, homologs pair preferentially across the genome but one chromosome segregates multisomically (Worthington et al., 2016). The Adriatic sturgeon shows tetrasomic segregation at 11 of 14 loci and disomic at three (Dalle Palle et al., 2022), and an AABB banana shows both disomic and polysomic

inheritance cytologically (Jeridi et al., 2012). Mixed inheritance within one genome is therefore expected, not exceptional. For ploidyspec this is why every reading is made per chromosome and per window.

Inheritance itself is measured from crosses, populations or cytology: segregation in mapping populations (Bicknell et al., 2023; Bourke et al., 2017), allele dosage in families (Delaval et al., 2024), approximate Bayesian computation on population samples (Roux & Pannell, 2015), and the population-genetic tools reviewed by Dufresne et al. (2014). One assembled individual cannot measure inheritance; it records the sequence consequences of past pairing.

2.3 Rediploidization: the time axis of the continuum

Xie et al. (2026) caught rediploidization near its start (commentary in Xie et al. (2026)). Snow carps (Schizothoracinae) share one autotetraploid origin. In *Schizopygopsis younghusbandi* most of the 25 chromosome quartets are still tetrasomic: their four copies are about equally similar (Ks ≈ 0.01). But on ten chromosomes, two of the four copies have fused with another chromosome (five fusions). On those chromosomes, fused copies pair with fused copies and unfused with unfused, so the quartet has split into two lineages that diverge. Divergence is highest at the fusion site and spreads outwards along the arms, and the fusions happened in three "waves" of different ages. A separate mechanism, a centromeric inversion, made the short arm of chromosome 17 disomic while the long arm stayed tetrasomic. The authors also note that downstream consequences of rediploidization, such as biased gene loss and expression between the two lineages, mimic subgenome dominance in allopolyploids.

At the other end of the time axis, salmonids show what a duplication about 80–95 million years old looks like. Their ancestor was autotetraploid, and rediploidization was slow and regional (Lien et al., 2016). About a quarter of each salmonid genome rediploidized only after the lineages had split (**lineage-specific ohnologue resolution**, LORe), more than 50 million years after the WGD, so these regions have diverged less than the rest (Robertson et al., 2017). In Atlantic salmon the most similar duplicated regions (above 90% sequence similarity, 25.6% of the anchored sequence) cluster in seven pairs of chromosome arms, each a large collinear block that includes a telomere (Lien et al., 2016); in resequencing data from charr, whitefish and salmon they are where deviant SNPs concentrate (Dallaire et al., 2023). The same pattern holds on deeper timescales: across 74 teleost genomes some duplicated regions stayed tetraploid for more than 60 million years after the teleost WGD, which points to an autopolyploid origin (Parey et al., 2022); and sturgeon and paddlefish share one WGD that was masked for decades because the two lineages separated before rediploidization was half done (Redmond et al., 2023). Rediploidization also reshapes karyotypes: the "pseudotriploid" *Cardamine cordifolia* (2n = 24) is a diploidized tetraploid whose chromosome number fell from 32 by terminal translocations (Mandáková et al., 2016), and the mangrove *Sonneratia* rearranged its chromosomes extensively after a whole-genome triplication (Feng et al., 2024). Gene loss during rediploidization proceeds mainly by DNA deletion rather than pseudogenization (Crombez et al., 2025), and allele loss is already measurable in a young autotetraploid (Yu et al., 2023); Li et al. (2021) review the patterns in plants.

This is the pattern ploidyspec was built to read: per chromosome, and along each chromosome, whether the copies still behave as one pool or have separated. The snow carp genomes are part of our panel, and we recover their fusions and their wave order without being told where to look (Section 9.2); three salmonids are in the panel too (Sections 9.4, 14.4).

2.4 Homoeologous exchange blurs things from the other side

Allopolyploids do not stay neatly separated either. Homoeologous exchange (HE) is recombination between chromosomes of different subgenomes (Deb et al., 2023; Gaeta & Chris Pires, 2010; Mason & Wendel, 2020). HE can be reciprocal, or "with replacement", where one subgenome's segment is overwritten by a copy of the other's, homogenizing that region. It is common in young allopolyploids, concentrated towards chromosome ends, often biased towards one subgenome, and leaves a mosaic (Deb et al., 2023). In a nascent wheat allotetraploid, HEs are biased to low-copy, subtelomeric regions, behave like homologous recombination hotspots, and often fall within genes (Zhang et al., 2020). Repeated rounds of HE with replacement act as a "polyploid ratchet" (Gaeta & Chris Pires, 2010) and produce presence/absence variation (Hurgobin et al., 2018); fixed exchanges can leave a genome that looks partly auto- and partly allopolyploid, a modern route to segmental

allopolyploidy (Mason & Wendel, 2020). Segmental allopolyploid genomes have been described in peanut (Leal-Bertioli et al., 2018), sour cherry (Wöhner et al., 2023) and the decaploid *Thinopyrum ponticum* (Deng et al., 2026); allotriploid AAB bananas carry frequent exchanges between A and B (Xie et al., 2024); and allopolyploid dodders show homoeologous recombination between subgenomes (Segawa et al., 2025). HE is not universal: three independent origins of *Brachypodium hybridum* show none (Mu et al., 2023), and the young allopolyploid *Arabidopsis suecica* shows no "genome shock" but gradual adaptation of meiosis (Burns et al., 2021; Burns et al., 2024).

For ploidyspec this means an allopolyploid can show local blocks where one homeolog carries the other's sequence. ploidyspec reads these as exchange candidates and as lineage switches, and keeps them apart from residual tetrasomy, where homeologs still pair in an old autopolyploid.

2.5 Repeats and centromeres turn over fastest

The fastest-changing parts of a genome are repeats. Centromeric satellites have diverged between the subgenomes of wheat (Su et al., 2019); in *Brachypodium*, centromeres turned over between homologous pairs from diploid to tetraploid (Chen et al., 2024); in cotton, centromeric repeats of the D progenitor invaded the A subgenome after polyploidization (Yan et al., 2025), and TE landscapes evolved asymmetrically between subgenomes (Shi et al., 2026). Repeats active in one progenitor before the merger are therefore a subgenome label that survives after unique sequence has diverged beyond alignment. This is the basis of TE-signature subgenome phasing (Gordon et al., 2019; Jia et al., 2022; Session & Rokhsar, 2023), and subgenome-specific TE content is used to support allopolyploid origins in new assemblies (Nieuwenhuis et al., 2024; Robbins et al., 2026; Song et al., 2024).

Two consequences for ploidyspec. In durum wheat our subgenome partition recovers the A and B subgenomes exactly from the same signal those methods use, and every chromosome is closer to the others of its own subgenome than to its homeolog. Homeolog pairs therefore only stand out against the cross-subgenome background, where all seven A–B pairs are found (Section 6.4). And shared satellites masquerade as near-identical sequence: Glechoma's apparent residual tetrasomy was a centromeric satellite (Section 6.10).

2.6 Divergence records isolation, not duplication

A preprint by Gaynor et al. (2026) (bioRxiv; not yet peer reviewed) makes an argument that applies to any divergence-based method, including ours. Using haplotype-phased polyploid assemblies, they find that synonymous divergence (Ks) builds up only between gene copies that do not recombine. Copies on chromosomes that pair at meiosis stay similar, whatever the ploidy. Duplicate-gene divergence therefore detects meiotic isolation, or divergence inherited from hybridizing parents, rather than genome doubling as such. That is why tetrasomic autopolyploid copies stay close. Simulations make the same point for Ks histograms: the peak can mark parental divergence, the last migration between parents, or the ancestral coalescence, and can be hidden at zero (Dunn & Sethuraman, 2025). Gene conversion between WGD duplicates is highest just after polyploidization and declines with time (Qiao et al., 2019).

ploidyspec's distances behave the same way, and we use this deliberately. Low distance among all copies of a chromosome means the copies still mix. Two clusters at a consistent depth mean two lineages have been isolated from each other. Divergence alone cannot say whether the isolation came from hybridization (allo) or evolved later (rediploidized auto). It also explains a blind spot: an old autopolyploidy that stayed tetrasomic leaves little divergence to find.

2.7 Other ways to tell auto from allo, and how ploidyspec differs

- **Gene-loss balance (P-index; Wang et al. (2019)).** After duplication, an allopolyploid tends to lose genes unequally between subgenomes (biased fractionation), and an autopolyploid equally. Wang et al. turn this into a 0–1 index from gene collinearity against an outgroup genome, with about 0.3 separating auto from allo. It needs gene annotation, synteny, and a suitable outgroup. Xie et al. found biased gene loss towards the fused copies in an autopolyploid, and Parey et al. (2022) biased retention in the autopolyploid teleost WGD, so biased fractionation can also arise after rediploidization.

- **Duplicated BUSCOs and gene clusters.** McHale et al. (2025) detect the ancient land snail and slug WGD from the proportion of duplicated BUSCO genes (16–21% in Stylommatophora versus 0.5–1.5% in other molluscs) and follow it in the patchwork retention of duplicated Hox clusters, mirroring vertebrates; Farhat et al. (2023) infer a WGD in neogastropods from synteny, Hox duplication and Ks. These approaches suit old duplications whose copies have diverged beyond the reach of k-mer comparison.
- **Gene trees and ancestral karyotypes** (Parey et al. (2022); Redmond et al. (2023); Robertson et al. (2017)) date when each duplicated region stopped recombining, across many species. They need annotated genomes of many related species.
- **k-mer spectra from reads** (GenomeScope 2.0 and Smudgeplot, Ranallo-Benavidez et al. (2020) Tetmer, Becher et al. (2020)) estimate ploidy, heterozygosity and subgenome divergence without an assembly, but summarize the whole genome in one number. Tetmer showed tetraploid *Euphrasia* to be allopolyploid with about 5% subgenome divergence (Becher et al., 2020); the same read-based profiling detects polyploidy across Microsporidia (Khalaf et al., 2024). Jenike et al. (2025) review k-mer methods for biodiversity genomics.
- **Subgenome phasing** assigns chromosomes of a known allopolyploid to subgenomes: by repeat signatures (Gordon et al., 2019; Session & Rokhsar, 2023), by subgenome-specific k-mers (SubPhaser, (Jia et al., 2022), which fails on autopolyploids by design), or by differential divergence against related genomes (M3, (Lv et al., 2025)). Zhang et al. (2023) benchmark these; Session (2024) reviews the field. They assume allopolyploidy and answer "which subgenome", not "auto or allo".
- **Allele dosage and segregation.** Gaynor et al. and Xie et al. use genotype classes (AAAB/ABBB versus AABB) or resequencing of many individuals to infer disomic versus polysomic inheritance directly.

ploidyspec sits between these. It uses chromosome-scale, haplotype-resolved assemblies of one individual, as produced at scale by the Darwin Tree of Life project, and reads structure chromosome by chromosome and along chromosomes, without annotation. It complements gene-based methods rather than replacing them. Section 14.1 sets them side by side.

2.8 Haplotype-resolved assemblies make this possible

Polyploid genomes were long assembled as collapsed mosaics (Kyriakidou et al., 2018). Long reads, Hi-C and new phasing methods now separate haplotypes in polyploids (Zhao & Shi, 2025): a tetraploid potato with four haplotypes, where about 50% of the genome turned out identical by descent and about 20% structurally rearranged (Sun et al., 2022); hexaploid sweet potato (Yang et al., 2017); the amphitriploid (AAABBB) gibel carp (Wang et al., 2022); AAB bananas (Xie et al., 2024); octoploid strawberry (Song et al., 2024); autopolyploid snow carps (Xie et al., 2026); a phased masu salmon (Wu et al., 2026). The Earth BioGenome Project aims to sequence all eukaryotes (Exposito-Alonso et al., 2020) and has set standards for phased reference assemblies (Howe et al., 2026). ploidyspec is designed for exactly this stream of assemblies, and treats the assembly as part of the measurement: reference-guided scaffolding (Alonge et al., 2019) can place a contig on the wrong homeolog, which is how masu's first residual-tetrasomy signal arose (Section 9.4).

2.9 What we are trying to achieve

2.9.1 The aim

Given a haplotype-resolved assembly, describe where the genome sits on two axes:

1. **Origin-like structure:** do the chromosome copies form one interchangeable pool (auto-like), or separate lineages at a consistent depth across the genome (allo-like), or something in between?
2. **Rediploidization:** has a once-tetrasomic genome started to split into lineages, where, by what mechanism (fusion, inversion, gradual divergence), and how far has it gone?

The tool should run on any species without tuning, report evidence rather than labels, and say when a question cannot be answered from the data.

2.9.2 Design principles

- **Reference-free and annotation-free.** Everything comes from k-mers of the assembled sequence, so a newly assembled species can be analysed the day its assembly is released.

- **Per chromosome, then per window.** Genome-wide summaries hide exactly the heterogeneity that rediploidization creates.
- **Descriptive states, not verdicts.** Each chromosome gets a state with the numbers that produced it.
- **Several independent lines of evidence.** Whole-chromosome distance, windowed distance, repeat markers, chromosome fusions and genome-wide partitions each fail differently, so agreement between them carries weight.
- **Validated on genomes with a known answer.** A simulator builds synthetic assemblies (diploid, auto, allo, rediploidizing, mislabelled) and an end-to-end test checks the whole pipeline against the truth.

3 Auto, allo and rediploidization: what the data can show

3.1 Two axes, not one label

A polyploid genome has two copies or more of a whole chromosome set. The classical labels sort polyploids by **origin**: an **autopolyploid** duplicates one genome, an **allopolyploid** joins two diverged genomes. They differ in what happens at meiosis. Autopolyploid copies are interchangeable and pair at random (**polysomic**, in a tetraploid **tetrasomic**, inheritance); allopolyploid copies pair with their own parent's partner (**disomic** inheritance), so the parental sets persist as **subgenomes**.

Twyford et al. (2025) argue this is a continuum, and their key point for us is that genomes move along it. After a duplication, an autopolyploid gradually **rediploidizes**: chromosomes stop pairing at random, region by region, and the copies diverge into two lineages that eventually look exactly like the subgenomes of an allopolyploid. So ploidyspec reports two separate axes:

1. **Origin-like structure**: today, do the copies of each chromosome form one interchangeable pool, or separate lineages at a consistent depth?
2. **Rediploidization**: is there evidence of the transition from the first state to the second: where, by what mechanism, and how far it has gone?

3.2 What each process leaves in the sequence

The table below is the conceptual core of the tool: every reading in ploidyspec is one of these signatures.

Process	What it leaves in one individual's assembly	Where ploidyspec sees it
Recent autopolyploidy, still tetrasomic	All copies of a chromosome about equally close; no lineage structure; repeat content shared.	`copy_state = tetrasomic_like` pairs tetrasomic-like
Allopolyploidy (two diverged parents)	Copies fall into two groups at a consistent depth on every chromosome; each group carries its parent's repeat bursts.	`resolved_lineages` throughout; homeolog pairs with low `pair_depth_cv`; TE split
Rediploidization by fusion (snow carps)	Some copies of a chromosome fused to another chromosome; fused and unfused copies form lineages; divergence highest at the fusion.	`fusions.tsv` (`status = fusion`); `fusion_lineages`; `lineage_divergence`
Rediploidization by inversion or gradual divergence	A lineage split along part of a chromosome only.	`partially_resolved` with `split_extent = regional`
Residual tetrasomy (salmonids)	In an old, mostly rediploidized duplication, a few regions where homeologs still pair and stay as close as alleles.	`residual_*` columns; `residual_tetrasomy.tsv`
Homeologous exchange (allopolyploids)	One copy carries its homeolog's sequence over a stretch: closer there to the homeolog than to its own homolog.	`exchange_*` columns
Karyotype change since the duplication	A chromosome whose arms are duplicated on two different chromosomes.	homeology map: `multi_partner_list`
Asynchronous resolution	Homeolog pairs or blocks diverged to different depths.	`pair_depth_cv`; `block_dist_cv`
Delayed (LORe-like) rediploidization	Homeolog blocks far less diverged than the rest but not near-allelic: arms that stopped recombining late (salmonid LORe).	homeology map: `n_lore_blocks`, `lore_bp`, `lore_pairs`; block `low_divergence`
An extra divergent set (AAB, AAAB)	One copy of every chromosome stands apart from the others.	`one_divergent_copy`; extra-set split (Section 6.3)

Table 1: Processes, their signatures, and the outputs that read them.

3.3 What one individual cannot tell you

Four limits shape every answer the tool gives.

1. **A rediploidized autopolyploid looks allopolyploid.** Once the lineages have separated everywhere, nothing in today's sequence records whether they started as one genome or two. ploidyspec says "allo-like" and adds that a long-rediploidized autopolyploid looks the same. Only leftovers of the transition (residual tetrasomy, fusions, a mix of states) point to an autopolyploid origin.
2. **Divergence records isolation, not duplication** (Gaynor et al., 2026): copies that still recombine stay similar. A tetrasomic autopolyploid leaves little to see beyond equal distances. A pair of number sets that is **not** detected is therefore one of two opposite things: copies so similar they still recombine (then they show as near-identical copies, not as missing pairs), or copies diverged beyond k-mer reach.
3. **Exchange and residual tetrasomy are mirror images.** Both make homeologs locally near-identical. The difference is which copies agree: in residual tetrasomy all copies are close to each other; in an exchange one copy carries its homeolog's sequence and is far from its own homolog. ploidyspec now tests both (Section 6.10) because the first version confused them.
4. **k-mers see homeology only up to about 8–10% divergence.** Glechoma's homeologs (0.042 apart) are mapped over 79% of the genome; the 90-My-old salmonid duplication over 16–30%, and Cerastium's homeologs (0.08 apart) are found by the whole-chromosome test but not by the window map. In durum wheat, whose genome is mostly fast-turning-over transposons, the seven A–B pairs (0.080–0.082 apart) are found only at whole-chromosome scale, and only against the cross-subgenome background; the window map finds no blocks. "Not detected" never means "not there".

> **Assembly artefacts look like biology**
> Missing sequence reads as divergence; a contig placed on the wrong homeolog reads as exchange or residual tetrasomy; one haplotype file numbering its chromosomes differently reads as a broken genome. Much of this guide's algorithmic detail is about guarding against these. The masu walkthrough (Section 9.4) shows what happens when a guard is missing.

3.4 How the four answers are built

Question	Built from
Ploidy	copies per number (assembly) × duplicated sets (homeolog pairs, genome partition, or homeology map), with odd copy numbers flagged
Origin-like structure	per-chromosome states; for two-copy genomes the pooled homeolog pairs, residual tetrasomy (≥ 2 pairs) and the partition; one-copy-apart patterns
TE markers	the within-genome contrast `te_split`, never the absolute fraction
Rediploidization	fusions; mixtures of states; regional splits; residual tetrasomy; exchange; rearrangements and block-divergence spread from the map; pair synchrony

Table 2: Section 6.11 gives the full decision logic.

4 Running ploidyspec

4.1 Input

ploidyspec needs, for one individual, every haplotype at chromosome scale, described by a **manifest**: a two-column table of FASTA path and haplotype label.

```
# fasta_path                          hap_label
data/sp/sp.hap1.1.primary.fa.gz       HAP1
data/sp/sp.hap2.1.primary.fa.gz       HAP2
```

Use `AUTO` instead of a label when one file holds several haplotypes named in the headers. Each chromosome-scale sequence (default ≥ 1 Mb) needs a chromosome number, read from the header: `chromosome: 4` in the description, or `SUPER_4` in the name. Other naming schemes are handled with `--chrom-regex` / `--hap-regex`. Shorter or unnumbered sequences are listed in `unplaced.tsv`, and the large ones are still tested as possible fused chromosomes.

> **What counts as 'a copy'**
> ploidyspec counts what the assembly contains. A tetraploid can be assembled as four haplotypes (four copies of each chromosome number) or as two haplotypes in which each base chromosome appears under two numbers (two copies each, arranged in homeolog pairs). Both are handled, and the answers to the four questions take the layout into account.

4.2 Installing and running

```
conda env create -f environment.yml && conda activate ploidyspec
pip install .
ploidyspec all --manifest manifests/sp.tsv --outdir results/sp --threads 8
ploidyspec all ... --with-te-markers --with-windowed-homeologs   # optional stages
ploidyspec panel --results results --outdir meta                 # many species
```

The only external tools are `samtools` and FastK (`FastK`, `Logex`, `Histex`, `Tabex`, `Profex`). Every stage can be run on its own, and each skips work already done, so an interrupted run resumes. `report.html` in the output directory collects tables and plots from whatever has run.

Resources. Memory scales with the longest chromosome, not with genome size: about 40 MB per Mb for the matrix stage, and about 14 bytes per base of the longest chromosome per thread for the windowed stage. Runtime on our panel was 0.2-13 hours per species. k-mer tables dominate disk use; `--cleanup` removes them and keeps every result.

4.3 The pipeline

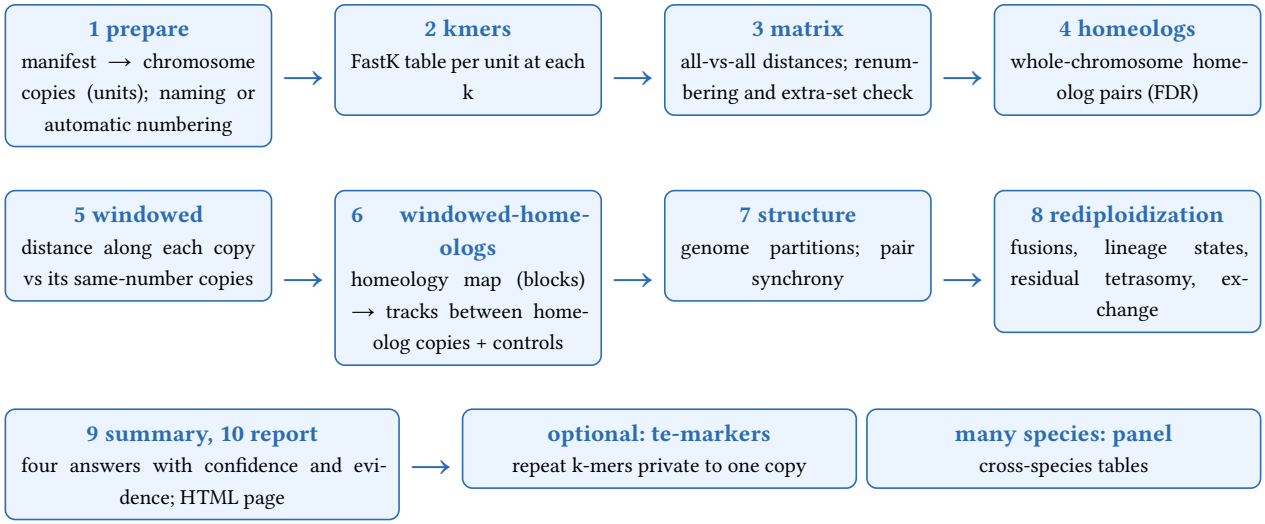


Figure 1: `ploidyspec all` runs these in order. Stage 6 runs with `--with-windowed-homeologs` (recommended), te-markers with `--with-te-markers`. Each stage reads only files written by earlier stages, so any stage can be re-run on its own.

4.4 Vocabulary used throughout

Unit one assembled copy of one chromosome, named `HAP2_chr05`: haplotype `HAP2`, chromosome number 5.

Copies the units sharing a chromosome number. Their count is set by the assembly, not inferred.

Same-number (allelic) distance distance between copies of one chromosome number. In a diploid this is heterozygosity.

Homeologs different chromosome numbers that descend from one ancestral chromosome through a duplication.

Duplicated sets chromosome numbers that fall into groups of homeologs (pairs, triples) across most of the genome.

Pooling reading a chromosome together with its homeolog partner when each number has only two copies.

State the per-chromosome reading (`tetrasomic_like`, `resolved_lineages`, ...), Section 6.9.

Block a stretch of a chromosome whose closest other chromosome is consistently the same one (Section 6.6).

5 Measurement primitives

Every number in ploidyspec comes from comparing sets of k-mers (words of k bases). There is no alignment, no annotation and no reference genome.

5.1 k-mer tables

FastK (Myers) counts every canonical k-mer of a sequence (a k-mer and its reverse complement count as one) into a sorted table. `kmers` builds one table per unit and per k; `Logex` intersects two tables; `Histex` counts distinct k-mers; `Profex` reads a per-position profile (below).

5.2 Whole-chromosome distance: Mash with a noise floor

Algorithm: pairwise distance between two units A and B

1. For each k in the sweep (default 15 and 23): n_A, n_B = distinct k-mers; s = shared k-mers; Jaccard $J = s/(n_A + n_B - s)$.
2. Mash distance $d = -\frac{1}{k}\ln(2J/(1+J))$, the per-base substitution rate under which a k-mer survives with probability $(1-d)^k$ (Ondov et al., 2016).
3. Chance floor: two unrelated sequences share $E = n_A n_B/\left(4^k/2\right)$ k-mers by chance. Resolution $z = (s-E)/\sqrt{E}$; k is usable if $z \geq 5$.
4. Report d at the **largest usable k** (most specific). If no k is usable, the pair is `resolution_limited` and reported without a distance.

On every one of the 53,309 chromosome pairs of the first panel, k = 23 was usable, so k = 23 is the working scale. Typical values: alleles 0.0001–0.04, homeologs 0.04–0.10, unrelated chromosomes 0.07–0.16.

5.3 Window distance: containment

For a window W of one copy and a whole other copy B, containment c = fraction of W's k-mer **positions** whose k-mer occurs anywhere in B, and $d = -\ln(c)/k$ (1.0 if nothing is shared). Containment is asymmetric and position-free: it asks whether W's sequence exists anywhere in B, not at the same coordinate.

5.4 Position-free windows

Algorithm: windowed track of copy A against copy B

1. Profile A's whole sequence against B's k-mer table (`FastK -p:B`): for every k-mer position of A, 1 if its k-mer is in B.
2. Profile A against its **own** table: the valid positions (Ns and gaps drop out of the denominator instead of reading as divergence).
3. `Profex -z` streams runs of present positions; an awk program sums them into fixed bins of gcd(window, step) positions, so the per-position profile never reaches Python.
4. Sum bins into windows (default 250 kb, step = window) along A's own coordinates; c = present / valid; $d = -\ln(c)/k$.

This replaced comparing windows at equal coordinates, which fell out of register after the first indel larger than a window (the guide's Section 12.1). Memory is about 14 bytes per base of B per concurrent profile.

5.5 Statistical tools used repeatedly

- **Robust z**: (value − median) / (1.4826 × MAD), with the MAD floored. Used for fusion containment and homeology-map windows; resistant to the very outliers being looked for.
- **Shuffle nulls**: labels (near-allelic, exchange, same-partner) are shuffled across the whole genome and the longest run recorded, 100–500 times; a real run must beat the 95th percentile. This adapts automatically to each species' noise level and window density.

- **Floors on nulls**: when a genome is nearly homozygous, near-allelic windows are rare and the shuffled maximum falls to 3–4 windows, so short clusters would pass. Run nulls are floored (8 windows by default).
- **Benjamini–Hochberg FDR** for homeolog pairs (Section 6.4).
- **Votes across copies**: real lineage structure or residual tetrasomy is seen from every copy; an assembly error is usually seen from the one or two copies it touches.

6 The stages, algorithm by algorithm

6.1 prepare: from a manifest to chromosome copies

Algorithm: prepare

1. Read the manifest: FASTA path and haplotype label (`AUTO` = read the haplotype from headers with `--hap-regex`).
2. Index each FASTA (`samtools faidx`; plain gzip must be recompressed with bgzip). Sequences shorter than `--min-len` (1 Mb) go to `unplaced.tsv`.
3. Unlocalised sequences (`unloc`, `_random`) are excluded: NCBI names them "chromosome N" too, which would give two units the same number.
4. Chromosome number from the header (`chromosome: N`, `SUPER_N`, or `--chrom-regex`). `--chrom-naming detect` falls back to automatic numbering when no header has a number.
5. **Automatic numbering**: per haplotype, keep sequences ≥ 10% of the median length of the longer half (chromosome-scale); the reference haplotype (most such sequences, excluding contig-level ones) is numbered by length; the others get provisional numbers (≥ 1001) that the matrix stage replaces by their closest reference chromosome.
6. Write `sequences.tsv` (one row per unit) and `unplaced.tsv`.

6.2 kmers

Extracts each unit to `chroms/` and builds its FastK table per k (`ktabs_kNN/`). Each table records its source file, size and modification time, and the sequence name and length; any change rebuilds it (a re-run on a renumbered assembly once silently reused old sequences).

6.3 matrix: distances, renumbering, extra sets

The all-vs-all distance matrix (Section 5.2), then three checks on chromosome numbering, all using that matrix and all logged in `matrix/chrom_label_corrections.tsv`. Every check renames the unit's files on disk, so all later stages see the corrected numbers.

Algorithm: renumbering between haplotype files

1. Reference = the source file with most units. Only other files are ever changed (correcting both sides of a swap independently can merge two chromosomes).
2. For each other unit: distance to its declared number's reference copies vs to the best other number. Ratio ≥ 1.5 makes it a candidate.
3. Match the file's units one-to-one to reference numbers, closest first (two units cannot claim one number; `ddLepDrab1` HAP4 needed this).
4. Apply a move when the matched number is ≥ 3× closer than the declared one; a mutual swap needs only ≥ 2× on both sides (`ddHesMatr1`, 3.1× and 2.3×).
5. A move into a slot whose occupant stays would duplicate a unit name. If the occupant is no closer to its own label than to the slot the matching gave it, it is **displaced** there (the label had no support). If the move vacates the occupant's matched slot, costs it ≤ 5% and improves the two units' summed distance ≥ 1.5×, the pair is **swapped** (evidence on one side only: `lpElePalu1` HAP2 chr12 is 3.2× closer to chr11 while HAP2 chr11 is tied). Otherwise the whole file's batch is rejected as `ambiguous`, never partly applied.

The displacement rule was added for the European whitefish (`fCorLav1`): its HAP2 numbering differs from HAP1 on 30 of 40 chromosomes in long cycles (7 → 9 → 10 → 11 → 13 → 12 → 7, at 10–32× evidence), but HAP2 chr33 has no allele in HAP1 at all (0.11–0.12 to everything), stayed put, blocked the cycle through chr33 and the whole correction was rejected. With displacement, 30 corrections plus 1 displacement apply cleanly.

Algorithm: extra chromosome set in one haplotype (odd ploidy)

1. Private numbers: numbers present in one haplotype only.

2. For each private unit: mean distance to every **shared** number's copies; background = the median of those.
3. Match private units one-to-one to shared numbers, closest first; keep a match if background / distance ≥ 1.3.
4. If at least 3 units and at least half the private units match, they are an extra set: they become a new haplotype (`HAP1` → `HAP1B`) carrying the matched numbers (`status = extra_set`).

The weevil `icStrMela3` (Section 9.3) is the case: HAP1 holds A as chr1–10 and B as chr11–20; each B chromosome is 0.047–0.066 from one A number against a median 0.093–0.099 to the rest (1.5–2.4×). A single private chromosome (the Monacha fusion leftover) is never a set.

6.4 homeologs: whole-chromosome pairs

Algorithm: homeolog pairs

1. For every pair of chromosome numbers (i, j): the mean distance over all copy combinations, so each number is one node whatever its copy count.
2. **Size adjustment**: regress these distances on log(L_i) + log(L_j) and |log L_i − log L_j| (three rounds, dropping pairs beyond 3 robust SD so homeologs do not shape the fit) and keep each pair's residual around the mean. Pairs are tested, and the partition and pooling are run, on these `adjusted_distance` values.
3. Fit the background: mean and SD of all pair distances, σ-clipping low values (2σ, 3 rounds) so a genome full of homeologs does not hide its own signal.
4. z = (d − mean) / SD; one-sided normal p; Benjamini–Hochberg q.
5. Accept pairs with q ≤ 0.05 **and** d ≤ 0.98 × median background (`--min-effect` 2%), greedily one partner per number, closest first.
6. **Subgenome partition**: cut the average-linkage tree of the raw number-to-number distances at two groups, and fit the size model with a same-set term added. The partition is accepted when the same-set coefficient (in robust SD) is at least 10 SD above 200 random splits of the same group sizes (z ≥ 10) and the smaller set holds ≥ 25% of the numbers.
7. **Cross-subgenome test**: with an accepted partition, remove only the size terms (keeping the set effect) and test the cross-set pairs against the cross-set background, as above. The result replaces the genome-wide test only if it finds more pairs (`pair_test` in `subgenome_partition.tsv`).

The size adjustment exists because whole-chromosome distance depends on the chromosomes' sizes: it correlates 0.8 with the size difference in Cerastium and in the diploid *Malva*. Without it, clustering sorted chromosomes into large and small; adjusted, Cerastium resolves into 18 pairs.

But the size adjustment alone also removed real subgenomes. In an allopolyploid each subgenome carries its own repeats, so non-homologous chromosomes of one subgenome can be closer than homeologs across them (Figure 2). Tested against every other pair in the genome, the homeologs then do not stand out. When one subgenome is also larger, the size regression absorbs the split. *Ranunculus repens* (2 × 16, 4x on x = 8 (Karbstein et al., 2025)) is the case: A (chr01–08, 4.7 Gb) and B (chr09–16, 3.8 Gb), within-set distances 0.064–0.068, homeologs 0.070–0.073, other A–B pairs 0.078–0.081. The genome-wide test found no pair. The subgenome partition (z = 109) and the cross-subgenome test find all eight (1–9, 2–15, 3–10, 4–12, 5–13, 6–11, 7–16, 8–14; z −13 to −17), and every A chromosome is larger than its homeolog. The same change recovers durum wheat's seven A–B pairs (sets exactly A and B, z = 179) and eight pairs in *Inula conyzae* (z = 136). No species without subgenomes reaches z = 5 (*Malva* 2.6), and Cerastium (z = 35, sets misplacing two chromosomes) keeps the genome-wide test's 18 pairs.

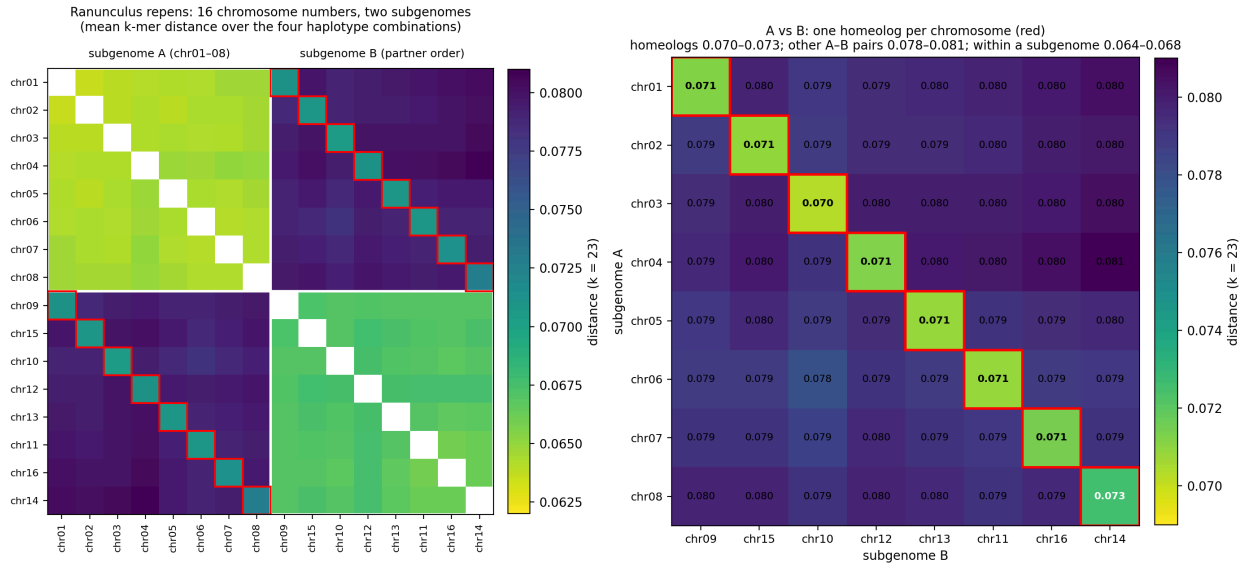


Figure 2: *Ranunculus repens*: number-to-number distances ordered by subgenome (white boxes) with homeolog pairs (red boxes), and the 8 × 8 A-versus-B block. Within a subgenome chromosomes are closer than homeologs across subgenomes; each A chromosome is closest to one B partner.

The 2% effect floor exists because a tight background makes tiny differences significant: on simulated diploids, a pair 1.25% below the median reached z = −4. Outputs: `homeolog_pairs.tsv`, every candidate in `homeolog_candidates_ranked.tsv`, and per chromosome in `ploidy_ancestry_summary.tsv` (`distance_ratio` = homeolog distance / own allelic distance).

6.5 windowed: along each chromosome, same number

For each chromosome number, every copy is windowed against every other copy of that number (Section 5.4, k = 15). These tracks carry the lineage splits along chromosomes (Section 6.9) and each copy's own-homolog distance per window, which the residual and exchange tests need (Section 6.10).

6.6 windowed-homeologs: the homeology map, then pair tracks

This stage first maps homeology genome-wide, then computes windowed tracks between the copies of every homeolog pair (whole-chromosome pairs plus block pairs from the map), plus control tracks.

Algorithm: homeology map

1. Take the haplotype with the most chromosomes. Window every chromosome (250 kb) against every **other** chromosome of that haplotype, at k = 23 (Section 5.4; n^2 profiles).
2. **Size correction**: a larger chromosome contains more of any window's k-mers. Fit one slope of window distance on log(target length) over all (window, target) pairs, dropping outliers, and subtract it.
3. **Cross-subgenome mode**: when the homeologs stage used the cross-subgenome test, each window is scored only against the other subgenome's chromosomes, so its own subgenome's shared repeats cannot win.
4. Per window: the closest chromosome, the median distance to all others, and robust z = (median − closest) / (1.4826 MAD). Homeolog-like if z ≥ 3.
5. Label each window with its closest chromosome if homeolog-like, else none. A **block** is a run of one label, bridging up to 2 windows of anything else.
6. Null: shuffle the labels over the whole genome (shuffling within a chromosome would keep a chromosome homeologous end to end as long as itself); a block must be longer than the 95th percentile of the longest shuffled run, at least 6 windows and `--min-segment-bp` (2 Mb).
7. **Reciprocity**: a block A ⟶ B counts only if B has a block back on A.
8. Pairs of numbers with ≥ 2 Mb of reciprocal blocks join the pair list; chromosomes with reciprocal blocks on two or more partners are reported.

9. **Low-divergence (LORe-like) blocks**: with at least 6 reciprocal blocks, a block is LORe-like when its median distance lies between 0.25× and 0.6× the median of all reciprocal blocks, and the pair is low in both directions. Below 0.25× it is `near_allelic`, left to the residual-tetrasomy test and its repeat guards.

Why the size correction: in the repeat-rich *Hesperis matronalis* (six chromosomes of up to 875 Mb, every window 0.022–0.026 from every chromosome) the closest chromosome was simply the largest, producing chr01/chr02 "blocks" over 21% of the genome; in Arctic charr, ten unmatched blocks pointed at its largest chromosome, which alignment showed shares only scattered short hits with them. A per-target baseline would also remove this, but absorbs real homeology where many chromosomes are related (*Lathraea*), so a single slope is used. Why k = 23: at k = 15 about 40% of a window's k-mers occur by chance in any 100 Mb chromosome, and salmonid homeologs barely stand out (masu 0.049 vs 0.057 for unrelated chromosomes). Why reciprocity: homeology runs both ways, repeats need not. Without it, Glechoma gained two short blocks onto chr16 at a third of the homeolog distance (shared repeats) and the diploid *Malva sylvestris* had seven blocks; with it, Glechoma has exactly its nine known pairs (78% of the genome) and *Malva* none.

Why a LORe-like class: in salmonids about a quarter of the genome rediploidized tens of millions of years after the rest (Lien et al., 2016; Robertson et al., 2017). Those arms stopped recombining late, so they are less diverged than other homeologs, but they are no longer near-identical, so the residual-tetrasomy test (Section 6.10) is blind to them by design. The map sees them: in whitefish 13 reciprocal blocks (243 Mb, 40% of the block span) sit at 0.011–0.026 against a median block distance of 0.045. The band's lower bound keeps near-allelic blocks out (they are residual tetrasomy or repeats: *Lathraea clandestina*'s chr03/chr04 blocks at 0.002 against a median of 0.016 are repeats, Section 6.10), and the two-way rule drops one-way low blocks. A second, younger duplication looks the same, so the reading names both.

Algorithm: pair tracks and controls

1. For each pair (i, j): window every copy of i against every copy of j and the reverse (k = 15), giving per window the distance to each partner copy.
2. Control: each paired chromosome against one unrelated chromosome (a paired number that is none of its partners), same windows.

6.7 structure: partitions and synchrony

Algorithm: genome partition

1. Cluster chromosome numbers by average linkage on the number-to-number distances, recording every k from n groups down to 2.
2. Separation = mean between-group / mean within-group distance.
3. Null: 500 random groupings with the same group sizes (seeded per species). z = (observed − null mean) / null SD; report k with $|z| \geq 2$.

A strong balanced two-group partition (`daInuConz1` 8 vs 8, z = 10.6) is a candidate subgenome structure; the rediploidization stage pools chromosomes across it (Section 6.9). For *Inula* the genome-wide pair test found nothing, but the size-independent subgenome partition of the homeologs stage (Section 6.4) gives the same sets and eight cross-set pairs. `pair_depth_cv` = SD / mean of the accepted pairs' distances: one duplication event diverges every pair to about the same depth (`daGleHede1` 0.042).

6.8 te-markers (optional)

For each pair of copies of a chromosome, at k = 13: high-copy k-mers (count ≥ 100, mostly repeats) and markers (high-copy k-mers ≥ 2× more abundant in one copy). `te_marker_fraction` = markers / high-copy k-mers of both copies. For chromosomes with three or more copies, the copies are split in two by distance and `split_ratio` = cross-group fraction / within-group fraction. The absolute fraction does not separate diploid, auto and allo (the diploid *Malva* spans 0.05–0.25); the within-genome split does.

6.9 rediploidization: fusions and per-chromosome states

6.9.1 Fusions

Algorithm: fusion detection

1. Candidates: placed copies ≥ 1.4× the median length of their sibling copies, and unplaced scaffolds ≥ half the median chromosome length.
2. For a candidate, containment of each **other** chromosome's best copy in it (shared k-mers / that copy's k-mers).
3. Background: containment between every pair of ordinary units of different numbers; robust z.
4. A component needs z ≥ 10, ≥ 2× the background median, and ≥ ¼ of the strongest hit.
5. Placed copies only: the partner must be ≥ 2.5× more enriched in the candidate than in its least-enriched unfused sibling (`sibling_excess`). Ancient homeology is shared by every copy; a fusion is not.
6. `status = fusion` when every partner number is **missing** from that haplotype (absorbed, not duplicated); otherwise `candidate_partner_present`.
7. `lineage_divergence` $= -\ln(c)/k$ of the partner containment: how long the fused lineage has been diverging.

A fusion on one haplotype in a genome with no other duplication (the land snail *Monacha*, chr23+chr24 joined in HAP1 only) is a fusion polymorphism or a scaffolding join, not rediploidization; the summary says so.

6.9.2 Lineage split within a chromosome number

Algorithm: split, extent and null (three or more copies)

1. Bipartition: seed with the two most distant copies, assign the rest to the closer seed. `dist_split` = mean cross-group / mean within-group distance; **balanced** if both groups have ≥ 2 copies.
2. Along every copy that has a partner in its own group: per window, mean distance to the other group / mean distance to its own group. Split windows have ratio ≥ 1.25; segments are runs (1-window gaps bridged) ≥ 2 Mb.
3. The **median copy**'s covered fraction is the reading (a real split shows from every copy). `whole` if ≥ 50%, `regional` if any segment.
4. Null: repeat for groupings that **cut across** the observed one (other balanced splits; or isolating each other copy), pooled genome-wide. The extent counts only if it beats that null's 95th percentile and every crossing grouping of its own chromosome.

6.9.3 Lineage switches

Algorithm: lineage switches (three or more copies)

1. Along each copy, label every window with its closest other copy of the same number; runs of one label (2-window gaps bridged) covering ≥ 10% of the chromosome and ≥ 2 Mb are kept.
2. Two consecutive runs with neighbours X then Y are a **switch** when over the first the copy is ≥ 2.5× closer to X than to Y (median over windows), over the second the reverse, and the far copy sits ≥ 3× the genome's allelic level away (per chromosome the most allelic copy's median closest distance; median over chromosomes).

The exchange readout compares different chromosome numbers; when subgenome copies share a number, an exchange shows as a switch. The weevil's HAP1 chr09 is A to 40 Mb and B after it. In polysomic genomes the nearest copy changes in short blocks between equally close copies, which the contrast and lineage-level rules exclude.

6.9.4 Divergence along fused chromosomes

Each confirmed fusion is windowed along the fused sequence against the unfused copies of its components (`d_unfused`, labelled by component) and the other fused copy (`d_fused`). The junction is the boundary that best separates the two components' windows; `junction_gradient` = median `d_unfused` within the inner 20% of each arm over the outer 20%. Above 1 means divergence peaks at the fusion site (Xie et al., 2026).

6.9.5 Pooling for two-copy genomes

With two copies per number nothing can be split. The chromosome is pooled with its whole-chromosome homeolog partner (four copies), or else with its reciprocal best match across a significant (z ≥ 10, balanced) two-group partition. Pooled groups are read on whole-chromosome distance only.

6.9.6 States

State	Rule
`fusion_lineages`	the chromosome is a component of a detected fusion
`resolved_lineages`	balanced split, `dist_split` ≥ 2.5 and extent `whole`; pooled: `dist_split` ≥ 2.5 (any two different numbers are far apart, so for pooled pairs this mostly says "these are homeologs, not alleles")
`partially_resolved`	two or more of: `dist_split` ≥ 1.25, `te_split` ≥ 2, a windowed segment; pooled: resolved but carrying residual tetrasomy
`candidate`	exactly one of those
`tetrasomic_like`	none: the copies are interchangeable
`one_divergent_copy`	one copy apart, with chromosome-wide (distance or TE) evidence; a windowed segment alone does not count
`not_assessable`	fewer than three copies and nothing to pool with

Table 3: Per-chromosome states (`rediploidization_by_chrom.tsv`).

6.10 Residual tetrasomy and homeologous exchange

Both read the pair tracks of Section 6.6 against each copy's own-homolog track from Section 6.5, for every homeolog pair (whole-chromosome or block).

Algorithm: residual tetrasomy

1. **Allelic level** per copy: median over windows of the distance to its closest own-number copy; per chromosome the median over its copies; capped at 1.5× the genome median (a chromosome carrying a large exchange must not raise its own yardstick).
2. Per window of a copy, **near-allelic** when all of:
   - closest partner copy < 3 × max(allelic level, 0.001);
   - closest partner copy < 0.25 × the species' median homeolog distance;
   - the copy's **own** homolog is also within 3 × allelic level (in residual tetrasomy every copy is close to every other);
   - the unrelated control is not as close (shared repeats are close to every chromosome carrying them);
   - the window is not low-complexity: its sequence compresses (zlib) to at least 0.75 of the genome's median ratio.
3. Runs of near-allelic windows (1-window gaps bridged) count when longer than the 95th percentile of the genome-wide longest run under shuffling (200 permutations), floored at 8 windows, and ≥ 2 Mb.
4. **Pair vote**: the pair has residual tetrasomy when more than half the copies of its two numbers show a counted run (3 of 4 with two haplotypes).
5. Per chromosome: the best-covered copy's segments, summed over partners; segments within 10% of an end are `start` / `end` .

Algorithm: homeologous exchange

1. Per window of a copy: **exchange-like** when the closest partner copy is less than half the distance to the copy's own homolog, and the own homolog is beyond 3 × allelic level.
2. Runs against their own shuffle null (same floor and length).
3. No pair vote: an exchange is usually carried by one copy; `exchange_support` says how many.

Why the complexity guard matters

Position-free containment cannot tell shared repeats from shared sequence. Glechoma's chr14/chr17 block

(16–22 Mb) read as near-allelic in all eight copy directions, but it is a centromeric satellite: it compresses to 0.06–0.13 against 0.32 for the rest of the chromosome and aligns at only 94–96%. Homeologous centromeres inherit one satellite family, so the unrelated control does not remove it. *Lathraea clandestina*'s chr03/chr04 segment is repeat-dense (0.19–0.27 against a median 0.29) and aligns at 98–99% against 99.9–100% for the alleles. Pilosella's segments, by contrast, align at the allelic level (99.6–99.8%) and are kept.

Why the own-homolog condition matters
The first version counted any window close to a homeolog copy. An exchange in one copy then passed the pair vote from the three copies that see it (the simulated `segmental_homeology` scenario failed this way), and so did contigs swapped between homeologs in masu's reference-scaffolded HAP2. 6 of masu's 8 earlier residual chromosomes rested on windows whose own homolog was 0.03–0.085 away, homeolog distance, not missing sequence.

6.11 summary: the decision logic

The summary reads the files above and answers four questions, each with a confidence and the evidence it used. In order of evaluation:

Ploidy. Copies = the modal copy number per chromosome (high confidence: it is what the assembly contains). Duplicated sets, first match wins: a genome partition into groups of ≥ 2 numbers (k ≥ n/4, z ≥ 10); homeolog pairs covering ≥ 50% of the numbers; a balanced k = 2 partition; reciprocal homeology-map blocks covering ≥ 30% of the genome with ≥ 2 pairs (then x is not read from the numbers, since the sets are reshuffled). With ≥ 3 pairs but no sets, an older duplication "shows on part of the genome". Odd copy numbers and split-out extra sets are stated.

Origin-like structure.

1. A fusion on one haplotype with no other duplication: not assessable.
2. Two copies and no duplication evidence (no sets, fewer than 3 whole- chromosome pairs at $|z| \geq 5$, and fewer than 3 map pairs): not assessable (gar and bowfin pass one pair each; Stagnicola one strong and two marginal).
3. One copy apart on most chromosomes: the same haplotype every time is read as an assembly or phasing problem, unless that haplotype is a split-out extra set, which reads as AAB-like (high); changing haplotype reads as A…AB-like.
4. Two-copy (pooled) genomes: tetrasomic-like on most pairs → auto-like; residual tetrasomy on ≥ 2 independent pairs → "auto-like origin, mostly rediploidized"; ≥ 60% split → allo-like (high if pair-depth CV ≤ 0.06).
5. Three or more copies: mostly tetrasomic → auto-like; mostly split → allo-like; a mixture → mixed.

TE markers. Only `te_split` is read: ≥ 2 on chromosomes that also split by distance means separate repeat histories.

Rediploidization. Fusions (between copies of a duplicated genome) → yes (high); mixtures of split and tetrasomic-like chromosomes → likely under way; two-copy genomes with residual tetrasomy → partly; no residual tetrasomy but chromosomes with two or more homeolog partners → advanced; otherwise diverged throughout. Exchange candidates, rearrangements and block-divergence spread (CV ≥ 0.15) and LORe-like blocks (at least one reciprocal pair, ≥ 4 Mb) are appended as "Also:". Residual tetrasomy ≥ 90% at chromosome ends is low confidence (homeolog-specific subtelomeric repeats pass the control), although real delayed rediploidization and homoeologous exchange are also biased towards chromosome ends (Lien et al., 2016; Zhang et al., 2020), so terminal signal is flagged, not discarded.

6.12 report, panel, simulate, cleanup

`report` writes one self-contained HTML page per species. `panel` stacks all species into `panel_summary.tsv`, `rediploidization_panel.tsv`, `species_summaries.tsv`. `simulate` writes synthetic assemblies with a known answer (Chapter 10). `cleanup` deletes k-mer tables and keeps every result.

7 What every output means

`OUTPUTS.md` in the repository documents every column. This chapter is the map: what each file is for and the columns you will actually read.

File	What to read in it
`sequences.tsv`	one row per unit: `unit_id`, `hap`, `chrom`, `seq_id`, `length`, `source`; `numbering` (`names` / `auto`)
`unplaced.tsv`	every excluded sequence and why (`below-min-len`, `unlocalised`, `no-chrom-match`, ...)
`matrix/whole_chrom_pairs.tsv`	per unit pair: `chosen_k`, k-mer counts, `jaccard_at_chosen_k`, `distance`, `resolution_limited`
`matrix/ploidy_summary.tsv`	copies per chromosome number and which haplotypes
`matrix/chrom_label_corrections.tsv`	`status` (`corrected`, `displaced`, `extra_set`, `ambiguous`), old/new number, the distances and ratio, `new_hap`
`matrix/*heatmap*.png`	the full matrix; the contrast version shows block structure
`homeologs/homeolog_pairs.tsv`	accepted pairs: `mean_distance`, `z_score`, `q_value`
`homeologs/ploidy_ancestry_summary.tsv`	per chromosome: partner, `distance_ratio`
`homeologs/subgenome_partition.tsv`	the two subgenomes, their z against random splits, and which pair test was used
`homeologs/homeology_map.tsv`	per window: `best`, `best_dist`, `median_dist`, `z`, `homeolog_like`
`homeologs/homeolog_blocks.tsv`	per block: `chrom`, `partner`, `start`–`end`, `median_dist`, `position`, `reciprocal`, `low_divergence`
`homeologs/homeology_map_summary.tsv`	`duplicated_frac`, `n_partner_pairs`, `multi_partner_list`, `block_dist_cv`, `n_lore_blocks`, `lore_bp`
`homeologs/homeology_map.png`	each chromosome painted by its partner; hatched = unmatched
`homeologs/windowed_chrAAxBB.png`	window distances between the copies of a homeolog pair
`windowed/windowed_all.tsv`	per window of `unit_a` vs `unit_b` (same number): `kmers_a`, `shared`, `distance`
`structure/genome_partition.tsv`	partitions with `group_sizes`, `separation_ratio`, `z_score`, `groups`
`structure/pair_synchrony.tsv`	`pair_depth_median`, `pair_depth_cv`
`subgenomes/te_marker_fraction_by_lineage.tsv`	`split_ratio` per chromosome
`rediploidization/fusions.tsv`	per tested scaffold: `components`, `containment`, `sibling_excess`, `lineage_divergence`, `status`
`rediploidization/rediploidization_by_chrom.tsv`	per chromosome: groups, `dist_split`, `te_split`, `split_extent`, `split_segments`, `copy_state`, `state_basis`, `pooled_with`, residual and exchange columns, `block_partners`
`rediploidization/residual_tetrasomy.tsv`	per copy and partner: windows, near-allelic and shared-with-control counts, `allele_level`, runs vs null, segments, exchange segments
`rediploidization/rediploidization_summary.tsv`	genome-level counts, residual and exchange totals, `map_*` metrics
`summary.tsv`, `summary.md`	the four answers, confidence, evidence

File	What to read in it
`report.html`	everything on one page

Reading a per-chromosome row in practice

Start with `copy_state` and `state_basis` (`copies`, `homeolog_pool`, `partition_pool`, `fusion`). For `copies`, look at `dist_split` and `split_extent`; for a pool, at `pooled_with` and `dist_split`. Then `residual_support` (e.g. `chr17:3/4` = 3 of the pair's 4 copies show a counted run), `residual_segments` (`HAP1:16.8-19.5Mb(interior)~chr17` = on HAP1, 16.8–19.5 Mb, mid-chromosome, against chr17), and the exchange columns the same way.

8 Reading the diagnostic plots

Every stage writes plots next to its tables, and `report.html` collects them. This chapter takes them one at a time: what is drawn, how to read it, the patterns that matter, and the traps. Examples are from real runs.

8.1 Conventions shared by every heatmap

- **Colour runs from yellow (close) to dark purple (far)** (matplotlib's `viridis_r`). Yellow always means "similar sequence".
- **The scale is stretched** to the 2nd–98th percentile of the values in that plot (stated in the title), so colours are comparable within a plot, not between plots. Read the colour bar.
- **Light grey means no data**: a copy shorter than the others ends earlier, or a cell is masked on purpose (the diagonal of a contrast heatmap).
- **Distances are k-mer distances**, not alignment identity: roughly the per-base divergence where sequence is shared, but containment-based window distances saturate in repeat-dense sequence (Section 6.10).

8.2 Whole-chromosome distance heatmap (`matrix/whole_chrom_distance_heatmap.png`)

What it shows. Every chromosome copy against every other, at the chosen k (Section 5.2), clustered so that similar copies sit together.

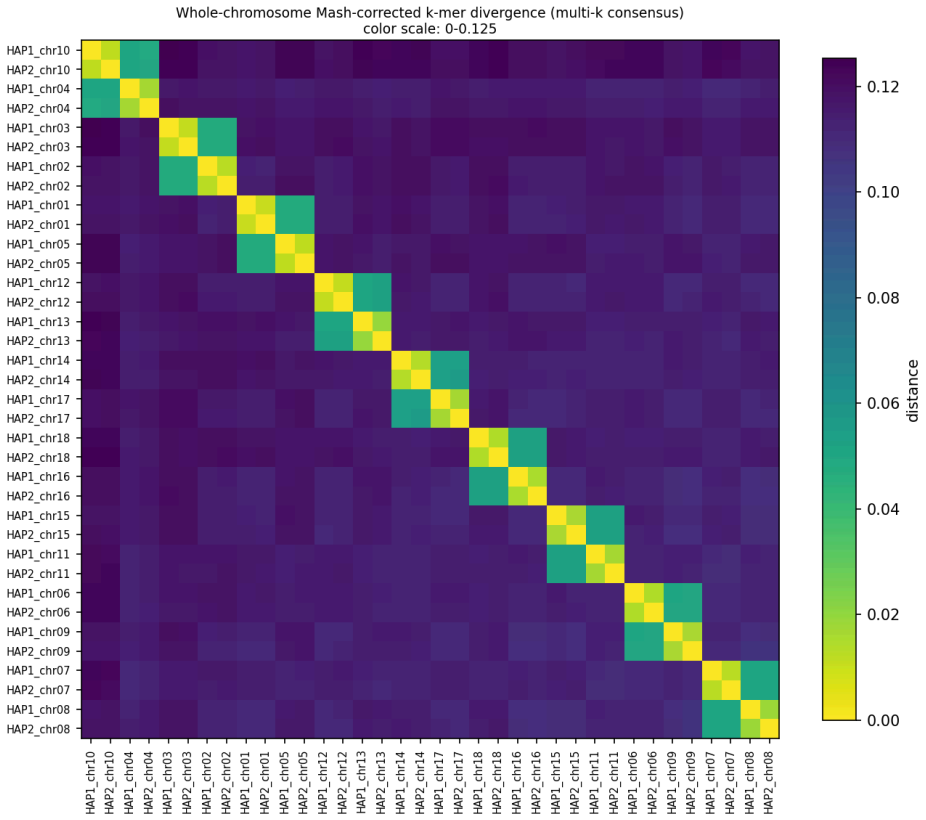


Figure 3: Glechoma (two haplotypes × 18 numbers). Yellow 2 × 2 squares on the diagonal: the two alleles of each chromosome (0.01). Teal 2 × 2 squares next to them: a homeolog pair, e.g. chr10 with chr04 (0.05). Everything else is purple: unrelated chromosomes (0.11–0.12).

How to read it. Look for the block sizes along the diagonal:

- **Yellow squares the size of the copy number** (2 × 2 with two haplotypes, 4 × 4 with four): the copies of each chromosome. In a tetrasomic genome the 4 × 4 block is uniformly yellow; in a genome with two lineages per chromosome it splits into two yellow 2 × 2 squares inside a lighter 4 × 4.
- **Teal blocks pairing two numbers**: homeologs. If most numbers have one, the genome has duplicated sets (Glechoma: nine).
- **A copy whose row is lighter or darker than its siblings'**: one divergent copy (AAAB-like, or an odd haplotype).

Traps. A yellow cell far from the diagonal between two different numbers means a numbering mismatch: check `chrom_label_corrections.tsv` (the matrix stage fixes most). Chromosome size also shifts the background colour slightly (Section 6.4), which is why the homeolog test uses adjusted distances.

8.3 Contrast heatmap (`whole_chrom_distance_heatmap_contrast.png`)

The same matrix with the diagonal masked (white) and the colour stretched over the remaining values. Masking the near-zero self cells lets the background use the whole colour range, so faint structure among unrelated chromosomes, for example two groups of chromosomes that share repeat families (wheat's A and B, Section 10.2), becomes visible. Read it for **structure in the background**, not for the copies themselves.

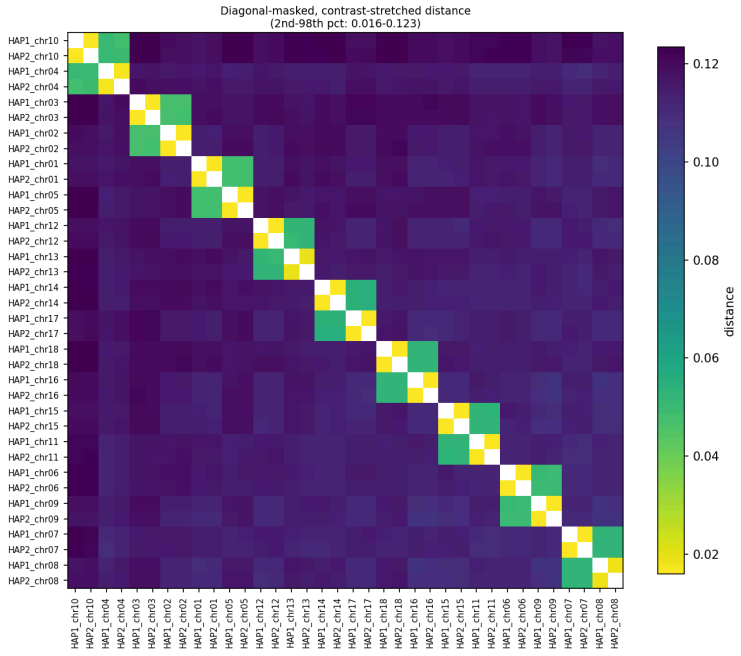


Figure 4: Glechoma, contrast version: the homeolog squares stand out against a flat background; there is no further grouping of chromosome numbers.

8.4 k-resolution diagnostic (`matrix/k_resolution_diagnostic.png`)

What it shows. For every pair of copies (one grey line each), the corrected distance at each k of the sweep; points are red where that k is below the chance-collision floor (Section 5.2).

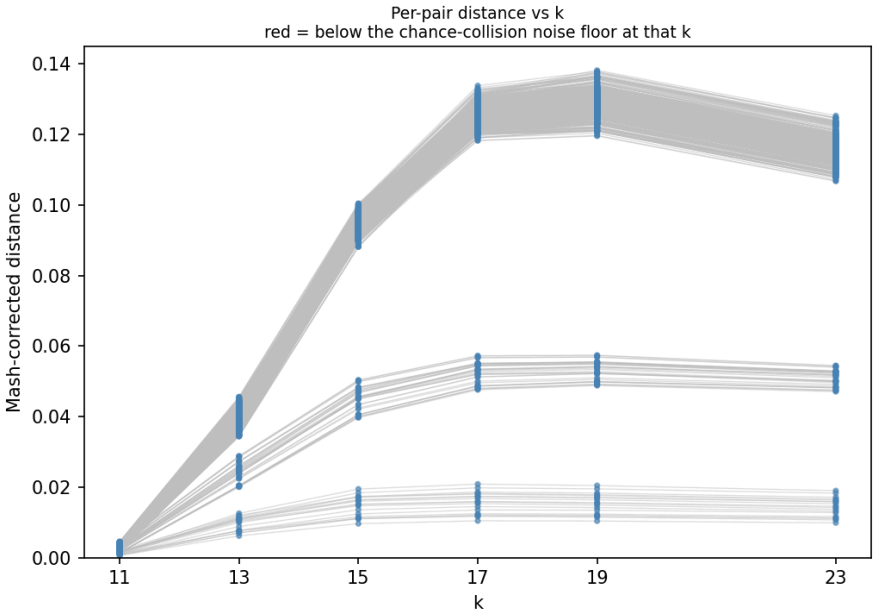


Figure 5: Glechoma. Three bands: alleles (0.01–0.02), homeologs (0.05) and unrelated chromosomes (0.11–0.13). At k = 11 everything collapses to 0: short k-mers are shared by chance, so k = 11 cannot separate anything.

How to read it. The distances should level off as k grows: the plateau (here k ≥ 17) is the real divergence, and the matrix uses the largest usable k. Distinct bands are the classes of comparison; a genome with homeologs has a middle band. Red points at large k mean those pairs are beyond what k-mers can measure.

8.5 Homeolog candidates (`homeologs/homeolog_pairs.png`)

What it shows. Every pair of chromosome numbers, ranked by mean distance; accepted homeolog pairs in red and labelled.

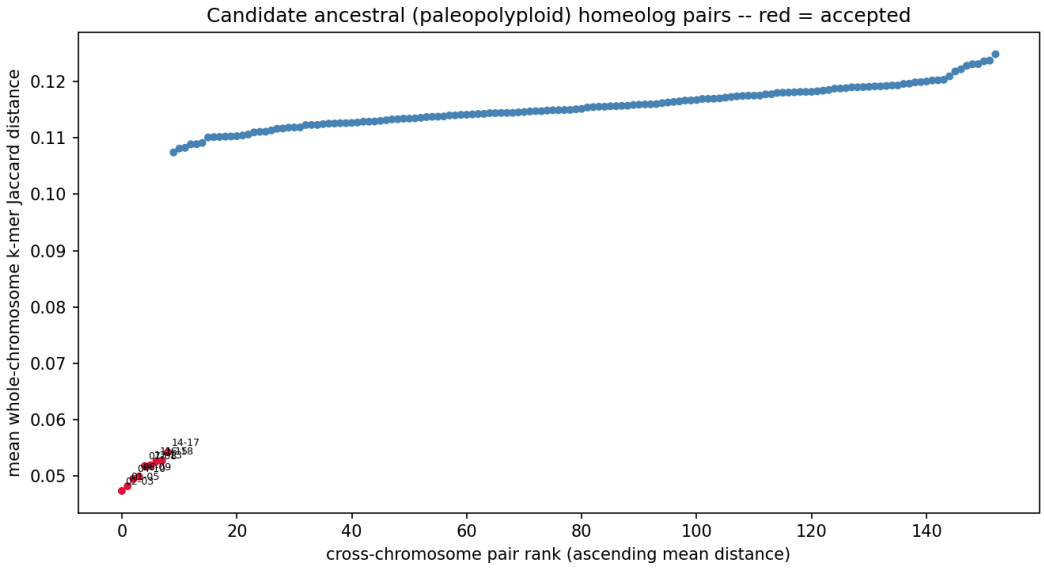


Figure 6: Glechoma: nine accepted pairs (red, 0.05) far below a continuous background (0.11–0.12). A gap like this is the clearest picture of a duplicated genome.

How to read it. A clean duplication shows a **gap** between a low cluster and the rest. A gradual slope without a gap, with a few red points at its foot, means weak evidence: read the z-scores in `homeolog_candidates_ranked.tsv`. The plot shows raw distances; acceptance is tested on size-adjusted ones (Section 6.4), so a red point can sit slightly above a blue one.

8.6 Reordered chromosome-number heatmap (`homeolog_pairs_reordered_heatmap.png`)

What it shows. Chromosome numbers (each averaged over its copies) clustered by average linkage, accepted pairs outlined in red.

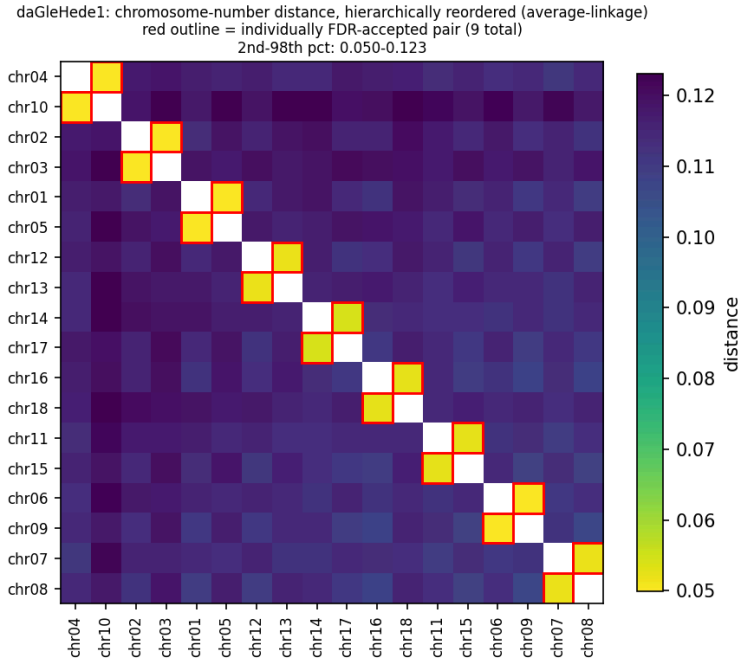


Figure 7: Glechoma: nine yellow 2 × 2 squares, each outlined, on a flat background: two sets of nine.

How to read it. The square sizes give the number of sets: 2 × 2 for sets of two (tetraploid on a two-haplotype assembly), 3 × 3 for three (hexaploid history, e.g. `drMyrSpic1`). Larger blocks of moderate colour without outlines are diffuse group structure (the genome partition, Section 6.7).

8.7 Homeology map (`homeologs/homeology_map.png`)

What it shows. Each chromosome of the mapped haplotype as a bar, painted along its length by the chromosome its homeolog block lies on (labels give the partner number). Grey: no homeolog detected. Hatched and faded: a block without a block back from the partner (not counted).

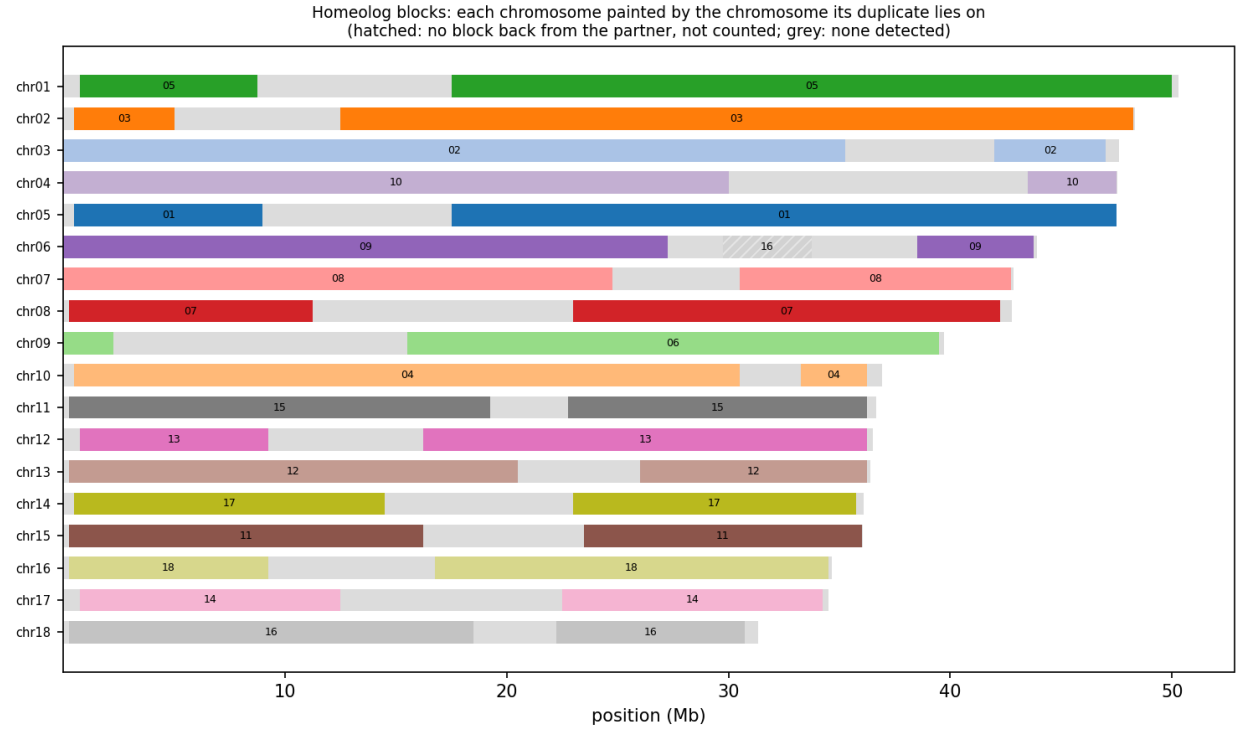


Figure 8: Glechoma: almost every chromosome painted by one partner over most of its length (chr01 by chr05, chr02 by chr03, ...): 79% of the genome in homeolog blocks, every chromosome with one partner.

How to read it. One colour per chromosome along most of its length: a duplicated genome with intact chromosomes (Glechoma, Salix, Trifolium). Two or more colours on one chromosome: its arms are duplicated on different chromosomes, i.e. fusions, fissions or translocations since the duplication (salmonids, Potamogeton). Mostly grey: no detectable duplication, or one too old for k-mers (Section 3.3). The charr map is in Section 9.4.

8.8 Windowed tracks along a chromosome (`windowed/windowed_chrNN.png`, `_heatmap.png`)

What it shows. Along each copy of one chromosome number, in 250 kb windows, the distance to each other copy (position-free, Section 5.4). The line plot draws one line per directed pair; the heatmap one row per directed pair ("HAP3 vs HAP1" = windows of HAP3 against the whole of HAP1).

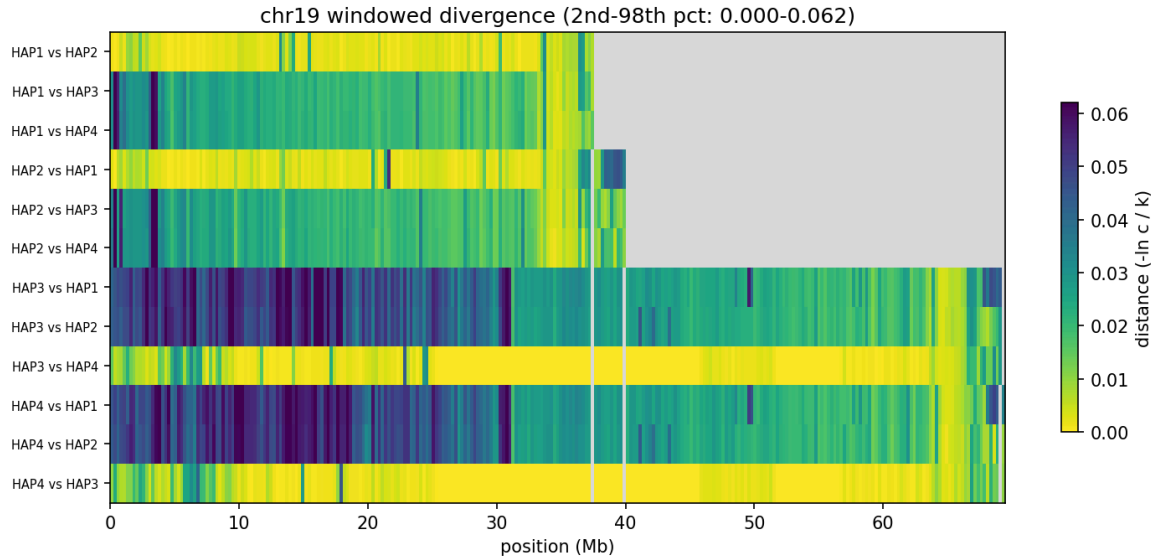


Figure 9: Snow carp `SchCurv1` chr19. HAP1 vs HAP2 and HAP3 vs HAP4 are yellow all along: two pairs of close copies. Every cross row (HAP1/HAP2 against HAP3/HAP4) is dark: two lineages over the whole chromosome. HAP3 and HAP4 run to 69 Mb because they carry the fused chr22; the HAP1 and HAP2 rows end at 38–40 Mb (grey).

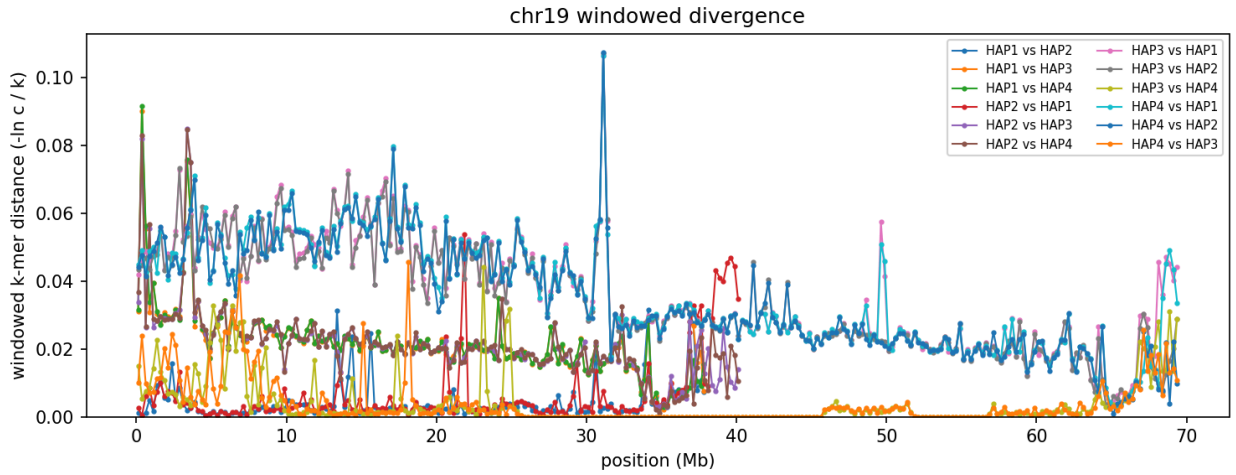


Figure 10: The same chromosome as lines: the cross pairs (0.04–0.06 on the chr19 part) sit far above the within-lineage pairs (near 0).

How to read it.

- **All rows similar and low**: interchangeable copies (tetrasomic-like).
- **Rows in two groups**, low within and high across, all along: two lineages (`resolved_lineages`, `fusion_lineages`).
- **Groups that appear only over part of the chromosome**: a regional split (snow carp chr17, the last 8 Mb: an inversion in HAP3).
- **One copy's rows dark everywhere**: one divergent copy.
- **A row that is low then high, while another goes high then low at the same point**: a copy changing lineage (a lineage switch, Section 6.9).

Traps. Isolated dark spikes in one row are usually sequence present in one copy and missing from the other (gaps, unplaced contigs), not divergence. Short scattered blocks in a polysomic genome are ordinary haplotype structure; the tool only counts splits that beat a null (Section 6.9).

8.9 Genome overview (`windowed/windowed_genome_overview*.png`)

Every chromosome's heatmap side by side on one shared colour scale, x axis in Mb. Use it to find the chromosomes worth opening.

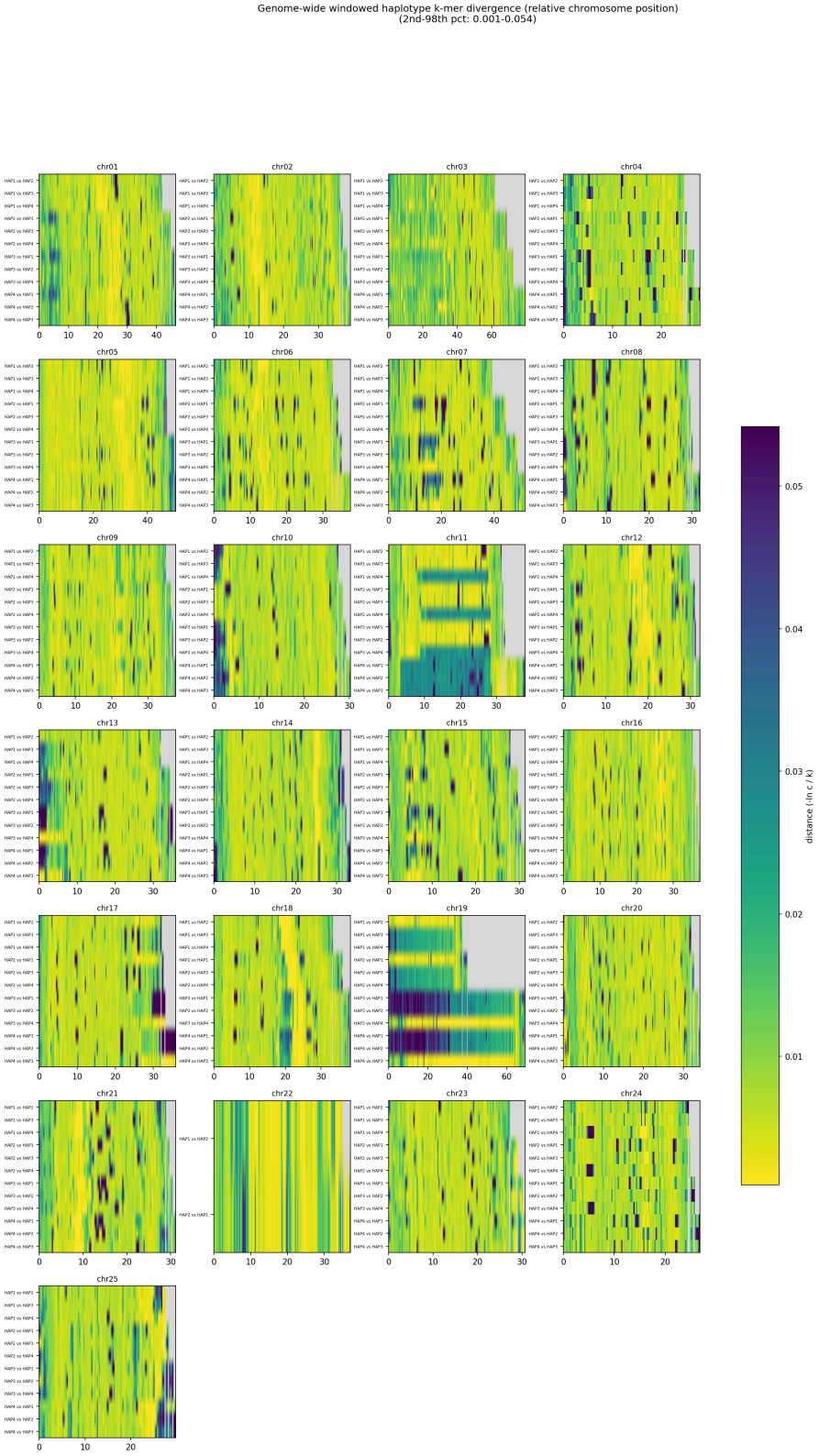


Figure 11: `SchCurv1` : most chromosomes are speckled yellow-green (four interchangeable copies with scattered haplotype blocks). Three stand out: chr19 (two lineages along its whole length, the fusion), chr11 (banded rows over 5–28 Mb) and chr17 (dark rows at its end, the inverted region).

How to read it. Look for **banding**: rows that stay dark or light over long stretches. Speckle is noise or ordinary polymorphism; bands are structure.

8.10 Tracks between homeologs (`homeologs/windowed_chrAAxBB*.png`, overview)

The same plots for the copies of a homeolog pair: windows of each copy of one number against each copy of the other. The level is the homeolog distance; a stretch where it falls to the allelic level is a residual-tetrasomy or exchange candidate (Section 6.10).

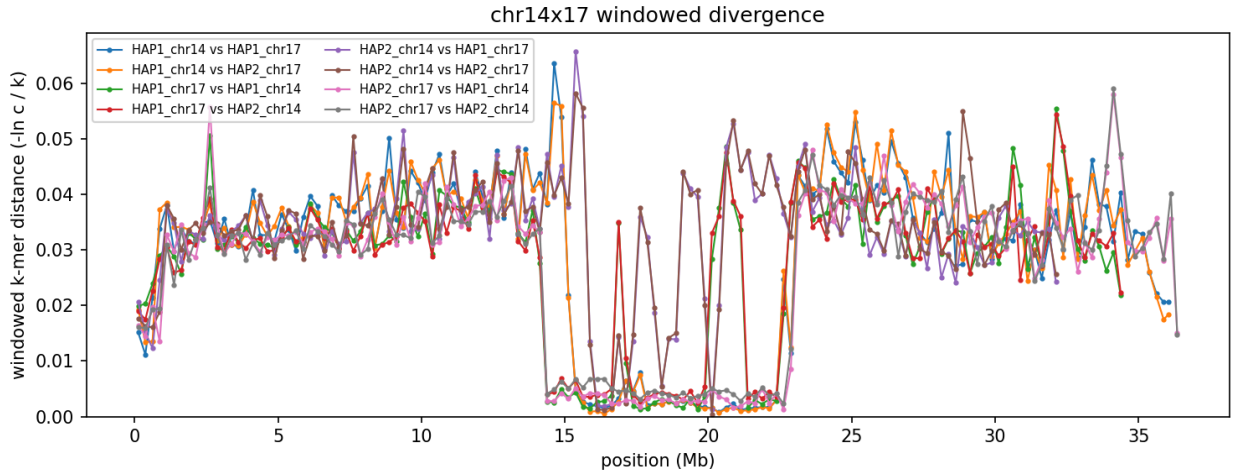


Figure 12: Glechoma chr14 × chr17: homeologs 0.03–0.05 apart, falling to 0.003 over 14–23 Mb in every direction. This is the trap: the block is a centromeric satellite shared by the two homeologs (it compresses 3–5× better than the rest and aligns at only 94–96%), not retained pairing.

How to read it. A drop seen from **all** copies of both numbers, in sequence that is not repeat-dense, is residual tetrasomy (Pilosella chr04 × chr05). A drop in one copy only, while its own homolog stays far, is an exchange. A drop exactly where the chromosome is repeat-rich, typically mid-chromosome or at the ends, is most likely shared repeats: the tool now excludes those windows, but the plot still shows them.

8.11 TE-marker painting (`subgenomes/te_markers_windowed_*.png`)

What it shows (with `--with-te-markers-windowed`). For a pair of copies, each window coloured by whose fossil-TE markers it matches (Section 6.8): blue for the first copy, red for the second, grey when markers of both or neither dominate (ambiguous), light grey when the window has no markers.

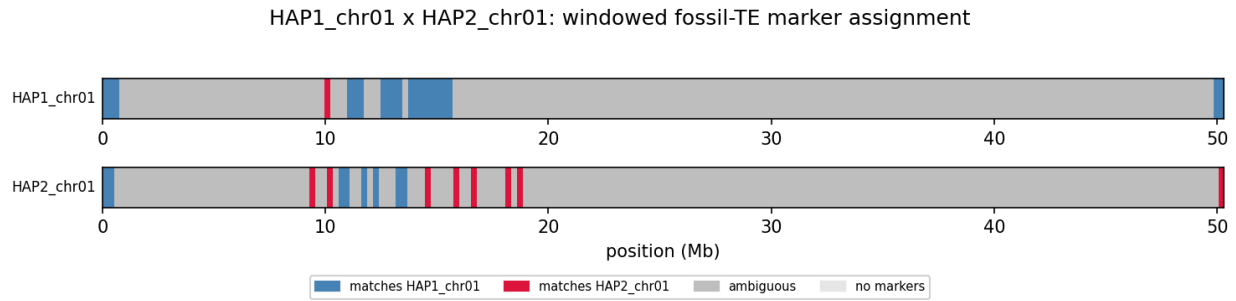


Figure 13: Glechoma chr01, its two alleles: almost all grey (ambiguous), with a few short blue and red patches. The two alleles share one repeat history, so this is what "no subgenome difference" looks like.

How to read it. Between copies from different subgenomes or lineages, each copy should be painted mostly in its own colour along long stretches: separate repeat histories. Between alleles of one lineage (as above) the painting is grey. Mixed painting with long switches between colours along one copy points to exchange between lineages. Read it together with `te_split`: the painting shows where, the split ratio how strong.

8.12 Fusion profile (`rediploidization/fusion_profile_<scaffold>.png`)

What it shows. Along a fused chromosome, each window's distance to the unfused copies of the component it belongs to (coloured by component), and to the other fused copy (grey). The title gives the junction and the gradient (Section 6.9).

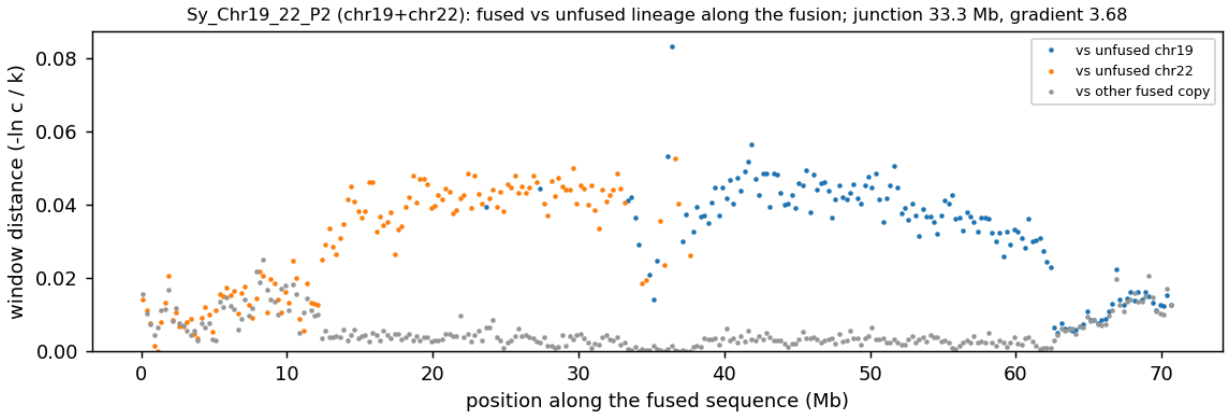


Figure 14: `SchYoun1`, the chr19+chr22 fused scaffold of HAP4. The chr22 part (orange) and the chr19 part (blue) meet at 33 Mb. Divergence from the unfused copies is 0.04–0.05 around the junction and falls to 0.01 at both ends (gradient 3.7); the two fused copies (grey) stay 0.003 apart throughout.

How to read it. A peak at the junction falling towards the ends: recombination with the unfused copies stopped first next to the fusion and divergence is spreading outwards (Xie et al. 2026). A flat profile: a young fusion (the youngest snow carp wave, gradient 0.6–0.7) or an old one whose divergence has already spread. The grey points show the fused lineage's own diversity.

8.13 Panel PCA (`meta/supplementary/poly_space_pca.png`)

What it shows. An exploratory principal component analysis of six numbers per species (allelic distance, homeology-map duplicated fraction, paired fraction, best partition z, resolved and tetrasomic fractions), each scaled to unit variance; species missing any number are left out.

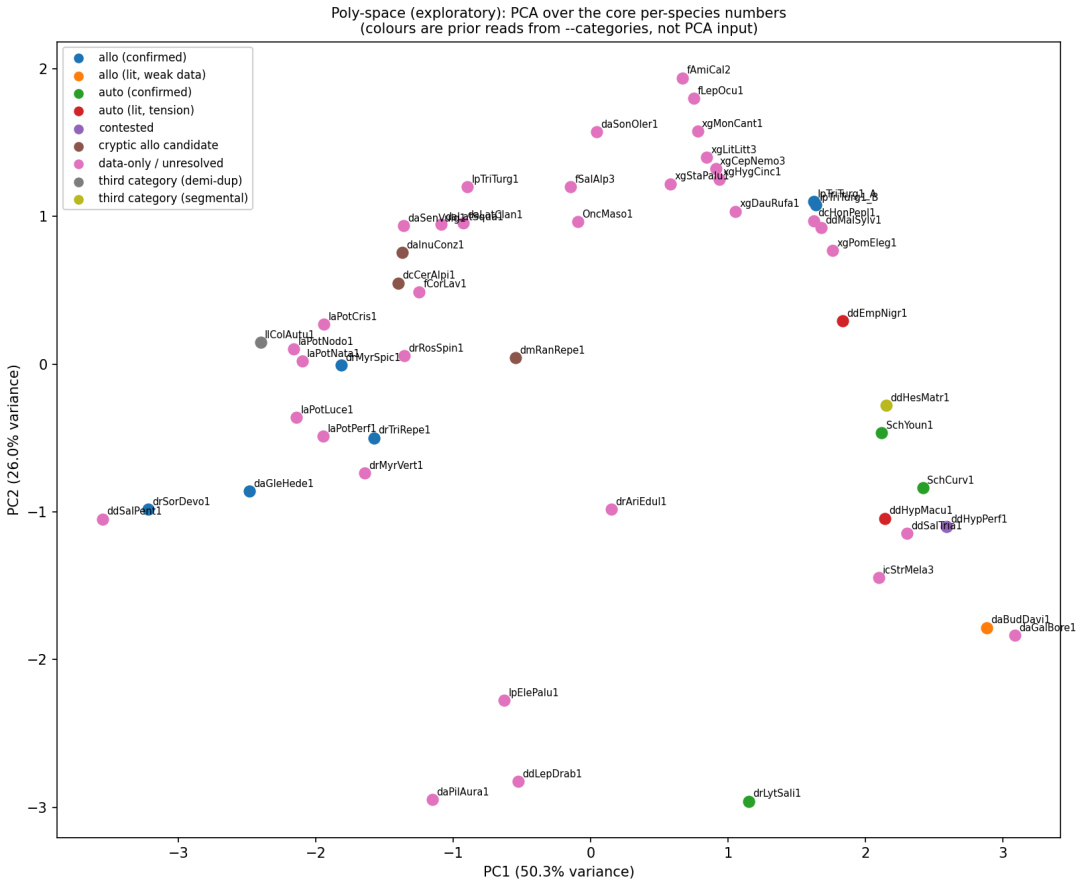


Figure 15: 52 species. Left: duplicated, paired, allo-like genomes (Salix, Sorbus, Glechoma, Potamogeton). Bottom right: assemblies with four copies where many chromosomes are tetrasomic-like (Buddleja, Galium, Lythrum). Top right: genomes without detectable duplication (snails, gar, bowfin, *Malva*).

How to read it. PC1 (50%) runs from duplicated genomes (negative: duplicated fraction, paired fraction and partition z all load −0.48 to −0.52) to undivided or tetrasomic ones (positive: tetrasomic fraction +0.25, allelic distance +0.21). PC2 (27%) separates genomes whose chromosomes split into lineages (positive: resolved fraction +0.44) from high-heterozygosity, tetrasomic ones (negative: allelic distance −0.56, tetrasomic fraction −0.50). It is a picture of where species sit relative to each other, not a classifier.

9 Walkthroughs

9.1 Glechoma hederacea: a two-haplotype allotetraploid

Input. Two haplotypes, 18 chromosomes each, numbered by name.

Matrix. HAP1 chr02 vs HAP2 chr02: 0.0128 (alleles). HAP1 chr02 vs HAP1 chr03: 0.0474. HAP1 chr01 vs HAP1 chr02: 0.113 (unrelated). So chr02 and chr03 are four times further apart than alleles but less than half as far as unrelated chromosomes: homeologs.

Homeologs. Nine pairs, all at q ≈ 0 and z between −18.6 and −20.7: chr02–03 (0.047), chr01–05, chr04–10, chr06–09, chr07–08, chr12–13, chr11–15, chr16–18, chr14–17 (0.054). `distance_ratio` 3.5 (median). Pair-depth CV 0.042: one synchronous event. The genome partition into nine pairs has z = 23.8.

Homeology map (Figure below). 78% of the genome in 39 reciprocal blocks, the same nine pairs, no chromosome with two partners, block CV 0.047.

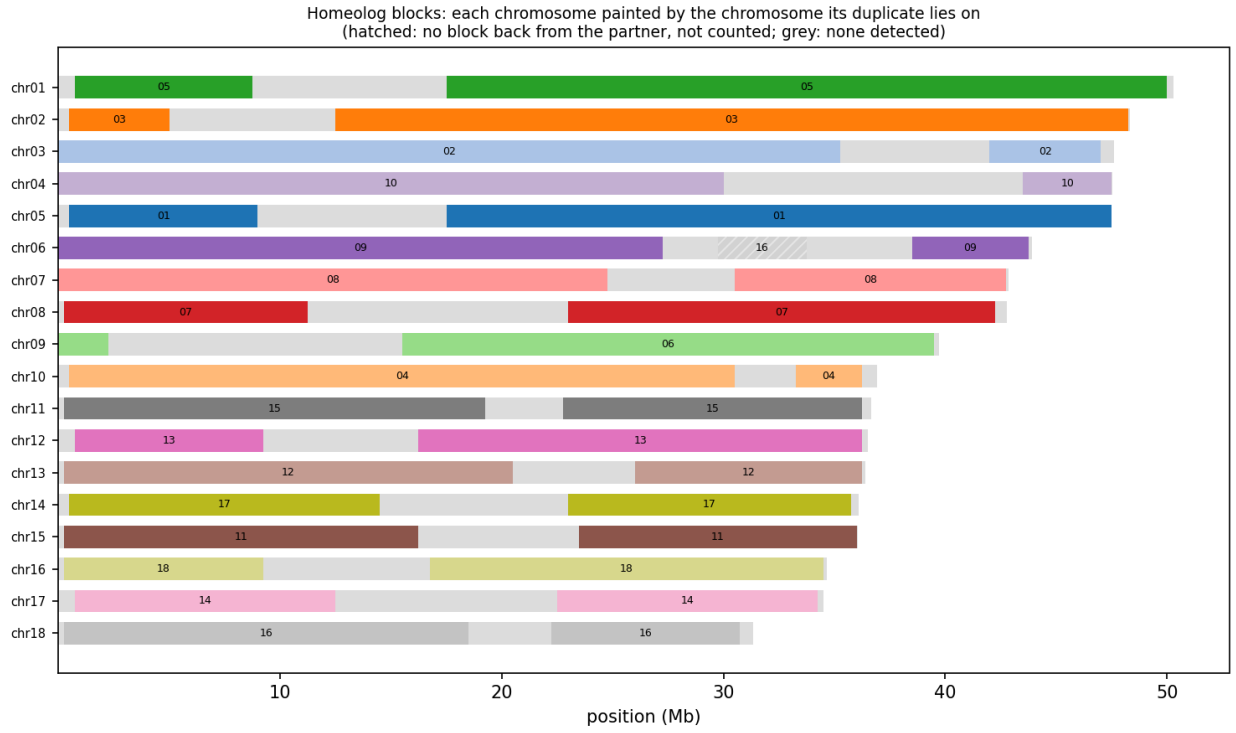


Figure 16: Glechoma: every chromosome is duplicated on exactly one partner along most of its length. Grey gaps are regions where the homeolog is not detected (diverged beyond k-mer reach, or lost).

Rediploidization stage. Each chromosome is pooled with its partner (four copies); every pool splits by `dist_split` 3.5–4.6, so all 18 are `resolved_lineages`. The residual test finds nothing. An earlier version called chr14/chr17 residual-tetrasomic over a mid-chromosome block (figure below); that block is a centromeric satellite and is now excluded (Section 6.10).

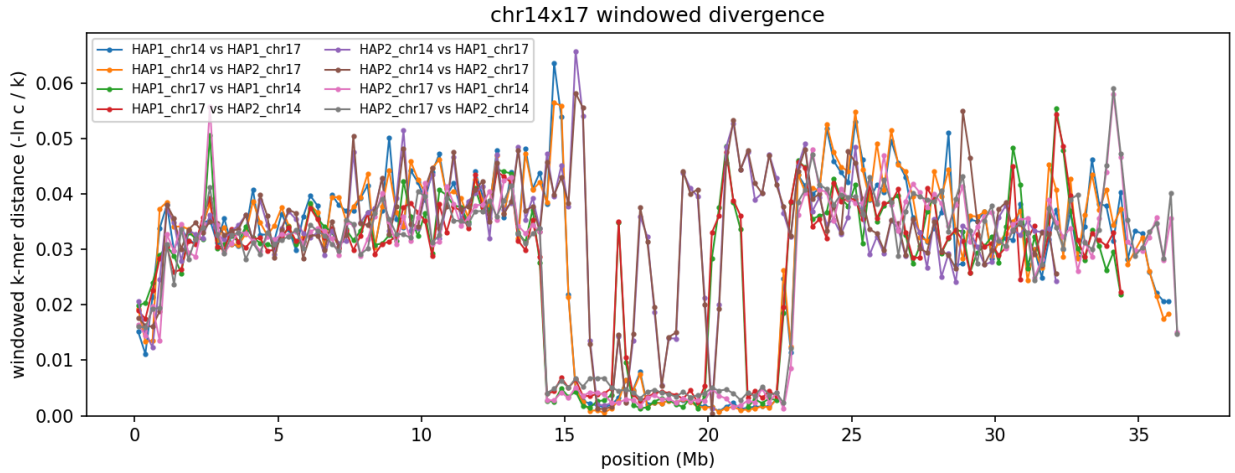


Figure 17: chr14 × chr17, all eight copy directions. Outside 14–23 Mb the homeologs are 0.03–0.05 apart; inside, k-mer containment says 0.003, as close as alleles. Alignment disagrees (94–96% identity there, for the allele as well as the homeolog), and the block compresses 3–5× better than the rest of the chromosome: a satellite array shared by the two homeologous centromeres, not retained pairing.

Summary. Ploidy: 2 copies × 18 numbers in sets of 2, x ≈ 9, up to 4x. Origin: allo-like (disomic-like), high. Rediploidization: diverged throughout.

What to take away. Glechoma is the textbook allo-like genome: a single synchronous duplication, every chromosome paired, no stretch where homeologs are as close as alleles. Its one apparent exception taught the tool to ignore repeat-dense windows.

9.2 Schizothorax curvilabiatus: rediploidization by fusion

Input. Four haplotypes × 25 chromosome numbers (an autotetraploid snow carp). HAP3 and HAP4 chr19 are 69.5 Mb, against 37.8 and 40.2 Mb for HAP1 and HAP2 chr19; chr22 is missing from HAP3 and HAP4.

Fusions. HAP3 chr19 and HAP4 chr19 are long-copy candidates. Each contains chr22's k-mers far above background (containment 0.452 and 0.437, z ≈ 48), 3.4× more than their unfused siblings do; chr22 is absent from both haplotypes: `status = fusion`, components chr19+chr22, `lineage_divergence` 0.035–0.036.

Lineages. chr19's copies split HAP1+HAP2 vs HAP3+HAP4 at `dist_split` 8.1 (0.007 within, 0.055 across), `te_split` 5.6, and along the whole chromosome (89% of windows, against a genome null of 7%): `fusion_lineages`. chr17 splits only over its last 8 Mb (24% covered, beats the null), `dist_split` 1.21, `te_split` 3.2: `partially_resolved`, regional. Most chromosomes (17 of 25) are `tetrasomic_like`.

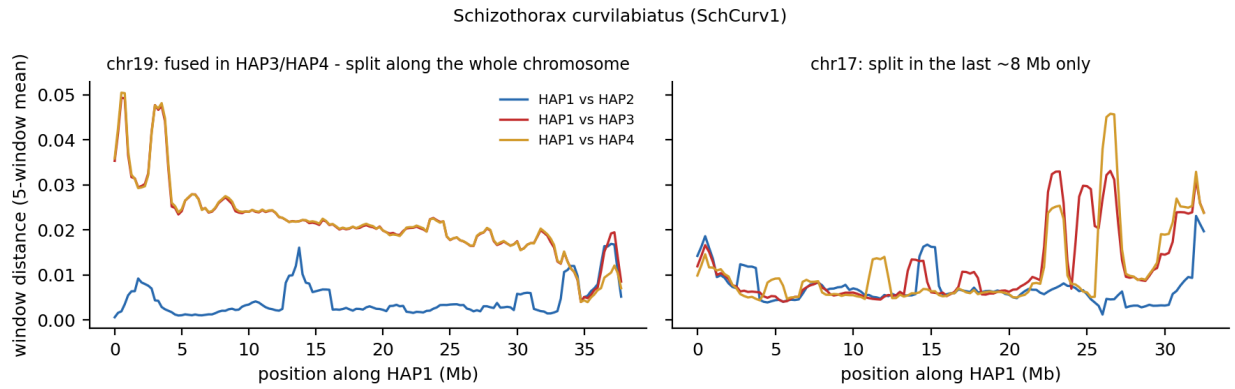


Figure 18: `SchCurv1` windows along HAP1. chr19: the fused copies stay far all along. chr17: the copies separate only towards the end.

Summary. 4 copies × 25, no older sets. Origin: mixed (tetrasomic-like and split chromosomes). Rediploidization: yes, one fusion between copies (the snow carp mechanism), high. This is the published result of Xie et al. (2026),

recovered without being told where to look. Aligning HAP3 chr17 to HAP1 shows why chr17 splits only at its end: HAP3 is inverted from 27 Mb to the end, the inversion mechanism Xie et al. describe for this chromosome.

9.3 Strophosoma melanogrammum: an AAB triploid in two files

Input. HAP1 has 20 chromosomes, HAP2 has 10. The ToL k-mer ploidy plot says triploid, AAB.

Matrix. chr1–10 are shared; chr11–20 are private to HAP1. Each private chromosome is closest to one shared number: HAP1 chr11 to chr01 at 0.049 against a median 0.098 to the other numbers (1.97×), chr12 to chr03 (1.99×), chr13 to chr04, … chr19 to chr09 (2.37×). Ten matches out of ten: an extra set. HAP1 chr11–20 become `HAP1B_chr01` – `HAP1B_chr10`.

Copies. Every chromosome now has three copies. On chr01, HAP1 and HAP2 are 0.011 apart (the two A copies); HAP1B is 0.049 from both (B). The bipartition isolates HAP1B with `dist_split` 4.6 and `te_split` 3.1: `one_divergent_copy`, odd haplotype HAP1B, on 9 of 10 chromosomes.

Summary. Ploidy: 3 copies × 10 numbers, HAP1B split out, odd copy number. Origin: AAB-like, one divergent chromosome set, high. Rediploidization: not assessable (a divergent set is not a lineage that split off).

The odd one out. On chr09 the outlier is HAP2, not HAP1B. Along HAP1 chr09 the windows show why: it matches the other A copy over 0–40 Mb (0.004 against 0.015 to B) and the B copy from 40 Mb to its end (0.004 against 0.018 to A), so chr09 reads AAB on one half and ABB on the other. The lineage-switch readout (Section 6.9) reports it as `HAP1:0.0-40.0Mb~HAP2;40.0-81.4Mb~HAP1B`: a homeologous exchange with replacement, or a phasing switch between the two sets held in HAP1′s file (Hi-C would tell).

9.4 Oncorhynchus masou: when a guard is missing

Input. Two haplotypes × 33. HAP2 was contig-level and was scaffolded on HAP1 for this project (`workflows/prep/scaffold_by_reference.py`): its contigs were placed where they align best on HAP1.

The first result. Residual tetrasomy on 8 of 10 paired chromosomes, 54 Mb, 62% terminal: an apparently textbook salmonid. Lien et al. (2016) describe such regions in Atlantic salmon.

What was wrong. In a region where two homeologs are similar, a HAP2 contig from chromosome B aligns almost as well to A and can be placed there. HAP1 A is then near-identical to "partner" HAP2 B (which carries A's sequence), and far from its own HAP2 A (which carries B's). Three of the four copies see a near-allelic stretch and the pair vote passed. The tell: those windows' own homolog distance was 0.03–0.085, homeolog distance.

The current result. With the own-homolog condition and the repeat guard: residual tetrasomy on one pair, chr17/chr29 (8.6 Mb, seen from all 4 copies); exchange candidates on 3 chromosomes, each in 1 of 2 copies (the swapped contigs, now read as what they are). The size-corrected homeology map finds 16.5% of the genome in blocks, 17 chromosome pairs, and 8 chromosomes with blocks on two or three others (chr32: chr01+chr04+chr33): the reshuffled salmonid karyotype. One residual pair is not enough to call an auto-like origin, so the origin reads allo-like (medium).

What to take away. Our own preprocessing produced the strongest "biological" signal in the panel. Masu remains a useful salmonid, but its residual numbers carry that history; Arctic charr and whitefish, assembled with both haplotypes scaffolded, are the cleaner salmonid tests. The k-mer map reaches 18–30% of their genomes; neither shows residual tetrasomy, but both show LORe-like blocks (Section 6.6): whitefish 243 Mb in five chromosome pairs, charr 8.5 Mb on one pair (chr09/chr11), at chromosome ends, as Lien et al. (2016) found the delayed regions of salmon to include telomeres.

Since then. A phased Hi-C assembly of masu with both haplotypes at chromosome level has been published (Wu et al., 2026) (GenBank GCA_052849675.1 and GCA_052849655.1). It replaces our self-scaffolded pair as `OncMaso1_wu2026` (`workflows/prep/masu_wu2026_prep.sh`). Its HAP2 numbers its chromosomes differently from HAP1: renumbering corrected 23, displaced 2 (HAP2 chr17 ⟶ 31, chr23 ⟶ 25) and left HAP2 chr32 ambiguous (2.9×). The first renumbering run blocked these corrections because a few chromosomes had no allele in the other haplotype (Section 6.3).

On the published assembly: residual tetrasomy on none of 28 tested chromosomes (the self-scaffolded chr17/chr29 signal does not reappear), no LORe-like blocks, and 6 near-allelic blocks that the residual test's guards reject. The homeology map finds 22% of the genome in 63 blocks, 17 chromosome pairs, and 7 chromosomes with blocks on two or three partners (chr08: chr07+chr16+chr26). Exchange candidates sit on 13 chromosomes (166 Mb), almost all in one of the two copies; on a Hi-C-phased assembly they are either real exchanges or phasing switches between homeologs, and need alignment before either reading. The origin reads allo-like (medium) and rediploidization advanced. Set against Atlantic salmon, where about a quarter of the genome rediploidized late (Lien et al., 2016), the k-mer map does not see masu's delayed regions as a distinct low-divergence class. Chapter 13 shows both masu assemblies.

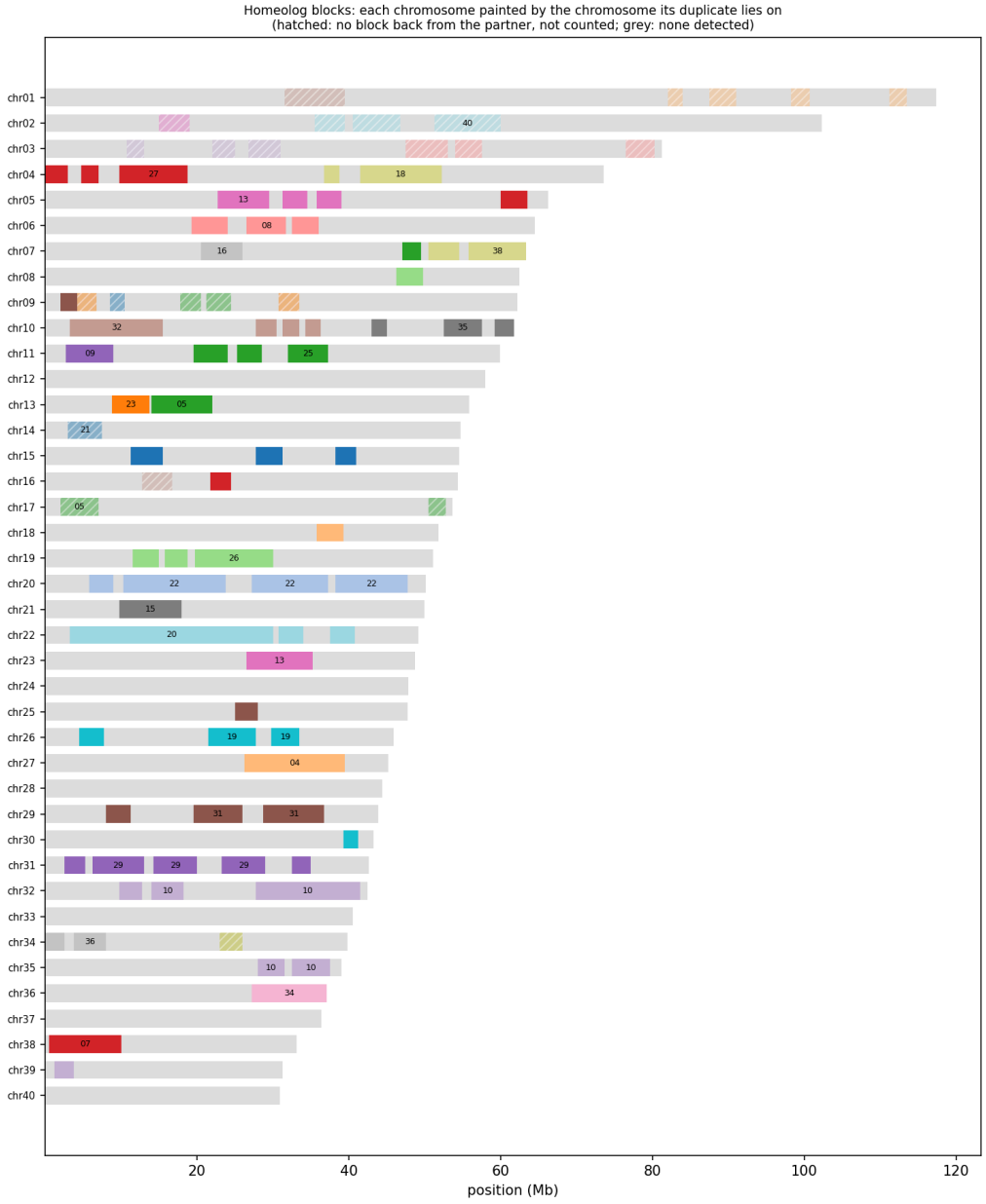


Figure 19: Arctic charr after the size correction: reciprocal blocks (solid) cover 18% of the genome in 18 chromosome pairs, and 6 chromosomes carry blocks on two homeologs (salmonid fusions since the duplication). Before the correction, ten unmatched blocks pointed at chr01, the largest chromosome; alignment showed only scattered short hits, a target-size artefact.

10 Validation

10.1 Simulated genomes

`ploidyspec simulate` writes random-sequence genomes with transposable-element families at high copy number; every haplotype except HAP1 carries a transposition and an inversion, so copies are not collinear. The end-to-end test (22 tests) runs the whole pipeline on each scenario.

Scenario	What it contains	Checked
diploid	two haplotypes	no pairs, no blocks
autotetraploid	four equivalent haplotypes	all `tetrasomic_like`, no blocks
autotetraploid, 2 hap	the same, each base chromosome numbered twice	pairs (i, i+n), pooled `tetrasomic_like`
allotetraploid	two progenitors, own TE bursts, 2 haplotypes	`resolved_lineages`, no residual tetrasomy
rediploidized	whole and 40% lineage splits, a placed and an unplaced fusion	states, both fusions, and the fusion junction at the middle of each fused copy
mislabelled	one file shifted by one, a swap in another	all relabelled
residual tetrasomy	diverged homeologs except the last 60% of two pairs; a satellite at every chromosome start	residual on exactly those four chromosomes; satellite excluded; “auto-like origin”
allotriploid, one file	AAB with A + B in HAP1′s file	HAP1B split out; AAB-like
segmental homeology	a fused chromosome whose arms are homeologous to two others; an exchange in one haplotype	two partners on chr01; exchange on HAP1 chr04 only

10.2 Real anchors

Anchor	Known	ploidyspec
`daGleHede1`	allopolyploid	9 pairs, CV 0.042, allo-like (high); map: exactly 9 pairs, 78%
`ddMalSylv1`	diploid	no pairs, 0% duplicated; not assessable
`fLepOcu1`, `fAmiCal2`	no duplication since 2R	one pair each, 0% in blocks; not assessable
`SchCurv1`, `SchYoun1`	autotetraploid snow carps, fusions (Xie et al., 2026)	all published fusions, in wave order; mixed with fusions
`drLytSali1`	tetrasomic by classical genetics	13/15 `tetrasomic_like`
`icStrMela3`	triploid, AAB (k-mer ploidy plot)	3 copies, AAB-like (high)
`lpTriTurg1`	durum wheat, AABB	the subgenome partition splits the 14 automatically numbered chromosomes into exactly the 7 A and 7 B (z = 179), and the cross-subgenome test finds all seven pairs, 1A–1B … 7A–7B (0.080–0.082, z −11 to −15), blind to the names; no window-map blocks; allo-like
`dmRanRepe1`	4x on x = 8 (Karbstein et al., 2025)	two subgenomes of 8 (z = 109, A 1.23× B), eight one-to-one pairs; window map 7 of 8 pairs, 6% of the genome; allo-like

10.3 Fusion profiles: divergence spreads from the fusion site

Xie et al. (2026) report that in snow carps divergence between fused and unfused copies is highest at the fusion site and spreads along the arms. The fusion profiles (Section 6.9) measure this directly, without being told where the junctions are. Every one of the eleven fused copies is placed with its junction found at 94–100% agreement, and the gradient follows fusion age:

Wave (Xie et al.)	Fusion	Overall divergence	Junction gradient
oldest	chr19+chr22 (`SchYoun1`, `SchCurv1`)	0.035–0.054	2.7–4.1
middle	chr04+15, chr08+16, chr20+23	0.026–0.029	2.1–2.6
youngest	chr11+chr14	0.015	0.6–0.7 (flat)
-	*Monacha* chr23+24 (one haplotype)	0.009	0.8 (flat)

Older fusions have both diverged further and concentrated that divergence at the junction; the youngest wave has not yet. The fused copies themselves stay within 0.001–0.008 of each other along their whole length.

11 What we got wrong, and fixed

Checking results by hand found problems in three rounds (7 October, 9–11 October, and 21 October). All are fixed and covered by tests; they are listed because earlier write-ups quoted numbers that depended on them.

Problem	What happened	Fix
Window registration	Windows were compared at equal coordinates. After the first indel larger than a window they compared different sequence, so across the panel only 0-34% of windows were usable. Every windowed-derived number was affected: regional splits, `split_extent`, run-length and windowed-CV metrics.	Position-free windows (FastK profiles). Panel re-run.
Chromosome relabelling	`ddLepDrab1` HAP4 numbers 14 of 16 chromosomes differently. The check rejected the whole batch because two units tied, so HAP4 stayed mislabelled. Its "strongest allo candidate" signals were all artefacts. `ddHesMatr1` has a HAP1 chr01/chr02 swap below the old threshold; its only homeolog pair was the swap.	One-to-one matching; mutual swaps at 2×. Both re-run.
Stale cache	Per-copy FASTAs and k-mer tables were reused by name. A re-run of `drLytSali1` on a new, renumbered assembly kept the old sequences.	Cache records its source; rebuilt on any change.
Window noise	After the windowed fix, chance runs of split windows and ordinary haplotype blocks were read as lineage splits, inflating `one_divergent_copy` and `candidate`.	Null from crossing groupings; one copy apart needs chromosome-wide evidence (Section 6.9).
Renumbering cycles	The whitefish's HAP2 numbering differs from HAP1 on 30 of 40 chromosomes in long cycles; one unit with no allele blocked a cycle and the whole correction was rejected. Eleocharis HAP2 chr11/chr12 were swapped with evidence on one side only.	Displacement and one-sided swaps (Section 6.3).
A haplotype holding two sets	The AAB weevil's HAP1 holds A and B under different numbers, so it read as 2 copies of 20 chromosomes and "a phasing problem".	Extra sets split out as a third copy (Section 6.3).
Residual tetrasomy from artefacts	Masu's 54 Mb rested on homeolog-swapped contigs in its reference-scaffolded HAP2; Glechoma's block was a centromeric satellite; Lathraea's was repeat-dense; one pair (Glechoma) was enough to call an auto-like origin.	Own homolog must be near; repeat-dense windows excluded; origin needs two pairs (Sections 6.10, 6.11).
Chromosome size	Whole-chromosome distance tracks chromosome size (correlation 0.8 with the size difference), so partitions sorted chromosomes by size; in the homeology map the largest chromosome won every window in repeat-rich genomes.	Size-adjusted pair distances; one size slope removed in the map (Sections 6.4, 6.6).
Size adjustment hiding subgenomes	The size regression absorbed a split into a larger and a smaller subgenome, and the genome-wide pair test compared homeologs against within-subgenome pairs that share repeats and are closer. Ranunculus, durum wheat and *Inula* lost every pair (the first version had found Ranunculus's eight).	Size-independent subgenome partition and a cross-subgenome pair test; map windows scored against the other subgenome only (Section 6.4).
Exchange between copies of one number	The exchange readout only compared different chromosome numbers.	Lineage switches (Section 6.9).
Silent on delayed rediploidization	Salmonid-style delayed (LORe) regions are less diverged than other homeologs but not near-allelic, so neither the residual test nor the summary mentioned them, although the map showed them (Chapter 14).	LORe-like block class in the homeology map (Section 6.6).

Problem	What happened	Fix
Masu assembly	Our masu HAP2 was contig-level and scaffolded on HAP1 by us; its swapped contigs produced most of the first residual signal.	Re-run on the published phased assembly (Wu et al., 2026) (Section 9.4).

Several inheritance-mode metrics from earlier work (`partition_consistency`, run length, `flip_rate`, the windowed CVs and a combined score) were retired, because labelling and window registration drove them. Their last values are archived in `meta/archive/` with a note.

12 Interpreting results, and the limits

12.1 A reading order for any species

1. **Is the assembly what you think?** `sequences.tsv`, `unplaced.tsv`, `chrom_label_corrections.tsv`, `ploidy_summary.tsv`. Many corrections, `ambiguous` rows, or uneven copy counts come first.
2. **Is there a duplication?** Homeolog pairs, the partition, the homeology map. If none: two-copy genomes are not assessable for origin.
3. **How do the copies relate?** States per chromosome; for two-copy genomes the pooled pairs and residual tetrasomy.
4. **What traces of transition?** Fusions, regional splits, residual tetrasomy, exchange, chromosomes with two partners, divergence spread.
5. **Then** the summary sentences, with their evidence.

12.2 Patterns and what they mean

Pattern	Most likely reading
all copies equally close; no lineages	tetrasomic: recent autopolyploid, not yet rediploidized
two lineages at one depth everywhere, synchronous pairs	allo-like: allopolyploid, or a long-rediploidized autopolyploid
mixture of tetrasomic and split chromosomes, fusions	rediploidization in progress (snow carps)
diverged pairs with near-allelic stretches on ≥ 2 pairs	autopolyploid origin with residual tetrasomy (salmonid-like); check terminal fraction and assembly history
chromosomes with blocks on two partners	karyotype rearranged since the duplication
homeolog blocks far less diverged than the rest, both directions	delayed (LORe-like) rediploidization of those regions, or a younger duplication
two subgenomes, chromosomes closer within a subgenome than to their homeologs	allopolyploid: each subgenome carries its progenitor's repeats (durum wheat, Ranunculus, *Inula*); pairs show only against the cross-subgenome background
a strong partition but no detectable homeolog pairs	subgenomes separated by repeat history after unique sequence diverged beyond k-mer reach
two copies, no pairs, no partition, in a known polyploid	the second set is diverged beyond k-mer reach, or the assembly holds one set per haplotype; not evidence of recombining copies
one copy apart, same haplotype always	assembly/phasing problem, or a split-out extra set (AAB)
one copy apart, haplotype changing	AAAB-like, a divergent extra genome
exchange in one copy	homeologous exchange, or a contig on the wrong homeolog

12.3 Limits

- One individual: structure, not inheritance; not origin.
- The assembly is part of the measurement (Section 3.3).
- Old duplications fade below k-mer detection (8–10% divergence).
- Thresholds are provisional (tuned on simulations, snow carps, Glechoma, *Malva*) and every one is a command-line flag (Appendix A).
- Holocentric genomes (*Eleocharis*) break and fuse chromosomes readily (Bureš et al., 2013; Hipp et al., 2009; Mandrioli & Manicardi, 2020), so chromosome numbers need not track ancestral units, and arm- and junction-based readings are weaker.
- Inheritance is per chromosome and can be partial or genotype-dependent (Bourke et al., 2017; Dalle Palle et al., 2022); "allo-like" on a chromosome means its copies do not mix today, not that the species is disomic everywhere.

- Repeat signals (TE split, partition in wheat) separate subgenomes by the same principle as TE-signature phasing (Session & Rokhsar, 2023); they are not independent of repeat turnover at centromeres (Chen et al., 2024; Su et al., 2019).

12.4 Where ploidyspec sits in poly-space

Twyford et al. (2025) propose placing polyploids in a multidimensional space of measurable genomic features. ploidyspec provides several axes from one assembly:

Twyford et al. axis	**ploidyspec measurement**
Heterozygosity / segregating variation	`allelic_distance_median`: distance between copies of the same chromosome number
Sequence divergence between duplicated sets	homeolog pair distance, `distance_ratio`, `dist_split`
Structural divergence	fusions between copies; regional lineage splits
Temporal heterogeneity (diploidization)	mix of states across chromosomes; `pair_depth_cv`; fusion `lineage_divergence`; LORe-like blocks
TE distribution	`te_split` (within-genome contrast)

`ploidyspec panel` builds an exploratory PCA over these numbers (Section 8.13). It is a picture, not a classifier.

12.5 Open questions

Settled on 10–11 October (by alignment, read k-mer spectra, or a fix): Glechoma chr14/chr17 (satellite), Lathraea chr03/chr04 (repeats; 98–99% vs 99.9% alleles), Pilosella chr04/chr05 (real: homeolog = allele identity), weevil chr09 (A to 40 Mb, B after: a lineage switch), snow carp chr17 (an inversion in HAP3 from 27 Mb), Monacha (HAP2 chr24 + chr23 joined at 37 Mb), drAriEdul1 chr01 (a 19 Mb divergent HAP4 block), *Salix triandra* (triploid) and *Rosa spinosissima* (tetraploid, AABB), Hesperis chr01/chr02 and charr chr01 (size artefacts), Cerastium (18 large–small homeolog pairs: subgenomes of different chromosome size). Settled on 21 October: Ranunculus (two subgenomes of 8 and eight homeolog pairs, hidden by the size adjustment and by subgenome-specific repeats until the cross-subgenome test).

Still open:

- Xie et al.'s exact coordinates for the `SchCurv1` chr17 inversion.
- Primary chromosome counts for Cerastium and Ranunculus.
- Hi-C to tell biology from assembly for the weevil chr09 switch and the Monacha join.
- Gene-based methods for duplications beyond k-mer reach (salmonids, snails).
- The open questions raised by the literature (Section 14.6).

13 Every species so far

Each card below is generated from the result files (`docs/guide/make_guide_species.py`). The table gives the overview; the cards give, per species: what is known, the assembly and any numbering fixes, distances, duplication by each method, states, fusions, residual tetrasomy and exchange with their support, the four answers with confidence and evidence, and caveats. “Dup. (map)” is the share of the genome in reciprocal homeolog blocks; “–” means the map has not run (one haplotype) or the run predates it.

Species	Name	Copies × nos.	Dup. (map)	Origin-like structure	Rediploidization
Two-haplotype genomes with duplicated sets (allo-like anchors and relatives)					
`daGleHede1`	*Glechoma hederacea*	2 × 18	79%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`drTriRepe1`	*Trifolium repens*	2 × 16	52%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`drSorDevo1`	*Sorbus devoniensis*	2 × 34	71%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`ddSalPent1`	*Salix pentandra*	2 × 38	78%	Allo-like (disomic-like) (high)	Advanced (medium)
`drMyrVert1`	*Myriophyllum verticillatum*	2 × 14	69%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`drMyrSpic1`	*Myriophyllum spicatum*	2 × 21	44%	Allo-like (disomic-like) (medium)	Advanced (medium)
`drRosSpin1`	*Rosa spinosissima*	2 × 14	35%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`daLatSqua1`	*Lathraea squamaria*	2 × 18	27%	Allo-like (disomic-like) (medium)	Advanced (medium)
`daLatClan1`	*Lathraea clandestina*	2 × 21	10%	Allo-like (disomic-like) (high)	Advanced (medium)
`daSenVulg1`	*Senecio vulgaris*	2 × 20	9%	Allo-like (disomic-like) (medium)	Diverged throughout the paired chromosomes (medium)
`laPotCris1`	*Potamogeton crispus*	2 × 26	52%	Allo-like (disomic-like) (medium)	Advanced (medium)
`laPotLuce1`	*Potamogeton lucens*	2 × 26	71%	Allo-like (disomic-like) (medium)	Advanced (medium)
`laPotNata1`	*Potamogeton natans*	2 × 25	59%	Allo-like (disomic-like) (medium)	Advanced (medium)
`laPotNodo1`	*Potamogeton nodosus*	2 × 26	59%	Allo-like (disomic-like) (medium)	Advanced (medium)
`laPotPerf1`	*Potamogeton perfoliatus*	2 × 26	60%	Allo-like (disomic-like) (medium)	Advanced (medium)
`daPilAura1`	*Pilosella aurantiaca*	2 × 18	82%	One copy apart on most chromosomes, changing haplotype between chromo… (medium)	Partly (medium)
`llColAutu1`	*Colchicum autumnale*	2 × 51	46%	Allo-like (disomic-like) (high)	Partly (medium)

Species	Name	Copies × nos.	Dup. (map)	Origin-like structure	Rediploidization
`lpElePalu1`		2 × 19	69%	Auto-like origin, mostly rediploidized (medium)	Partly (medium)
`lpTriTurg1`	*Triticum turgidum*	2 × 14	0%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
Cryptic or partial duplication in two-haplotype genomes					
`daInuConz1`	*Inula conyza (now Pentanema conyzae)*	2 × 16	32%	Allo-like (disomic-like) (high)	Advanced (medium)
`dmRanRepe1`	*Ranunculus repens*	2 × 16	6%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`dcCerAlpi1`	*Cerastium alpinum*	2 × 36	0%	Allo-like (disomic-like) (high)	Diverged throughout the paired chromosomes (medium)
`daSonOler1`	*Sonchus oleraceus*	2 × 16	18%	Allo-like (disomic-like) (medium)	Advanced (medium)
`dcHonPepl1`	*Honckenya peploides*	2 × 34	3%	Not assessable (not assessable)	Advanced (medium)
Genomes assembled with three or more copies					
`SchCurv1`	*Schizothorax curvilabiatus*	4 × 25	2%	Mixed (medium)	Yes (high)
`SchYoun1`	*Schizopygopsis younghusbandi*	4 × 25	0%	Mixed (medium)	Yes (high)
`drLytSali1`	*Lythrum salicaria*	4 × 15	42%	Auto-like (tetrasomic-like) (high)	No sign yet (medium)
`daBudDavi1`	*Buddleja davidii*	4 × 19	0%	Auto-like (tetrasomic-like) (medium)	Little sign (low)
`daGalBore1`	*Galium boreale*	4 × 11	0%	Auto-like (tetrasomic-like) (high)	Little sign (low)
`ddLepDrab1`	*Lepidium draba*	4 × 16	68%	Mixed (medium)	Little sign (low)
`ddEmpNigr1`	*Empetrum nigrum*	4 × 13	2%	Allo-like (disomic-like) (medium)	Split throughout (medium)
`drAriEdul1`	*Aria edulis*	4 × 17	24%	Mixed (medium)	Likely under way (medium)
`ddHesMatr1`	*Hesperis matronalis*	4 × 6	0%	One copy apart on most chromosomes, changing haplotype between chromo… (medium)	Little sign (low)
`ddHypMacu1`	*Hypericum maculatum*	4 × 8	0%	One copy apart on most chromosomes, nearly always the same haplotype … (medium)	Not assessable (not assessable)
`ddHypPerf1`	*Hypericum perforatum*	4 × 8	0%	One copy apart on most chromosomes, changing haplotype between chromo… (medium)	Likely under way (medium)
`ddSalTria1`	*Salix triandra*	3 × 19	0%	Auto-like (tetrasomic-like) (medium)	No sign yet (medium)
`icStrMela3`	*Strophosoma melanogrammum*	3 × 10	0%	AAB-like (high)	Not assessable (not assessable)
Salmonids					

Species	Name	Copies × nos.	Dup. (map)	Origin-like structure	Rediploidization
`OncMaso1`	*Oncorhynchus masou*	2 × 33	16%	Allo-like (disomic-like) (medium)	Partly (medium)
`OncMaso1_wu2026`	*Oncorhynchus masou*	2 × 33	22%	Allo-like (disomic-like) (medium)	Advanced (medium)
`fSalAlp3`	*Salvelinus alpinus*	2 × 40	18%	Allo-like (disomic-like) (high)	Advanced (medium)
`fCorLav1`	*Coregonus lavaretus*	2 × 40	30%	Allo-like (disomic-like) (medium)	Advanced (medium)
Diploid-like and negative controls					
`ddMalSylv1`	*Malva sylvestris*	2 × 21	1%	Not assessable (not assessable)	Not assessable (not assessable)
`fLepOcu1`	*Lepisosteus oculatus*	2 × 29	0%	Not assessable (not assessable)	Not assessable (not assessable)
`fAmiCal2`	*Amia calva*	2 × 23	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgCepNemo3`	*Cepaea nemoralis*	2 × 22	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgHygCinc1`	*Hygromia cinctella*	2 × 21	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgDauRufa1`	*Daudebardia rufa*	2 × 28	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgLitLitt3`	*Littorina littorea*	2 × 17	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgMonCant1`	*Monacha cantiana*	2 × 24	0%	Not assessable (not assessable)	Not read as rediploidization (medium)
`xgPomEleg1`	*Pomatias elegans*	2 × 13	0%	Not assessable (not assessable)	Not assessable (not assessable)
`xgStaPalu1`	*Stagnicola palustris*	2 × 18	2%	Not assessable (not assessable)	Not assessable (not assessable)
`lpTriTurg1_A`	*Triticum turgidum, A subgenome*	2 × 7	0%	Not assessable (not assessable)	Not assessable (not assessable)
`lpTriTurg1_B`	*Triticum turgidum, B subgenome*	2 × 7	0%	Not assessable (not assessable)	Not assessable (not assessable)
One usable haplotype					
`daEupConf1`	*Euphrasia confusa*	1 × 22	24%	Not assessable from this assembly (not assessable)	Not assessable from this assembly (not assessable)
`ddAraThal4`	*Arabidopsis thaliana*	1 × 5	–	Not assessable from this assembly (not assessable)	Not assessable (not assessable)
`ddPopNigr1`		1 × 19	23%	Not assessable from this assembly (not assessable)	Not assessable from this assembly (not assessable)
`ddSalCine1`		1 × 18	0%	Not assessable from this assembly (not assessable)	Not assessable (not assessable)
`drIngLaur1`		1 × 13	0%	Not assessable from this assembly (not assessable)	Not assessable (not assessable)
`drTriDubi3`		1 × 15	35%	Not assessable from this assembly (not assessable)	Not assessable from this assembly (not assessable)
`drUrtDioi1`		1 × 13	0%	Not assessable from this assembly (not assessable)	Not assessable (not assessable)

13.1 Two-haplotype genomes with duplicated sets (allo-like anchors and relatives)

daGleHede1 *Glechoma hederacea*

Known: polyploid (level unspecified); origin: allo (user-provided)

Assembly	HAP1 (18), HAP2 (18); 18 chromosome numbers, 2 copies each; 599 sequences unplaced
Numbering fixes	none
Distances	same number 0.0144, different numbers 0.1151
Homeolog pairs	9 (median 0.052, CV 0.0424)
Homeology map	79% of the genome in 36 blocks; 9 chromosome pairs; block CV 0.047
States	resolved lineages 18
Fusions	none
Residual tetrasomy	0 of 18 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 18 chromosome numbers; the numbers form sets of 2 (9 groups of mostly 2 chromosome numbers), so x ≈ 9 and up to 4x relative to x. copies from the assembly (2 per number); 9 groups of mostly 2 chromosome numbers (genome partition z = 24.5); homeolog distance 0.052 vs unrelated 0.115
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.04). This is also what a long-rediploidized autopolyploid looks like. 18 chromosome numbers assessed: 18 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2234
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/18 homeolog-paired chromosome numbers; pair-depth CV 0.04; homeology map: 79% of the genome in 36 homeolog blocks, 9 chromosome pairs, block-divergence CV 0.05

drTriRepe1 *Trifolium repens*

Known: allotetraploid 2n=4x=32; 2n=32; origin: allo (Griffiths et al. 2019 Plant Cell; Williams et al. 2012 BMC Plant Biol; Santangelo et al. 2023 Genome Biol & Evol; Kuo et al. 2024 BMC Biology; Olsen et al. 2021 New Phytologist; Abberton 2007; Ansari et al. 1999 Annals of Botany; Ullah et al. 2022 Frontiers; Ellison et al. 2006)

Assembly	HAP1 (16), HAP2 (16); 16 chromosome numbers, 2 copies each; 349 sequences unplaced
Numbering fixes	none
Distances	same number 0.0187, different numbers 0.1054
Homeolog pairs	8 (median 0.061, CV 0.0321)
Homeology map	52% of the genome in 57 blocks; 8 chromosome pairs; block CV 0.042
States	resolved lineages 16
Fusions	none
Residual tetrasomy	0 of 16 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 16 chromosome numbers; the numbers form sets of 2 (8 groups of mostly 2 chromosome numbers), so x ≈ 8 and up to 4x relative to x. copies from the assembly (2 per number); 8 groups of mostly 2 chromosome numbers (genome partition z = 15.6); homeolog distance 0.061 vs unrelated 0.105
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like. 16 chromosome numbers assessed: 16 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1692
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/16 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 52% of the genome in 57 homeolog blocks, 8 chromosome pairs, block-divergence CV 0.04

drSorDevo1 *Sorbus devoniensis*

Known: tetraploid, 2n=4x=68 (x=17); 2n=68; origin: allo -- hybridogenic, formed via a triploid-bridge mechanism from S. torminalis (sexual diploid, confirmed MATERNAL parent via chloroplast/microsatellite data) x members of subgenus Aria/S. aria s.l. (paternal,

pollen donor) (Hamston 2017 (thesis); Chester et al. 2007 Bot J Linn Soc; Hamston et al. 2018 Frontiers in Plant Science; Sell 1989 Watsonia; Fay et al. 2022 Curtis's Bot Mag; Hamston, Pellicer & Fay 2015 New J Botany; Proctor, Proctor & Groenhof 1989 New Phytologist)

Assembly	HAP1 (34), HAP2 (34); 34 chromosome numbers, 2 copies each; 198 sequences unplaced
Numbering fixes	none
Distances	same number 0.0112, different numbers 0.1104
Homeolog pairs	17 (median 0.049, CV 0.0299)
Homeology map	71% of the genome in 80 blocks; 17 chromosome pairs; block CV 0.045
States	resolved lineages 34
Fusions	none
Residual tetrasomy	0 of 34 tested, 0.0 Mb
Exchange candidates	chr30 5.9 Mb (chr31:1/2)

Ploidy	high	2 haplotype copies of each of 34 chromosome numbers; the numbers form sets of 2 (17 groups of mostly 2 chromosome numbers), so x ≈ 17 and up to 4x relative to x. copies from the assembly (2 per number); 17 groups of mostly 2 chromosome numbers (genome partition z = 42.7); homeolog distance 0.049 vs unrelated 0.110
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like. 34 chromosome numbers assessed: 34 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1528
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. Also: homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 0/34 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 71% of the genome in 80 homeolog blocks, 17 chromosome pairs, block-divergence CV 0.04; exchange: chr30(chr31:1/2)

ddSalPent1 *Salix pentandra*

Known: polyploid group ("C5"/section Pentandrae) hypothesized to have arisen from allopolyploid offspring via crosses between the Salix and Vetrix clades; not directly stated; origin: allo -- suggested, hedged ("might have arisen") (Gulyaev et al. 2022 Annals of Botany (whole-genome resequencing phylogeny))

Assembly	HAP1 (38), HAP2 (38); 38 chromosome numbers, 2 copies each; 308 sequences unplaced
Numbering fixes	none
Distances	same number 0.0088, different numbers 0.1481
Homeolog pairs	19 (median 0.075, CV 0.0384)
Homeology map	78% of the genome in 62 blocks; 20 chromosome pairs; block CV 0.086; two or more partners: chr03:chr09+chr16, chr16:chr03+chr04
States	resolved lineages 38
Fusions	none
Residual tetrasomy	0 of 38 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 38 chromosome numbers; the numbers form sets of 2 (19 groups of mostly 2 chromosome numbers), so x ≈ 19 and up to 4x relative to x. copies from the assembly (2 per number); 19 groups of mostly 2 chromosome numbers (genome partition z = 46.5); homeolog distance 0.075 vs unrelated 0.148
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.04). This is also what a long-rediploidized autopolyploid looks like. 38 chromosome numbers assessed: 38 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2000
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 2 chromosome(s) carry blocks homeologous to two or more others (chr03:chr09+chr16, chr16:chr03+chr04): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/38 homeolog-paired chromosome numbers; pair-depth CV 0.04; homeology map: 78% of the genome in 62 homeolog blocks, 20 chromosome pairs, block-divergence CV 0.09

drMyrVert1 *Myriophyllum verticillatum*

Known: tetraploid, 2n=28 (x=7) reported, though even the tetraploid classification itself was historically disputed (one source explicitly rejected interpreting it as part of a polyploid series); 2n=28; origin: unresolved -- this species is used almost exclusively as an OUTGROUP in phylogenetic studies of M. spicatum/M. sibiricum; no source found addresses its own origin (Lukosiunas 2024 (thesis); Lu et al. 2017 Aquatic Botany (used only as outgroup); Aiken 1981 Brittonia (dismisses a polyploid-series interpretation); Kavova 2019 (thesis, "probably represented only by cytotype 2n=28"))

Assembly	HAP1 (14), HAP2 (14); 14 chromosome numbers, 2 copies each; 2197 sequences unplaced
Numbering fixes	none
Distances	same number 0.0204, different numbers 0.1488
Homeolog pairs	7 (median 0.098, CV 0.0255)
Homeology map	69% of the genome in 27 blocks; 7 chromosome pairs; block CV 0.033
States	resolved lineages 14
Fusions	none
Residual tetrasomy	0 of 14 tested, 0.0 Mb
Exchange candidates	none

Ploidy	medium	2 haplotype copies of each of 14 chromosome numbers; the numbers form sets of 2 (7 groups of mostly 2 chromosome numbers), so x ≈ 7 and up to 4x relative to x. copies from the assembly (2 per number); 7 groups of mostly 2 chromosome numbers (genome partition z = 11.7); homeolog distance 0.098 vs unrelated 0.149
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like. 14 chromosome numbers assessed: 14 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.5213
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/14 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 69% of the genome in 27 homeolog blocks, 7 chromosome pairs, block-divergence CV 0.03

drMyrSpic1 *Myriophyllum spicatum*

Known: allohexaploid, AABBCC (three subgenomes), stepwise (AB)C origin -- an A-B allopolyploid lineage formed first, then hybridized with a more divergent C-like lineage; 2n=42 (x=7); origin: allo -- directly confirmed via chromosome-scale genome assembly (Wang et al. 2026 The Plant Journal (chromosome-scale genome, AABBCC); Lu et al. 2017 Aquatic Botany (G3pdh evidence); Pashnick 2021 (thesis, disomic inheritance confirmed by GBS); Hannay et al. 2026 Evolutionary Applications; Volkova, Ivanova & Bobrov 2024 Aquatic Botany)

Assembly	HAP1 (21), HAP2 (21); 21 chromosome numbers, 2 copies each; 1497 sequences unplaced
Numbering fixes	none
Distances	same number 0.0090, different numbers 0.1346
Homeolog pairs	8 (median 0.061, CV 0.3318)
Homeology map	44% of the genome in 38 blocks; 9 chromosome pairs; block CV 0.129; two or more partners: chr01:chr02+chr07, chr02:chr01+chr07, chr07:chr01+chr02
States	resolved lineages 16, not assessable 5
Fusions	none
Residual tetrasomy	0 of 17 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 21 chromosome numbers; the numbers form sets of 3 (7 groups of mostly 3 chromosome numbers), so x ≈ 7 and up to 6x relative to x. copies from the assembly (2 per number); 7 groups of mostly 3 chromosome numbers (genome partition z = 26.0); homeolog distance 0.061 vs unrelated 0.135
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.33). This is also what a long-rediploidized autopolyploid looks like. 16 chromosome numbers assessed: 16 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3590
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 3 chromosome(s) carry blocks homeologous to two or more others (chr01:chr02+chr07, chr02:chr01+chr07, chr07:chr01+chr02): fusions, fissions or translocations since the duplication.

residual tetrasomy on 0/17 homeolog-paired chromosome numbers; pair-depth CV 0.33; homeology map: 44% of the genome in 38 homeolog blocks, 9 chromosome pairs, block-divergence CV 0.13

drRosSpin1 *Rosa spinosissima*

Known: CONTESTED: majority of older cytogenetic/phylogenomic literature reports allotetraploid, 2n=4x=28 (x=7); but a 2026 chromosome-level genome paper reports DIPLOID, 2n=14; 2n=14 (2026 genome paper) vs 2n=28 (majority of older sources) -- direct contradiction; origin: allo-leaning per most sources (phylogenomic hybrid signal, isozyme data, rDNA loci pattern) IF the tetraploid reading is correct; the 2026 genome paper reporting diploid does not address auto/allo since it argues the species isn't polyploid at all (Debray et al. 2022 Systematic Biology (50-50 intersectional hybrid, phylogenomics); Mayland-Quellhorst et al. 2021 Nordic J Botany (allopolyploid, 2n=4x=28, citing Fougere-Danezan et al. 2014); Wissemann & Ritz 2005 Bot J Linn Soc (isozyme evidence); Vozarova 2023 (thesis, rDNA loci); Roberts 1977 Bot J Linn Soc (meiosis ambiguous between allo/auto-then-stabilized); Ruhsam et al. 2026 Wellcome Open Research (genome paper, reports diploid 2n=14))

Assembly	HAP1 (14), HAP2 (14); 14 chromosome numbers, 2 copies each; 408 sequences unplaced
Numbering fixes	none
Distances	same number 0.0127, different numbers 0.1065
Homeolog pairs	7 (median 0.054, CV 0.0290)
Homeology map	35% of the genome in 52 blocks; 7 chromosome pairs; block CV 0.076
States	resolved lineages 14
Fusions	none
Residual tetrasomy	0 of 14 tested, 0.0 Mb
Exchange candidates	none

Ploidy	medium	2 haplotype copies of each of 14 chromosome numbers; the numbers form sets of 2 (7 groups of mostly 2 chromosome numbers), so x ≈ 7 and up to 4x relative to x. copies from the assembly (2 per number); 7 groups of mostly 2 chromosome numbers (genome partition z = 14.2); homeolog distance 0.054 vs unrelated 0.107
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like. 14 chromosome numbers assessed: 14 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1215
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/14 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 35% of the genome in 52 homeolog blocks, 7 chromosome pairs, block-divergence CV 0.08

daLatSqua1 *Lathraea squamaria*

Known: hexaploid, 2n=42, documented gene (PHYA) duplication; also reported with variable genome-size/ploidy peaks (2C/4C/8C/16C) suggesting further cytotype variation within populations; 2n=42; origin: origin (auto vs allo) not addressed by any source found -- ploidy level and duplication evidence are well confirmed, but no source discusses hybrid vs within-species origin (Zonneveld 2019 (genome size); Bennett & Mathews 2006 American J Botany (PHYA gene duplication); Atkinson & Atkinson 2020 J Ecology (Biological Flora account, chromosome counts from Gates & Latter 1927 and Dangeard 1937, explicitly does not address auto/allo))

Assembly	HAP1 (18), HAP2 (18); 18 chromosome numbers, 2 copies each; 335 sequences unplaced
Numbering fixes	none
Distances	same number 0.0003, different numbers 0.1171
Homeolog pairs	8 (median 0.091, CV 0.0653)
Homeology map	27% of the genome in 35 blocks; 12 chromosome pairs; block CV 0.061; two or more partners: chr03:chr02+chr16, chr05:chr06+chr10, chr10:chr05+chr13, chr11:chr13+chr17, chr13:chr10+chr11, chr14:chr15+chr18, chr18:chr04+chr14
States	resolved lineages 16, not assessable 2
Fusions	none
Residual tetrasomy	0 of 17 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 18 chromosome numbers; the numbers form sets of 2 (8 homeolog pairs covering 89% of numbers), so x ≈ 9 and up to 4x relative to x. copies from the assembly (2 per number); 8 homeolog pairs covering 89% of numbers; homeolog distance 0.091 vs unrelated 0.117

Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.07). This is also what a long-rediploidized autopolyploid looks like. 16 chromosome numbers assessed: 16 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3198
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 7 chromosome(s) carry blocks homeologous to two or more others (chr03:chr02+chr16, chr05:chr06+chr10, chr10:chr05+chr13, chr11:chr13+chr17, chr13:chr10+chr11, chr14:chr15+chr18, chr18:chr04+chr14): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/17 homeolog-paired chromosome numbers; pair-depth CV 0.07; homeology map: 27% of the genome in 35 homeolog blocks, 12 chromosome pairs, block-divergence CV 0.06

daLatClan1 *Lathraea clandestina*

Known: occurs in two ploidies (genome sizes 2.18 and 4.18 pg reported); a separate table entry lists 2C=2.20pg and chromosome number 42 (hexaploid, matching L. squamaria); shares the same PHYA gene-duplication evidence as L. squamaria; 2n=42 (hexaploid, per one source) though genome-size data suggests cytotype variation; origin: origin not addressed by any source found, same gap as its congener L. squamaria (Zonneveld 2019; Bennett & Mathews 2006 (PHYA gene duplication detected in this species too))

Assembly	HAP1 (21), HAP2 (21); 21 chromosome numbers, 2 copies each; 5484 sequences unplaced
Numbering fixes	none
Distances	same number 0.0033, different numbers 0.1079
Homeolog pairs	10 (median 0.092, CV 0.0317)
Homeology map	10% of the genome in 25 blocks; 10 chromosome pairs; block CV 0.859; two or more partners: chr04:chr03+chr15, chr05:chr07+chr17, chr11:chr01+chr02, chr15:chr04+chr18
States	resolved lineages 20, not assessable 1
Fusions	none
Residual tetrasomy	0 of 20 tested, 0.0 Mb
Exchange candidates	chr07 3.2 Mb (chr05:1/2)

Ploidy	medium	2 haplotype copies of each of 21 chromosome numbers; the numbers form sets of 2 (9 groups of mostly 2 chromosome numbers), so x ≈ 9 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 9 groups of mostly 2 chromosome numbers (genome partition z = 10.8); homeolog distance 0.092 vs unrelated 0.108
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like. 20 chromosome numbers assessed: 20 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2535
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 4 chromosome(s) carry blocks homeologous to two or more others (chr04:chr03+chr15, chr05:chr07+chr17, chr11:chr01+chr02, chr15:chr04+chr18): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.86), as when regions resolve at different times; 2 low-divergence homeolog blocks (5 Mb, 6% of the block span, mostly at chromosome ends; chr05xchr07): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication; homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 0/20 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 10% of the genome in 25 homeolog blocks, 10 chromosome pairs, block-divergence CV 0.86; LORe-like blocks: 2, 5.2 Mb; exchange: chr07(chr05:1/2)

daSenVulg1 *Senecio vulgaris*

Known: tetraploid, 2n=40; 2n=40; origin: genuinely contested historically, but more recent molecular evidence leans allotetraploid; older autotetraploid-from-S.-vernalis hypothesis (Kadereit 1984) was specifically tested and refuted by later isozyme work (Kadereit 1984 Plant Syst Evol (proposes auto, from S. vernalis); Harris & Ingram 1992 Heredity (no definitive evidence either way); Ashton & Abbott 1992 Plant Syst Evol (isozyme evidence refutes the S.-vernalis-autotetraploid hypothesis, favors allo); Kim et al. 2008

Science (states allotetraploid, hybridization with S. squalidus + unknown parent); Chapman & Abbott 2010 New Phytologist (RAY2b homeolog evidence for allotetraploidy); Abbott & Lowe 2004 Biol J Linn Soc (frames as an explicitly unresolved auto-vs-allo question as of that paper))

Assembly	HAP1 (20), HAP2 (20); 20 chromosome numbers, 2 copies each; 767 sequences unplaced
Numbering fixes	none
Distances	same number 0.0001, different numbers 0.0944
Homeolog pairs	10 (median 0.067, CV 0.0739)
Homeology map	9% of the genome in 15 blocks; 5 chromosome pairs; block CV 0.236
States	resolved lineages 20
Fusions	none
Residual tetrasomy	0 of 20 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 20 chromosome numbers; the numbers form sets of 2 (10 groups of mostly 2 chromosome numbers), so x ≈ 10 and up to 4x relative to x. copies from the assembly (2 per number); 10 groups of mostly 2 chromosome numbers (genome partition z = 18.3); homeolog distance 0.067 vs unrelated 0.094
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.07). This is also what a long-rediploidized autopolyploid looks like. 20 chromosome numbers assessed: 20 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0413
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. Also: homeolog blocks diverged to different depths (CV 0.24), as when regions resolve at different times; 2 low-divergence homeolog blocks (7 Mb, 6% of the block span; chr15xchr19): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication. residual tetrasomy on 0/20 homeolog-paired chromosome numbers; pair-depth CV 0.07; homeology map: 9% of the genome in 15 homeolog blocks, 5 chromosome pairs, block-divergence CV 0.24; LORe-like blocks: 2, 7.0 Mb

`laPotCris1` *Potamogeton crispus*

Known: highly variable ploidy: 2n=50, 52, 78 reported by one cytological review; 2n=12x=84 reported by another (an extreme, unusually high count) -- greater cytotype diversity than the other four Potamogeton species checked; 2n=50/52/78/84 (multiple conflicting reports); origin: no direct origin statement found for P. crispus specifically, though it sits within a genus-wide pattern of extensive documented hybridization (e.g. P. x cooperi has P. crispus as one parent) and a base tetraploid (2n=52, x=13-ish by some countings) condition widely treated as ancestral across the genus (Hollingsworth, Preston & Gornall 1998 Aquatic Botany; Ganie, Reshi & Wafai 2020 Tropical Ecology; Wan et al. 2012 Plant Syst Evol; Hollingsworth, Preston & Barrett 1995 Bot J Linn Soc)

Assembly	HAP1 (26), HAP2 (26); 26 chromosome numbers, 2 copies each; 3030 sequences unplaced
Numbering fixes	none
Distances	same number 0.0021, different numbers 0.1303
Homeolog pairs	12 (median 0.098, CV 0.0807)
Homeology map	52% of the genome in 45 blocks; 17 chromosome pairs; block CV 0.047; two or more partners: chr01:chr22+chr24, chr02:chr10+chr21+chr23, chr05:chr09+chr21+chr25, chr09:chr05+chr19, chr10:chr02+chr17, chr19:chr09+chr25, chr21:chr02+chr05, chr25:chr05+chr19
States	resolved lineages 24, not assessable 2
Fusions	none
Residual tetrasomy	0 of 26 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 26 chromosome numbers; the numbers form sets of 2 (12 groups of mostly 2 chromosome numbers), so x ≈ 12 and up to 4x relative to x. copies from the assembly (2 per number); 12 groups of mostly 2 chromosome numbers (genome partition z = 18.6); homeolog distance 0.098 vs unrelated 0.130
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.08). This is also what a long-rediploidized autopolyploid looks like. 24 chromosome numbers assessed: 24 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3356

Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 8 chromosome(s) carry blocks homeologous to two or more others (chr01:chr22+chr24, chr02:chr10+chr21+chr23, chr05:chr09+chr21+chr25, chr09:chr05+chr19, chr10:chr02+chr17, chr19:chr09+chr25, chr21:chr02+chr05, chr25:chr05+chr19): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/26 homeolog-paired chromosome numbers; pair-depth CV 0.08; homeology map: 52% of the genome in 45 homeolog blocks, 17 chromosome pairs, block-divergence CV 0.05

`laPotLuce1` *Potamogeton lucens*

Known: tetraploid, 2n=52 (x=13-ish); documented parent in multiple named interspecific hybrids (P. x salicifolius with P. perfoliatus, P. x fluitans with P. natans); 2n=52; origin: explicitly stated as "probably allotetraploid" by one dedicated phylogenetic study (Wang, Zhang & Wang 2007 Plant Syst Evol (ITS phylogeny, "probably allotetraploids" for natans and lucens); Kaplan 2007 Plant Biosystems (P. x salicifolius, isozyme evidence); Fant & Preston 2001/2004 (isozyme evidence for P. x fluitans and P. x salicifolius parentage))

Assembly	HAP1 (26), HAP2 (26); 26 chromosome numbers, 2 copies each; 2119 sequences unplaced
Numbering fixes	none
Distances	same number 0.0094, different numbers 0.1173
Homeolog pairs	12 (median 0.080, CV 0.1097)
Homeology map	71% of the genome in 57 blocks; 19 chromosome pairs; block CV 0.054; two or more partners: chr01:chr06+chr15+chr16+chr22, chr03:chr12+chr19+chr24, chr06:chr01+chr22, chr12:chr03+chr21, chr15:chr01+chr18, chr16:chr01+chr20, chr19:chr03+chr21, chr21:chr12+chr19, chr22:chr01+chr06
States	resolved lineages 24, not assessable 2
Fusions	none
Residual tetrasomy	0 of 26 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 26 chromosome numbers; the numbers form sets of 2 (12 groups of mostly 2 chromosome numbers), so x ≈ 12 and up to 4x relative to x. copies from the assembly (2 per number); 12 groups of mostly 2 chromosome numbers (genome partition z = 19.5); homeolog distance 0.080 vs unrelated 0.117
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.11). This is also what a long-rediploidized autopolyploid looks like. 24 chromosome numbers assessed: 24 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2458
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 9 chromosome(s) carry blocks homeologous to two or more others (chr01:chr06+chr15+chr16+chr22, chr03:chr12+chr19+chr24, chr06:chr01+chr22, chr12:chr03+chr21, chr15:chr01+chr18, chr16:chr01+chr20, chr19:chr03+chr21, chr21:chr12+chr19, chr22:chr01+chr06): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/26 homeolog-paired chromosome numbers; pair-depth CV 0.11; homeology map: 71% of the genome in 57 homeolog blocks, 19 chromosome pairs, block-divergence CV 0.05

`laPotNata1` *Potamogeton natans*

Known: tetraploid, 2n=52; also an octoploid cytotype (2n=8x=104) reported as a first record by one source; 2n=52 (also 2n=104 reported); origin: explicitly stated as "probably allotetraploid" by the same dedicated phylogenetic study as P. lucens; also a well-documented parent of several named hybrids (P. x schreberi with P. nodosus, P. x fluitans with P. lucens) (Wang, Zhang & Wang 2007 Plant Syst Evol; Ganie, Reshi & Wafai 2020 (octoploid first record); Hollingsworth, Preston & Preston 1995 (P. x schreberi hybrid evidence); Fant, Preston & Barrett 2001 (P. x fluitans))

Assembly	HAP1 (25), HAP2 (25); 25 chromosome numbers, 2 copies each; 1100 sequences unplaced
Numbering fixes	none
Distances	same number 0.0045, different numbers 0.1220
Homeolog pairs	12 (median 0.077, CV 0.1263)
Homeology map	59% of the genome in 51 blocks; 17 chromosome pairs; block CV 0.059; two or more partners: chr01:chr03+chr15+chr17+chr20, chr03:chr01+chr15, chr10:chr16+chr24, chr15:chr01+chr03, chr16:chr10+chr21, chr17:chr01+chr22, chr20:chr01+chr13
States	resolved lineages 24, not assessable 1
Fusions	none
Residual tetrasomy	0 of 25 tested, 0.0 Mb

Exchange candidates	none	
Ploidy	high	2 haplotype copies of each of 25 chromosome numbers; the numbers form sets of 2 (12 groups of mostly 2 chromosome numbers), so x ≈ 12 and up to 4x relative to x. copies from the assembly (2 per number); 12 groups of mostly 2 chromosome numbers (genome partition z = 19.9); homeolog distance 0.077 vs unrelated 0.122
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.13). This is also what a long-rediploidized autopolyploid looks like. 24 chromosome numbers assessed: 24 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2498
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 7 chromosome(s) carry blocks homeologous to two or more others (chr01:chr03+chr15+chr17+chr20, chr03:chr01+chr15, chr10:chr16+chr24, chr15:chr01+chr03, chr16:chr10+chr21, chr17:chr01+chr22, chr20:chr01+chr13): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/25 homeolog-paired chromosome numbers; pair-depth CV 0.13; homeology map: 59% of the genome in 51 homeolog blocks, 17 chromosome pairs, block-divergence CV 0.06

laPotNodo1 *Potamogeton nodosus*

Known: tetraploid, n=26 (2n=52); 2n=52; origin: no direct origin statement found for P. nodosus specifically, though documented as a hybridization partner (P. x schreberi with P. natans) within the same genus-wide allotetraploid-leaning pattern (Hollingsworth, Preston & Gornall 1998 Aquatic Botany; Hollingsworth, Preston & Barrett 1995 (P. x schreberi hybrid))

Assembly	HAP1 (26), HAP2 (26); 26 chromosome numbers, 2 copies each; 3550 sequences unplaced
Numbering fixes	none
Distances	same number 0.0025, different numbers 0.1239
Homeolog pairs	13 (median 0.094, CV 0.0872)
Homeology map	59% of the genome in 51 blocks; 19 chromosome pairs; block CV 0.054; two or more partners: chr03:chr07+chr11+chr17+chr19, chr07:chr03+chr25, chr08:chr17+chr20, chr11:chr03+chr19+chr25, chr12:chr16+chr26, chr16:chr12+chr22, chr17:chr03+chr08, chr19:chr03+chr11, chr25:chr07+chr11
States	resolved lineages 26
Fusions	none
Residual tetrasomy	0 of 26 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 26 chromosome numbers; the numbers form sets of 2 (13 groups of mostly 2 chromosome numbers), so x ≈ 13 and up to 4x relative to x. copies from the assembly (2 per number); 13 groups of mostly 2 chromosome numbers (genome partition z = 19.4); homeolog distance 0.094 vs unrelated 0.124
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.09). This is also what a long-rediploidized autopolyploid looks like. 26 chromosome numbers assessed: 26 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2838
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 9 chromosome(s) carry blocks homeologous to two or more others (chr03:chr07+chr11+chr17+chr19, chr07:chr03+chr25, chr08:chr17+chr20, chr11:chr03+chr19+chr25, chr12:chr16+chr26, chr16:chr12+chr22, chr17:chr03+chr08, chr19:chr03+chr11, chr25:chr07+chr11): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/26 homeolog-paired chromosome numbers; pair-depth CV 0.09; homeology map: 59% of the genome in 51 homeolog blocks, 19 chromosome pairs, block-divergence CV 0.05

laPotPerf1 *Potamogeton perfoliatus*

Known: tetraploid, 2n=52 (x=13); "consistent with a presumed allopolyploid nature, although definitive evidence is needed" per one dedicated review; also documented as a parent in at least 3 named hybrids (P. x salicifolius, P. x torssanderi, P. intortusifolius); 2n=52; origin: presumed allopolyploid, but explicitly flagged by the source itself as not definitively confirmed (Ganie, Reshi & Wafai 2020

Tropical Ecology ("presumed allopolyploid... definitive evidence needed"); Kaplan 2007; Kaplan & Fehrer 2007 Annals of Botany (P. x torssanderi triple hybrid); Du et al. 2009 Aquatic Botany (P. intortusifolius))

Assembly	HAP1 (26), HAP2 (26); 26 chromosome numbers, 2 copies each; 1704 sequences unplaced
Numbering fixes	none
Distances	same number 0.0145, different numbers 0.1202
Homeolog pairs	13 (median 0.088, CV 0.0848)
Homeology map	60% of the genome in 58 blocks; 20 chromosome pairs; block CV 0.054; two or more partners: chr03:chr10+chr13+chr15+chr20+chr26, chr09:chr13+chr24, chr10:chr03+chr20+chr26, chr11:chr16+chr25, chr13:chr03+chr09, chr15:chr03+chr26, chr16:chr11+chr21, chr20:chr03+chr10, chr26:chr03+chr10+chr15
States	resolved lineages 26
Fusions	none
Residual tetrasomy	0 of 26 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 26 chromosome numbers; the numbers form sets of 2 (13 groups of mostly 2 chromosome numbers), so x ≈ 13 and up to 4x relative to x. copies from the assembly (2 per number); 13 groups of mostly 2 chromosome numbers (genome partition z = 19.3); homeolog distance 0.088 vs unrelated 0.120
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.08). This is also what a long-rediploidized autopolyploid looks like. 26 chromosome numbers assessed: 26 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3552
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 9 chromosome(s) carry blocks homeologous to two or more others (chr03:chr10+chr13+chr15+chr20+chr26, chr09:chr13+chr24, chr10:chr03+chr20+chr26, chr11:chr16+chr25, chr13:chr03+chr09, chr15:chr03+chr26, chr16:chr11+chr21, chr20:chr03+chr10, chr26:chr03+chr10+chr15): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/26 homeolog-paired chromosome numbers; pair-depth CV 0.08; homeology map: 60% of the genome in 58 homeolog blocks, 20 chromosome pairs, block-divergence CV 0.05

`daPilAura1` *Pilosella aurantiaca*

Known: tetraploid, 2n=36 (also pentaploid 2n=45 and rare heptaploid cytotypes reported), facultatively apomictic (apospory); 2n=36; origin: suggestive of hybrid origin at the genus level ("extensive reticulate evolution and common polyploidy, often of hybrid origin" -- Majeskyet al. 2017) and P. aurantiaca is documented as a PARENT of further allopolyploid hybrids (e.g. P. rubra, ~4 sets P. aurantiaca + 2 sets P. officinarum), but no source found directly confirms P. aurantiaca's OWN origin as opposed to its role in producing hybrid derivatives (Krahulcova, Krahulec & Rosenbaumova 2011 Sexual Plant Reprod; Krahulcova, Raabe & Krahulec 2012 Kochia; Okada et al. 2011 Plant Cell Physiol; Krahulcova, Rotreklova & Krahulec 2014 Folia Geobot; Gloríková 2025 (thesis); Krahulec et al. 2011 Preslia; Majesky, Krahulec & Vasut 2017 Taxon)

Assembly	HAP1 (18), HAP2 (18); 18 chromosome numbers, 2 copies each; 966 sequences unplaced
Numbering fixes	ambiguous 2
Distances	same number 0.0386, different numbers 0.0821
Homeolog pairs	9 (median 0.036, CV 0.1051)
Homeology map	82% of the genome in 211 blocks; 10 chromosome pairs; block CV 0.323; two or more partners: chr01:chr07+chr14, chr07:chr01+chr06
States	one divergent copy 18
Fusions	none
Residual tetrasomy	4 of 18 tested, 79.0 Mb, 0% terminal; chr04 28.8 Mb (chr05:3/4); chr05 34.2 Mb (chr04:3/4); chr09 9.8 Mb (chr11:3/4); chr11 6.2 Mb (chr09:3/4)
Exchange candidates	chr02 3.2 Mb (chr03:1/2)

Ploidy	high	2 haplotype copies of each of 18 chromosome numbers; the numbers form sets of 2 (9 groups of mostly 2 chromosome numbers), so x ≈ 9 and up to 4x relative to x. copies from the assembly (2 per number); 9 groups of mostly 2 chromosome numbers (genome partition z = 23.5); homeolog distance 0.036 vs unrelated 0.082
Auto- or allo-like	medium	One copy apart on most chromosomes, changing haplotype between chromosomes: AAB-like (one divergent genome copy). 18 chromosome numbers assessed: 0 split into lineages, 0 tetrasomic-like, 18 one divergent copy, 0 candidate

TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2074
Rediploidization	medium	Partly: 4 of 18 homeolog-paired chromosome numbers keep stretches of residual tetrasomy (79.0 Mb, 0% of it at chromosome ends) where the homeologs are still as close as alleles; elsewhere they have diverged. Also: 2 chromosome(s) carry blocks homeologous to two or more others (chr01:chr07+chr14, chr07:chr01+chr06): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.32), as when regions resolve at different times; 5 low-divergence homeolog blocks (115 Mb, 4% of the block span; chr01xchr07, chr08xchr13): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication; homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 4/18 homeolog-paired chromosome numbers (2 pair(s)), 79.0 Mb, 0% of it terminal; pair-depth CV 0.11; homeology map: 82% of the genome in 211 homeolog blocks, 10 chromosome pairs, block-divergence CV 0.32; LORe-like blocks: 5, 115.5 Mb; exchange: chr02(chr03:1/2)

Caveats: 2 chromosome-numbering mismatch(es) left ambiguous (not applied)

`llColAutu1` *Colchicum autumnale*

Known: tetraploid, 2n=4x=36 (x=9), part of a Central/South European polyploid complex; genus-wide, Colchicum ranges from tetraploid to 24-ploid with ~75% of species polyploid; 2n=36; origin: explicitly unresolved -- the most authoritative source found states directly that "the extent to which past allopolyploidy or autopolyploidy explains the lability of Colchicum chromosome numbers remains an open question, as there are no experimental crosses or other studies addressing hybridization" (Chacon & Renner 2014) (Persson et al. 2011 Taxon; Fridlender et al. 2002 Plant Cell Reports; Chacon, Cusimano & Renner 2014 Systematic Botany; Feinbrun 1958 Evolution; Nordenstam 1998; D'Amato 1957)

Assembly	HAP1 (51), HAP2 (51); 51 chromosome numbers, 2 copies each; 1998 sequences unplaced
Numbering fixes	none
Distances	same number 0.0001, different numbers 0.1118
Homeolog pairs	24 (median 0.069, CV 0.0517)
Homeology map	46% of the genome in 295 blocks; 30 chromosome pairs; block CV 0.103; two or more partners: chr04:chr10+chr12+chr49, chr06:chr15+chr27+chr51, chr10:chr04+chr17, chr20:chr44+chr48, chr27:chr06+chr37, chr29:chr36+chr43, chr43:chr29+chr33
States	partially resolved 2, resolved lineages 46, not assessable 3
Fusions	none
Residual tetrasomy	2 of 51 tested, 5.0 Mb, 0% terminal; chr04 2.8 Mb (chr10:4/4;chr12:0/4;chr49:0/4); chr10 2.2 Mb (chr04:4/4;chr17:0/4)
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 51 chromosome numbers; the numbers form sets of 2 (23 groups of mostly 2 chromosome numbers), so x ≈ 23 and up to 4x relative to x. copies from the assembly (2 per number); 23 groups of mostly 2 chromosome numbers (genome partition z = 30.8); homeolog distance 0.069 vs unrelated 0.112
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.05). This is also what a long-rediploidized autopolyploid looks like. 48 chromosome numbers assessed: 48 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0500
Rediploidization	medium	Partly: 2 of 51 homeolog-paired chromosome numbers keep stretches of residual tetrasomy (5.0 Mb, 0% of it at chromosome ends) where the homeologs are still as close as alleles; elsewhere they have diverged. Also: 7 chromosome(s) carry blocks homeologous to two or more others (chr04:chr10+chr12+chr49, chr06:chr15+chr27+chr51, chr10:chr04+chr17, chr20:chr44+chr48, chr27:chr06+chr37, chr29:chr36+chr43, chr43:chr29+chr33): fusions, fissions or translocations since the duplication; 2 low-divergence homeolog blocks (15 Mb, 1% of the block span; chr29xchr43): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication. residual tetrasomy on 2/51 homeolog-paired chromosome numbers (1 pair(s)), 5.0 Mb, 0% of it terminal; pair-depth CV 0.05; homeology map: 46% of the genome in 295 homeolog blocks, 30 chromosome pairs, block-divergence CV 0.10; LORe-like blocks: 2, 15.0 Mb

lpElePalu1

Assembly	HAP1 (19), HAP2 (19); 19 chromosome numbers, 2 copies each; 1116 sequences unplaced
Numbering fixes	corrected 1, swapped 1
Distances	same number 0.0184, different numbers 0.0905
Homeolog pairs	9 (median 0.020, CV 0.5434)
Homeology map	69% of the genome in 108 blocks; 14 chromosome pairs; block CV 0.476; two or more partners: chr01:chr02+chr18, chr02:chr01+chr03+chr07+chr18, chr03:chr02+chr18, chr07:chr02+chr08, chr13:chr06+chr17, chr18:chr01+chr02+chr03
States	tetrasomic like 8, candidate 4, resolved lineages 4, one divergent copy 2, not assessable 1
Fusions	none
Residual tetrasomy	10 of 19 tested, 310.0 Mb, 52% terminal; chr04 38.0 Mb (chr05:4/4); chr05 39.5 Mb (chr04:4/4); chr07 3.0 Mb (chr02:0/4;chr08:3/4); chr08 7.8 Mb (chr07:3/4); chr09 32.0 Mb (chr10:4/4); chr10 33.5 Mb (chr09:4/4); +4 more
Exchange candidates	chr01 14.5 Mb (chr18:2/2); chr02 15.2 Mb (chr03:1/2;chr07:2/2;chr18:1/2); chr03 11.0 Mb (chr02:1/2); chr07 10.8 Mb (chr02:2/2;chr08:1/2); chr08 6.8 Mb (chr07:1/2); chr10 2.8 Mb (chr09:1/2); +5 more

Ploidy — medium

2 haplotype copies of each of 19 chromosome numbers; the numbers form sets of 2 (7 groups of mostly 2 chromosome numbers), so x ≈ 7 and up to 4x relative to x.

copies from the assembly (2 per number); 7 groups of mostly 2 chromosome numbers (genome partition z = 15.6); homeolog distance 0.020 vs unrelated 0.091

Auto- or allo-like — medium

Auto-like origin, mostly rediploidized: the duplicated sets have separated along most of their length, but 10 of 19 paired chromosome numbers keep stretches where the homeologs are as close as alleles (residual tetrasomy). That is expected after an autopolyploidization; recent homeologous exchanges in an allopolyploid can leave similar stretches.

18 chromosome numbers assessed: 4 split into lineages, 8 tetrasomic-like, 2 one divergent copy, 4 candidate; residual tetrasomy on 10/19 homeolog-paired chromosome numbers (5 pair(s)), 310.0 Mb, 52% of it terminal

TE markers — not assessable

Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it.

te_marker_fraction mean 0.0819

Rediploidization — medium

Partly: 10 of 19 homeolog-paired chromosome numbers keep stretches of residual tetrasomy (310.0 Mb, 52% of it at chromosome ends) where the homeologs are still as close as alleles; elsewhere they have diverged. Also: 6 chromosome(s) carry blocks homeologous to two or more others (chr01:chr02+chr18, chr02:chr01+chr03+chr07+chr18, chr03:chr02+chr18, chr07:chr02+chr08, chr13:chr06+chr17, chr18:chr01+chr02+chr03): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.48), as when regions resolve at different times; homeologous-exchange candidates on 11 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome).

residual tetrasomy on 10/19 homeolog-paired chromosome numbers (5 pair(s)), 310.0 Mb, 52% of it terminal; pair-depth CV 0.54; homeology map: 69% of the genome in 108 homeolog blocks, 14 chromosome pairs, block-divergence CV 0.48; exchange: chr01(chr18:2/2);chr02(chr03:1/2;chr07:2/2;chr18:1/2);chr03(chr02:1/

Caveats: Holocentric chromosomes (Eleocharis): fission and fusion are frequent, so chromosome numbers need not reflect ancestral units. HAP2 chr11/chr12 were swapped in the assembly (corrected; confirmed collinear by alignment).

lpTriTurg1 *Triticum turgidum*

Known: 2n = 4x = 28 (AABB), allotetraploid (known)

Assembly	HAP1 (14), HAP2 (14); 14 chromosome numbers, 2 copies each; 1876 sequences unplaced
Numbering fixes	corrected 14
Distances	same number 0.0004, different numbers 0.0812
Homeolog pairs	7 (median 0.081, CV 0.0107)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 14
Fusions	none
Residual tetrasomy	0 of 14 tested, 0.0 Mb
Exchange candidates	none

Ploidy — high

2 haplotype copies of each of 14 chromosome numbers; the numbers form sets of 2 (7 homeolog pairs covering 100% of numbers), so x ≈ 7 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication.

copies from the assembly (2 per number); 7 homeolog pairs covering 100% of numbers; homeolog distance 0.081 vs unrelated 0.081

Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.01). This is also what a long-rediploidized autopolyploid looks like. The chromosome numbers form two subgenomes (7 and 7 numbers, size-independent z = 179) and every homeolog pair crosses them. Non-homologous chromosomes within a subgenome are closer (0.069) than homeologs across them (0.081): each subgenome carries its own repeats, as when two diverged genomes merge. 14 chromosome numbers assessed: 14 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate; subgenomes chr01,chr02,chr07,chr08,chr09,chr11,chr12 | chr03,chr04,chr05,chr06,chr10,chr13,chr14 (z = 178.8); homeolog pairs tested across them
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0137
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/14 homeolog-paired chromosome numbers; pair-depth CV 0.01; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: Chromosomes numbered automatically (by length); the k=2 partition is exactly the A (1A-7A) and B (1B-7B) subgenomes. No A-B homeolog pair is detectable by k-mers.

13.2 Cryptic or partial duplication in two-haplotype genomes

daInuConz1 *Inula conyza (now Pentanema conyzae)*

Known: tetraploid, 2n=4x=32, base number x=8 -- confirmed across at least 7 independent karyological studies spanning 1977-2024, single cytotype reported (no mixed-ploidy populations found for this species specifically, unlike some congeners); 2n=32; origin: allo-leaning, from clade-level context rather than a direct genomic study of this species: the Pentanema conyzae clade (Inula's Inuleae relatives, this species' genus following 2018 recircumscription) shows "reticulate patterns... best explained by hybridization and introgression" (Gutierrez-Larruscain et al. 2018) and allopolyploidization is explicitly proposed as a diversification mechanism within this exact clade (Gutierrez-Larruscain et al. 2019, AFLP fingerprinting, for congeners P. langeanum/P. maletii); ploidy across the clade ranges 2n=2x=16 (diploid) to hexaploid on the same x=8 base -- this species' tetraploid level is consistent with being an early/founding polyploidization step in that documented reticulate history (Siljak-Yakovlev, Garnatje, Perez-Lorenzo & Valles 2024; Garnatje et al. 2024 Flora Mediterranea; Santos-Vicente, Rico & Martinez-Ortega 2009 Taxon (IAPT/IOPB data 8); Morton 1977 Watsonia; Marhold et al. 2009/2013 Taxon; Zonneveld 2019; Chalup & Seijo 2013 Taxon; Gutierrez-Larruscain et al. 2018 Taxon (Pentanema phylogeny); Gutierrez-Larruscain et al. 2019 Bot J Linn Soc (AFLP))

Assembly	HAP1 (16), HAP2 (16); 16 chromosome numbers, 2 copies each; 6169 sequences unplaced
Numbering fixes	none
Distances	same number 0.0007, different numbers 0.0948
Homeolog pairs	8 (median 0.095, CV 0.0160)
Homeology map	32% of the genome in 188 blocks; 9 chromosome pairs; block CV 0.070; two or more partners: chr01:chr10+chr12, chr10:chr01+chr15
States	resolved lineages 16
Fusions	none
Residual tetrasomy	0 of 16 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 16 chromosome numbers; the numbers form sets of 2 (8 homeolog pairs covering 100% of numbers), so x ≈ 8 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 8 homeolog pairs covering 100% of numbers; homeolog distance 0.095 vs unrelated 0.095
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.02). This is also what a long-rediploidized autopolyploid looks like. The chromosome numbers form two subgenomes (8 and 8 numbers, size-independent z = 136) and every homeolog pair crosses them. Non-homologous chromosomes within a subgenome are closer (0.079) than homeologs across them (0.095): each subgenome carries its own repeats, as when two diverged genomes merge. One subgenome is 1.35x the size of the other. 16 chromosome numbers assessed: 16 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate; subgenomes chr01,chr02,chr03,chr04,chr05,chr06,chr09,chr15 | chr07,chr08,chr10,chr11,chr12,chr13,chr14,chr16 (z = 136.5); homeolog pairs tested across them

TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0166
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 2 chromosome(s) carry blocks homeologous to two or more others (chr01:chr10+chr12, chr10:chr01+chr15): fusions, fissions or translocations since the duplication. residual tetrasomy on 0/16 homeolog-paired chromosome numbers; pair-depth CV 0.02; homeology map: 32% of the genome in 188 homeolog blocks, 9 chromosome pairs, block-divergence CV 0.07

dmRanRepe1 *Ranunculus repens*

Known: not externally checked for ploidy level; origin: no longer treated as diploid -- see notes (none)

Assembly	HAP1 (16), HAP2 (16); 16 chromosome numbers, 2 copies each; 1112 sequences unplaced
Numbering fixes	none
Distances	same number 0.0230, different numbers 0.0710
Homeolog pairs	8 (median 0.071, CV 0.0089)
Homeology map	6% of the genome in 170 blocks; 7 chromosome pairs; block CV 0.125
States	resolved lineages 16
Fusions	none
Residual tetrasomy	0 of 16 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 16 chromosome numbers; the numbers form sets of 2 (8 homeolog pairs covering 100% of numbers), so x ≈ 8 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 8 homeolog pairs covering 100% of numbers; homeolog distance 0.071 vs unrelated 0.071
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.01). This is also what a long-rediploidized autopolyploid looks like. The chromosome numbers form two subgenomes (8 and 8 numbers, size-independent z = 109) and every homeolog pair crosses them. Non-homologous chromosomes within a subgenome are closer (0.066) than homeologs across them (0.071): each subgenome carries its own repeats, as when two diverged genomes merge. One subgenome is 1.23x the size of the other. 16 chromosome numbers assessed: 16 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate; subgenomes chr01,chr02,chr03,chr04,chr05,chr06,chr07,chr08 | chr09,chr10,chr11,chr12,chr13,chr14,chr15,chr16 (z = 108.7); homeolog pairs tested across them
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0658
Rediploidization	medium	Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start. residual tetrasomy on 0/16 homeolog-paired chromosome numbers; pair-depth CV 0.01; homeology map: 6% of the genome in 170 homeolog blocks, 7 chromosome pairs, block-divergence CV 0.12

dcCerAlpi1 *Cerastium alpinum*

Known: not externally checked for ploidy level; origin: no longer treated as diploid -- see notes (none)

Assembly	HAP1 (36), HAP2 (36); 36 chromosome numbers, 2 copies each; 4061 sequences unplaced
Numbering fixes	none
Distances	same number 0.0076, different numbers 0.0983
Homeolog pairs	18 (median 0.083, CV 0.0314)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 36
Fusions	none
Residual tetrasomy	0 of 36 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 36 chromosome numbers; the numbers form sets of 2 (18 groups of mostly 2 chromosome numbers), so x ≈ 18 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 18 groups of mostly 2 chromosome numbers (genome partition z = 25.8); homeolog distance 0.083 vs unrelated 0.098
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.03). This is also what a long-rediploidized autopolyploid looks like.

36 chromosome numbers assessed: 36 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate

TE markers not assessable — Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it.
te_marker_fraction mean 0.1979

Rediploidization medium — Diverged throughout the paired chromosomes: no stretch where homeologs are as close as alleles. Rediploidization is complete, or the genome was allo-like from the start.
residual tetrasomy on 0/36 homeolog-paired chromosome numbers; pair-depth CV 0.03; homology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: Each of its 18 homeolog pairs joins a large and a small chromosome: subgenomes of different chromosome size (found after the size adjustment).

daSonOler1 *Sonchus oleraceus*

Known: not formally verified this session; general recollection of 2n=18 (n=9) flagged as uncertain, conflicts with our n=16; uncertain, possible discrepancy; origin: n/a (appears diploid from our own data) (none formally checked)

Assembly	HAP1 (16), HAP2 (16); 16 chromosome numbers, 2 copies each; 460 sequences unplaced
Numbering fixes	none
Distances	same number 0.0000, different numbers 0.1103
Homeolog pairs	1 (median 0.082, CV –)
Homeology map	18% of the genome in 27 blocks; 9 chromosome pairs; block CV 0.133; two or more partners: chr01:chr10+chr13, chr06:chr07+chr14, chr07:chr06+chr16
States	resolved lineages 2, not assessable 14
Fusions	none
Residual tetrasomy	0 of 15 tested, 0.0 Mb
Exchange candidates	none

Ploidy high — 2 haplotype copies of each of 16 chromosome numbers; an older duplication shows on part of the genome (1 whole-chromosome homeolog pairs, 18% of the genome in homeolog blocks), too little to read duplicated sets from.
copies from the assembly (2 per number); 1 whole-chromosome homeolog pairs, 18% of the genome in homeolog blocks

Auto- or allo-like medium — Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs. This is also what a long-rediploidized autopolyploid looks like.
2 chromosome numbers assessed: 2 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate

TE markers not assessable — Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it.
te_marker_fraction mean 0.0392

Rediploidization medium — Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 3 chromosome(s) carry blocks homeologous to two or more others (chr01:chr10+chr13, chr06:chr07+chr14, chr07:chr06+chr16): fusions, fissions or translocations since the duplication.
residual tetrasomy on 0/15 homeolog-paired chromosome numbers; homeology map: 18% of the genome in 27 homeolog blocks, 9 chromosome pairs, block-divergence CV 0.13

dcHonPepl1 *Honckenya peploides*

Known: not externally checked; origin: n/a (none)

Assembly	HAP1 (34), HAP2 (34); 34 chromosome numbers, 2 copies each; 446 sequences unplaced
Numbering fixes	none
Distances	same number 0.0023, different numbers 0.0748
Homeolog pairs	0
Homeology map	3% of the genome in 22 blocks; 10 chromosome pairs; block CV 0.150; two or more partners: chr02:chr29+chr34, chr03:chr16+chr25, chr04:chr11+chr26, chr06:chr05+chr13+chr22
States	not assessable 34
Fusions	none
Residual tetrasomy	0 of 15 tested, 0.0 Mb
Exchange candidates	none

Ploidy high — 2 haplotype copies of each of 34 chromosome numbers; an older duplication shows on part of the genome (3% of the genome in homeolog blocks), too little to read duplicated sets from.
copies from the assembly (2 per number); 3% of the genome in homeolog blocks

Auto- or allo-like	not assessable	Not assessable: duplicated blocks are detected (homeology map), but no whole-chromosome homeolog pairs to read the copies across. homeology map: 3% of the genome in blocks, 10 chromosome pairs; no whole-chromosome pairs
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0282
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 4 chromosome(s) carry blocks homeologous to two or more others (chr02:chr29+chr34, chr03:chr16+chr25, chr04:chr11+chr26, chr06:chr05+chr13+chr22): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.15), as when regions resolve at different times. residual tetrasomy on 0/15 homeolog-paired chromosome numbers; homeology map: 3% of the genome in 22 homeolog blocks, 10 chromosome pairs, block-divergence CV 0.15

13.3 Genomes assembled with three or more copies

SchCurv1 *Schizothorax curvilabiatus*

Known: 4n = 98, autotetraploid snow carp (Xie et al. 2026)

Assembly	HAP1 (25), HAP2 (25), HAP3 (24), HAP4 (24); 25 chromosome numbers, 4 copies each; 2006 sequences unplaced
Numbering fixes	none
Distances	same number 0.0129, different numbers 0.1249
Homeolog pairs	2 (median 0.099, CV 0.1586)
Homeology map	2% of the genome in 4 blocks; 1 chromosome pairs; block CV 0.186
States	tetrasomic like 17, candidate 2, partially resolved 2, fusion lineages 2, one divergent copy 2
Fusions	chr19+chr22 (HAP3), chr19+chr22 (HAP4)
Residual tetrasomy	0 of 4 tested, 0.0 Mb
Exchange candidates	chr10 2.2 Mb (chr03:1/4)

Ploidy	high	4 haplotype copies of each of 25 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	Mixed: tetrasomic-like and split chromosomes in one genome (see rediploidization). 25 chromosome numbers assessed: 4 split into lineages, 17 tetrasomic-like, 2 one divergent copy, 2 candidate
TE markers	low	Repeat history tracks the lineages on 4 chromosome(s) only. TE split >= 2 on 4/24 chromosomes (median 1.10); absolute te_marker_fraction 0.2089 (reference only)
Rediploidization	high	Yes: 1 chromosome fusion(s) between copies, which separate fused and unfused lineages (the snow carp mechanism). Also: homeolog blocks diverged to different depths (CV 0.19), as when regions resolve at different times; homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). 1 fusion(s): chr19+chr22; 4 split / 17 tetrasomic-like chromosomes; 4 regional split(s); residual tetrasomy on 0/4 homeolog-paired chromosome numbers; pair-depth CV 0.16; homeology map: 2% of the genome in 4 homeolog blocks, 1 chromosome pairs, block-divergence CV 0.19; exchange: chr10(chr03:1/4)

Caveats: copy number varies between chromosomes: 2 copies x1, 4 copies x24

SchYoun1 *Schizopygopsis younghusbandi*

Known: 4n = 90, autotetraploid snow carp (Xie et al. 2026)

Assembly	HAP1 (25), HAP2 (15), HAP3 (25), HAP4 (15); 25 chromosome numbers, 4 copies each; 10 sequences unplaced
Numbering fixes	none
Distances	same number 0.0168, different numbers 0.1331
Homeolog pairs	1 (median 0.120, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 14, candidate 1, fusion lineages 10
Fusions	chr04+chr15 (HAP2), chr04+chr15 (HAP4), chr08+chr16 (HAP2), chr08+chr16 (HAP4), chr11+chr14 (HAP2), chr11+chr14 (HAP4), chr19+chr22 (HAP2), chr19+chr22 (HAP4), chr20+chr23 (HAP2), chr20+chr23 (HAP4)
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 25 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	Mixed: tetrasomic-like and split chromosomes in one genome (see rediploidization). 25 chromosome numbers assessed: 10 split into lineages, 14 tetrasomic-like, 0 one divergent copy, 1 candidate
TE markers	medium	Repeat history does not track the copies' lineages (no within-genome TE contrast). TE split >= 2 on 0/15 chromosomes (median 1.06); absolute te_marker_fraction 0.2227 (reference only)
Rediploidization	high	Yes: 5 chromosome fusion(s) between copies, which separate fused and unfused lineages (the snow carp mechanism). 5 fusion(s): chr04+chr15, chr08+chr16, chr11+chr14, chr19+chr22, chr20+chr23; 10 split / 14 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: copy number varies between chromosomes: 2 copies x10, 4 copies x15

drLytSali1 *Lythrum salicaria*

Known: autotetraploid with tetrasomic inheritance; unspecified; origin: auto (user-provided, not yet formally cited)

Assembly	HAP1 (15), HAP2 (15), HAP3 (15), HAP4 (15); 15 chromosome numbers, 4 copies each; 614 sequences unplaced
Numbering fixes	corrected 8
Distances	same number 0.0248, different numbers 0.1254
Homeolog pairs	5 (median 0.097, CV 0.0465)
Homeology map	42% of the genome in 31 blocks; 8 chromosome pairs; block CV 0.057; two or more partners: chr04:chr05+chr07, chr05:chr04+chr10, chr07:chr04+chr10, chr10:chr05+chr07, chr13:chr12+chr14
States	tetrasomic like 13, candidate 1, one divergent copy 1
Fusions	none
Residual tetrasomy	0 of 11 tested, 0.0 Mb
Exchange candidates	none

Ploidy	medium	4 haplotype copies of each of 15 chromosome numbers; the numbers form sets of 2 (5 homeolog pairs covering 67% of numbers), so x ≈ 8 and up to 8x relative to x. copies from the assembly (4 per number); 5 homeolog pairs covering 67% of numbers; homeolog distance 0.097 vs unrelated 0.125
Auto- or allo-like	high	Auto-like (tetrasomic-like): on most chromosomes all copies are about equally close. 15 chromosome numbers assessed: 0 split into lineages, 13 tetrasomic-like, 1 one divergent copy, 1 candidate
TE markers	low	Repeat history tracks the lineages on 1 chromosome(s) only. TE split >= 2 on 1/15 chromosomes (median 1.05); absolute te_marker_fraction 0.3070 (reference only)
Rediploidization	medium	No sign yet: copies interchangeable throughout. Also: 5 chromosome(s) carry blocks homeologous to two or more others (chr04:chr05+chr07, chr05:chr04+chr10, chr07:chr04+chr10, chr10:chr05+chr07, chr13:chr12+chr14): fusions, fissions or translocations since the duplication. 0 split / 13 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/11 homeolog-paired chromosome numbers; pair-depth CV 0.05; homeology map: 42% of the genome in 31 homeolog blocks, 8 chromosome pairs, block-divergence CV 0.06

daBudDavi1 *Buddleja davidii*

Known: tetraploid 2n=76; 2n=76 (x=19); origin: allo (Yang et al. 2023, Annals of Botany (ASTRAL/PhyloNet))

Assembly	HAP1 (19), HAP2 (19), HAP3 (19), HAP4 (19); 19 chromosome numbers, 4 copies each; 239 sequences unplaced
Numbering fixes	none
Distances	same number 0.0270, different numbers 0.1029
Homeolog pairs	1 (median 0.099, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 15, candidate 1, partially resolved 1, one divergent copy 2
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 19 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	Auto-like (tetrasomic-like): on most chromosomes all copies are about equally close. 19 chromosome numbers assessed: 1 split into lineages, 15 tetrasomic-like, 2 one divergent copy, 1 candidate

TE markers	low	Repeat history tracks the lineages on 1 chromosome(s) only. TE split >= 2 on 1/19 chromosomes (median 1.11); absolute te_marker_fraction 0.1668 (reference only)
Rediploidization	low	Little sign: 1 chromosome(s) partly split among tetrasomic-like ones. 1 split / 15 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

daGalBore1 *Galium boreale*

Known: tetraploid 2n=44 (Eurasian taxon); hexaploid 2n=66 in the American-Asiatic taxon (sometimes split as G. septentrionale), part of a wider 2n=22/44/66(+) polyploid complex; 2n=44 (n=11); origin: unresolved -- literature is cytology-only (chromosome counts, pollen-size/ploidy correlation), none address auto vs allo directly (Love & Love 1954; Kliphuis 1973/1986; Schanzer & Elkordy 2014; Natali, Manen & Ehrendorfer 1995 (Rubieae phylogeny); Zonneveld 2019)

Assembly	HAP1 (11), HAP2 (11), HAP3 (11), HAP4 (11); 11 chromosome numbers, 4 copies each; 338 sequences unplaced
Numbering fixes	none
Distances	same number 0.0244, different numbers 0.0996
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 10, partially resolved 1
Fusions	none
Residual tetrasomy	0 of 4 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 11 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	high	Auto-like (tetrasomic-like): on most chromosomes all copies are about equally close. 11 chromosome numbers assessed: 1 split into lineages, 10 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	medium	Repeat history does not track the copies' lineages (no within-genome TE contrast). TE split >= 2 on 0/11 chromosomes (median 1.05); absolute te_marker_fraction 0.1289 (reference only)
Rediploidization	low	Little sign: 1 chromosome(s) partly split among tetrasomic-like ones. 1 split / 10 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/4 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

ddLepDrab1 *Lepidium draba*

Known: polyploid series, base x=8: octoploid in Europe (2n=64), hexaploid-decaploid in western Asia; also cited as n=16,32; 2n=64 (octoploid, Europe); origin: allo-leaning at the tribe level (Lepidieae, which contains L. draba, is described as falling within the "hybrid or highly polyploid classes"; tribe-wide pattern attributed to "rampant hybridization" that "often precedes allopolyploidization"), but no experimentally-confirmed origin statement specific to L. draba itself (Walden, Kiefer & Koch 2026 Systematic Biology; Kagale et al. 2014 Plant Cell ("Lepidium event" WGD shared by L. densiflorum/meyenii/sativum); Francis & Warwick 2008 (chromosome counts); Hendriks et al. 2023 Current Biology (Brassicaceae 1000-gene phylogeny))

Assembly	HAP1 (16), HAP2 (16), HAP3 (16), HAP4 (16); 16 chromosome numbers, 4 copies each; 311 sequences unplaced
Numbering fixes	corrected 41
Distances	same number 0.0214, different numbers 0.1429
Homeolog pairs	8 (median 0.069, CV 0.0617)
Homeology map	68% of the genome in 34 blocks; 8 chromosome pairs; block CV 0.132
States	tetrasomic like 8, candidate 1, partially resolved 1, one divergent copy 6
Fusions	none
Residual tetrasomy	0 of 16 tested, 0.0 Mb
Exchange candidates	chr03 3.8 Mb (chr06:1/4)

Ploidy	high	4 haplotype copies of each of 16 chromosome numbers; the numbers form sets of 2 (8 groups of mostly 2 chromosome numbers), so x ≈ 8 and up to 8x relative to x. copies from the assembly (4 per number); 8 groups of mostly 2 chromosome numbers (genome partition z = 19.4); homeolog distance 0.069 vs unrelated 0.143
Auto- or allo-like	medium	Mixed: tetrasomic-like and split chromosomes in one genome (see rediploidization). 16 chromosome numbers assessed: 1 split into lineages, 8 tetrasomic-like, 6 one divergent copy, 1 candidate
TE markers	low	Repeat history tracks the lineages on 1 chromosome(s) only. TE split >= 2 on 1/16 chromosomes (median 1.25); absolute te_marker_fraction 0.5465 (reference only)

Rediploidization	low	Little sign: 1 chromosome(s) partly split among tetrasomic-like ones. Also: homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). 1 split / 8 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/16 homeolog-paired chromosome numbers; pair-depth CV 0.06; homeology map: 68% of the genome in 34 homeolog blocks, 8 chromosome pairs, block-divergence CV 0.13; exchange: chr03(chr06:1/4)

ddEmpNigr1 *Empetrum nigrum*

Known: two cytotypes: diploid E. nigrum s.str. (2n=26) vs tetraploid E. hermaphroditum (2n=52); our assembly (4 copies, n=13) matches the tetraploid; 2n=52 (n=13); origin: auto (Faegri 1937 (explicit); Meirmans & Kolar 2025 PNAS (uses this species pair as a clean autopolyploid model specifically because it is not confounded by hybridization))

Assembly	HAP1 (13), HAP2 (13), HAP3 (13), HAP4 (13); 13 chromosome numbers, 4 copies each; 745 sequences unplaced
Numbering fixes	none
Distances	same number 0.0197, different numbers 0.1073
Homeolog pairs	0
Homeology map	2% of the genome in 2 blocks; 1 chromosome pairs; block CV –
States	partially resolved 8, resolved lineages 2, one divergent copy 3
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 13 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	Allo-like (disomic-like): most chromosomes split into two lineages. A long-rediploidized autopolyploid would look the same. 13 chromosome numbers assessed: 10 split into lineages, 0 tetrasomic-like, 3 one divergent copy, 0 candidate
TE markers	medium	Repeat history tracks the lineages on several chromosomes: separate TE histories in the split copies. TE split >= 2 on 4/13 chromosomes (median 1.73); absolute te_marker_fraction 0.2386 (reference only)
Rediploidization	medium	Split throughout: either rediploidization is complete or the genome was allo-like from the start; these data cannot tell which. Also: lineage switches on 2 chromosome(s) (chr05;chr10): a copy that is one lineage's sequence along part of its length and another's along the rest (homeologous exchange, or a phasing switch between copies). 10 split / 0 tetrasomic-like chromosomes; 2 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 2% of the genome in 2 homeolog blocks, 1 chromosome pairs

drAriEdul1 *Aria edulis*

Known: diploid 2n=34 per modern (flow cytometry) sources -- CONTRADICTS our assembly's 4-copy structure; 2n=34 (x=17); base number matches our 17 chromosome numbers exactly, only the copy count disagrees; origin: n/a for true diploid A. edulis; if our sample is actually a tetraploid relative (cf. newly-described A. wyensis), allo per that source (Green 2024's own hypothesis: A. wyensis probably derived indirectly from a cross between diploid A. edulis and a member of the A. porrigentiformis group -- an allotetraploid origin if that hypothesis holds) (Green 2024 (A. wyensis); Andersen et al. 2024; Kelly & Hanley 2020; King et al. 2023; Pellicer et al. 2012 Annals of Botany (Sorbus cytotype survey); Dickinson 2018 Taxon)

Assembly	HAP1 (17), HAP2 (17), HAP3 (17), HAP4 (17); 17 chromosome numbers, 4 copies each; 182 sequences unplaced
Numbering fixes	none
Distances	same number 0.0138, different numbers 0.1095
Homeolog pairs	8 (median 0.099, CV 0.0279)
Homeology map	24% of the genome in 29 blocks; 10 chromosome pairs; block CV 0.056; two or more partners: chr01:chr06+chr14, chr13:chr15+chr16, chr15:chr11+chr13
States	tetrasomic like 3, candidate 7, partially resolved 3, one divergent copy 4
Fusions	none
Residual tetrasomy	0 of 17 tested, 0.0 Mb
Exchange candidates	none

Ploidy	medium	4 haplotype copies of each of 17 chromosome numbers; the numbers form sets of 2 (8 groups of mostly 2 chromosome numbers), so x ≈ 8 and up to 8x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (4 per number); 8 groups of mostly 2 chromosome numbers (genome partition z = 10.1); homeolog distance 0.099 vs unrelated 0.109

Auto- or allo-like	medium	Mixed: tetrasomic-like and split chromosomes in one genome (see rediploidization). 17 chromosome numbers assessed: 3 split into lineages, 3 tetrasomic-like, 4 one divergent copy, 7 candidate
TE markers	low	Repeat content differs between the split copies on several chromosomes, but the split isolates the same odd haplotype throughout, so this may reflect that haplotype's assembly rather than separate TE histories. TE split >= 2 on 6/17 chromosomes (median 1.47); absolute te_marker_fraction 0.2194 (reference only)
Rediploidization	medium	Likely under way: some chromosomes have split into lineages while others stay tetrasomic-like. Also: 3 chromosome(s) carry blocks homeologous to two or more others (chr01:chr06+chr14, chr13:chr15+chr16, chr15:chr11+chr13): fusions, fissions or translocations since the duplication. 3 split / 3 tetrasomic-like chromosomes; 5 regional split(s); residual tetrasomy on 0/17 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 24% of the genome in 29 homeolog blocks, 10 chromosome pairs, block-divergence CV 0.06

ddHesMatr1 *Hesperis matronalis*

Known: highly cytologically variable: n=7,8,12,14,24 and 2n=14,16,24,26,28 reported across populations; genome very large (~8Gb), one detailed cytological study specifically describes it as a SEGMENTAL ALLOTETRAPLOID; 2n=24 (most commonly cited tetraploid count); origin: segmental allotetraploid -- a genuine third category, distinct from both clean auto and clean allo: Francis et al. 2009 reports BOTH bivalent (disomic/allo-like) AND quadrivalent (tetrasomic/auto-like) chromosome associations at meiosis in the same cytological study, i.e. partial homology between subgenomes rather than a clean split either way (Francis, Cavers & Warwick 2009 Canadian J Plant Science (segmental allotetraploid, bivalent+quadrivalent); Raina 1987 Cytologia (suggests autopolyploid-consistent multivalent formation, some tension with Francis et al.); Manton 1932 Annals of Botany (aneuploid-loss origin hypothesis); Hohmann et al. 2015 Plant Cell / Mandakova et al. 2017 Plant Cell (phylogenomics: no past mesopolyploidization event detected despite the huge genome size); Rifkin et al. 2026 bioRxiv (both diploid and tetraploid cytotypes found in wild populations, new diploid-accession genome))

Assembly	HAP1 (6), HAP2 (6), HAP3 (6), HAP4 (6); 6 chromosome numbers, 4 copies each; 2373 sequences unplaced
Numbering fixes	corrected 2
Distances	same number 0.0256, different numbers 0.0723
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 1, resolved lineages 1, one divergent copy 4
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 6 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	One copy apart on most chromosomes, changing haplotype between chromosomes: AAAB-like (one divergent genome copy). 6 chromosome numbers assessed: 1 split into lineages, 1 tetrasomic-like, 4 one divergent copy, 0 candidate
TE markers	medium	Repeat history tracks the lineages on several chromosomes: separate TE histories in the split copies. TE split >= 2 on 1/6 chromosomes (median 1.22); absolute te_marker_fraction 0.0836 (reference only)
Rediploidization	low	Little sign: 1 chromosome(s) partly split among tetrasomic-like ones. Also: lineage switches on 1 chromosome(s) (chr06): a copy that is one lineage's sequence along part of its length and another's along the rest (homeologous exchange, or a phasing switch between copies). 1 split / 1 tetrasomic-like chromosomes; 0 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

ddHypMacu1 *Hypericum maculatum*

Known: subsp. obtusiusculum tetraploid 2n=32; 2n=32 (x=8); origin: auto (Koch et al. 2013; Robson & Adams 1968; Robson 2002)

Assembly	HAP1 (8), HAP2 (8), HAP3 (8), HAP4 (8); 8 chromosome numbers, 4 copies each; 759 sequences unplaced
Numbering fixes	none
Distances	same number 0.0423, different numbers 0.1273
Homeolog pairs	1 (median 0.113, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	one divergent copy 8
Fusions	none

Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	chr01 19.0 Mb (chr03:1/4); chr03 2.5 Mb (chr01:1/4)

Ploidy	high	4 haplotype copies of each of 8 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	One copy apart on most chromosomes, nearly always the same haplotype (HAP1 (8 chromosomes)): most likely an assembly or phasing problem in that haplotype, unless the assembly was phased by subgenome. 8 chromosome numbers assessed: 0 split into lineages, 0 tetrasomic-like, 8 one divergent copy, 0 candidate
TE markers	low	Repeat content differs between the split copies on several chromosomes, but the split isolates the same odd haplotype throughout, so this may reflect that haplotype's assembly rather than separate TE histories. TE split >= 2 on 5/8 chromosomes (median 2.17); absolute te_marker_fraction 0.4445 (reference only)
Rediploidization	not assessable	Not assessable: the copies split as one divergent set (AAAB-like), not into lineages within a duplicated genome. Also: lineage switches on 1 chromosome(s) (chr06): a copy that is one lineage's sequence along part of its length and another's along the rest (homeologous exchange, or a phasing switch between copies); homeologous-exchange candidates on 2 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). 0 split / 0 tetrasomic-like chromosomes; 0 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs; exchange: chr01(chr03:1/4);chr03(chr01:1/4)

ddHypPerf1 *Hypericum perforatum*

Known: tetraploid 2n=32; 2n=32 (x=8); origin: genuinely unresolved/contested in the field -- both hypotheses actively cited in recent sources (Martonfi 2008; Nurk 2011; Koch et al. 2013; Corral et al. 2011 ("whether it is allo- or autopolyploid remains unclear"))

Assembly	HAP1 (8), HAP2 (8), HAP3 (8), HAP4 (8); 8 chromosome numbers, 4 copies each; 418 sequences unplaced
Numbering fixes	none
Distances	same number 0.0362, different numbers 0.1269
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 2, partially resolved 2, one divergent copy 4
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	4 haplotype copies of each of 8 chromosome numbers; no older duplicated sets detected. copies from the assembly (4 per number)
Auto- or allo-like	medium	One copy apart on most chromosomes, changing haplotype between chromosomes: AAAB-like (one divergent genome copy). 8 chromosome numbers assessed: 2 split into lineages, 2 tetrasomic-like, 4 one divergent copy, 0 candidate
TE markers	medium	Repeat history tracks the lineages on several chromosomes: separate TE histories in the split copies. TE split >= 2 on 2/8 chromosomes (median 1.69); absolute te_marker_fraction 0.4078 (reference only)
Rediploidization	medium	Likely under way: some chromosomes have split into lineages while others stay tetrasomic-like. 2 split / 2 tetrasomic-like chromosomes; 1 regional split(s); residual tetrasomy on 0/2 homeolog-paired chromosome numbers (no unrelated-chromosome control; shared repeats not excluded); homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

ddSalTria1 *Salix triandra*

Known: historically reported as DIPLOID (haploid n=19 or n=22, two forms) -- CONTRADICTS our assembly's triploid (3-copy) structure; haploid n=19 or n=22 (diploid) per a 1924 source -- no modern triploid count found; origin: no information found on auto/allo origin either way (Blackburn & Harrison 1924 Annals of Botany)

Assembly	HAP1 (19), HAP2 (19), HAP3 (19); 19 chromosome numbers, 3 copies each; 154 sequences unplaced
Numbering fixes	none
Distances	same number 0.0155, different numbers 0.1619
Homeolog pairs	3 (median 0.140, CV 0.0328)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	tetrasomic like 13, one divergent copy 6
Fusions	none

Residual tetrasomy	0 of 6 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	3 haplotype copies of each of 19 chromosome numbers; an odd copy number (triploid-like or higher odd ploidy); no older duplicated sets detected. copies from the assembly (3 per number)
Auto- or allo-like	medium	Auto-like (tetrasomic-like): on most chromosomes all copies are about equally close. 19 chromosome numbers assessed: 0 split into lineages, 13 tetrasomic-like, 6 one divergent copy, 0 candidate
TE markers	low	Repeat history tracks the lineages on 2 chromosome(s) only. TE split >= 2 on 2/19 chromosomes (median 1.22); absolute te_marker_fraction 0.2473 (reference only)
Rediploidization	medium	No sign yet: copies interchangeable throughout. Also: lineage switches on 1 chromosome(s) (chr03): a copy that is one lineage's sequence along part of its length and another's along the rest (homeologous exchange, or a phasing switch between copies). 0 split / 13 tetrasomic-like chromosomes; 0 regional split(s); residual tetrasomy on 0/6 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

`icStrMela3` *Strophosoma melanogrammum*

Known: triploid (AAB) by the ToL k-mer ploidy plot (ToL genomic_data k31 ploidy plot)

Assembly	HAP1 (10), HAP1B (10), HAP2 (10); 10 chromosome numbers, 3 copies each; 653 sequences unplaced
Numbering fixes	extra_set 10
Distances	same number 0.0489, different numbers 0.0940
Homeolog pairs	2 (median 0.086, CV 0.0302)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	one divergent copy 10
Fusions	none
Residual tetrasomy	0 of 4 tested, 0.0 Mb
Exchange candidates	chr08 34.0 Mb (chr02:1/3)

Ploidy	high	3 haplotype copies of each of 10 chromosome numbers (HAP1B split out of one haplotype file holding two chromosome sets); an odd copy number (triploid-like or higher odd ploidy); no older duplicated sets detected. copies from the assembly (3 per number); extra set(s) matched one-to-one to the shared numbers
Auto- or allo-like	high	AAB-like: one divergent chromosome set (HAP1B, assembled as a second set in HAP1), the other copies as close as alleles. 10 chromosome numbers assessed: 0 split into lineages, 0 tetrasomic-like, 10 one divergent copy, 0 candidate
TE markers	low	Repeat content differs between the split copies on several chromosomes, but the split isolates the same odd haplotype throughout, so this may reflect that haplotype's assembly rather than separate TE histories. TE split >= 2 on 9/10 chromosomes (median 3.45); absolute te_marker_fraction 0.1659 (reference only)
Rediploidization	not assessable	Not assessable: the copies split as one divergent set (AAB-like), not into lineages within a duplicated genome. Also: lineage switches on 1 chromosome(s) (chr09): a copy that is one lineage's sequence along part of its length and another's along the rest (homeologous exchange, or a phasing switch between copies); homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). 0 split / 0 tetrasomic-like chromosomes; 0 regional split(s); residual tetrasomy on 0/4 homeolog-paired chromosome numbers; pair-depth CV 0.03; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs; exchange: chr08(chr02:1/3)

Caveats: a haplotype file held a second chromosome set; split out as HAP1B; HAP1 holds two chromosome sets (A as chr1-10, B as chr11-20); the matrix stage splits B out as HAP1B. HAP1 chr09 sits between A and B (0.030 to HAP2 chr09, 0.027 to B).

13.4 Salmonids

`OncMaso1` *Oncorhynchus masou*

Known: salmonid whole-genome duplication (Ss4R, ~90 Mya), autotetraploid origin; residual tetrasomy mapped in Atlantic salmon (Lien et al. 2016)

Assembly	HAP1 (33), HAP2 (33); 33 chromosome numbers, 2 copies each; 427 sequences unplaced
Numbering fixes	none
Distances	same number 0.0063, different numbers 0.1242

Homeolog pairs	7 (median 0.098, CV 0.0909)
Homeology map	16% of the genome in 48 blocks; 17 chromosome pairs; block CV 0.241; two or more partners: chr01:chr31+chr32, chr09:chr12+chr19, chr10:chr12+chr13, chr12:chr09+chr10, chr30:chr02+chr31, chr31:chr01+chr30, chr32:chr01+chr04+chr33, chr33:chr08+chr32
States	partially resolved 2, resolved lineages 12, not assessable 19
Fusions	none
Residual tetrasomy	2 of 25 tested, 8.6 Mb, 39% terminal; chr17 5.2 Mb (chr29:4/4); chr29 3.3 Mb (chr17:4/4)
Exchange candidates	chr16 2.8 Mb (chr06:1/2); chr29 3.5 Mb (chr17:1/2); chr30 2.8 Mb (chr02:1/2)

Ploidy	high	2 haplotype copies of each of 33 chromosome numbers; an older duplication shows on part of the genome (7 whole-chromosome homeolog pairs, 16% of the genome in homeolog blocks), too little to read duplicated sets from. copies from the assembly (2 per number); 7 whole-chromosome homeolog pairs, 16% of the genome in homeolog blocks
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.09). This is also what a long-rediploidized autopolyploid looks like. 14 chromosome numbers assessed: 14 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2029
Rediploidization	medium	Partly: 2 of 25 homeolog-paired chromosome numbers keep stretches of residual tetrasomy (8.6 Mb, 39% of it at chromosome ends) where the homeologs are still as close as alleles; elsewhere they have diverged. Also: 8 chromosome(s) carry blocks homeologous to two or more others (chr01:chr31+chr32, chr09:chr12+chr19, chr10:chr12+chr13, chr12:chr09+chr10, chr30:chr02+chr31, chr31:chr01+chr30, chr32:chr01+chr04+chr33, chr33:chr08+chr32): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.24), as when regions resolve at different times; homeologous-exchange candidates on 3 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 2/25 homeolog-paired chromosome numbers (1 pair(s)), 8.6 Mb, 39% of it terminal; pair-depth CV 0.09; homeology map: 16% of the genome in 48 homeolog blocks, 17 chromosome pairs, block-divergence CV 0.24; exchange: chr16(chr06:1/2);chr29(chr17:1/2);chr30(chr02:1/2)

Caveats: HAP2 was contig-level and was scaffolded on HAP1 (workflows/prep/scaffold_by_reference.py); positions along HAP2 are borrowed from HAP1, and exchange candidates can be scaffolding placements.

`OncMaso1_wu2026` *Oncorhynchus masou*

Known: salmonid whole-genome duplication (Ss4R, ~90 Mya); published Hi-C phased assembly (Wu et al. 2026)

Assembly	HAP1 (33), HAP2 (33); 33 chromosome numbers, 2 copies each; 184 sequences unplaced
Numbering fixes	ambiguous 1, corrected 23, displaced 2
Distances	same number 0.0093, different numbers 0.1242
Homeolog pairs	8 (median 0.104, CV 0.1283)
Homeology map	22% of the genome in 63 blocks; 17 chromosome pairs; block CV 0.425; two or more partners: chr02:chr21+chr31, chr07:chr08+chr11, chr08:chr07+chr16+chr26, chr09:chr13+chr22, chr11:chr07+chr12, chr16:chr03+chr08, chr21:chr02+chr30
States	candidate 2, resolved lineages 10, one divergent copy 4, not assessable 17
Fusions	none
Residual tetrasomy	0 of 28 tested, 0.0 Mb
Exchange candidates	chr01 27.2 Mb (chr25:1/2); chr02 21.8 Mb (chr21:1/2;chr31:1/2); chr09 4.2 Mb (chr13:1/2); chr14 13.2 Mb (chr29:1/2); chr16 7.4 Mb (chr03:1/2); chr17 2.8 Mb (chr04:1/2); +7 more

Ploidy	high	2 haplotype copies of each of 33 chromosome numbers; an older duplication shows on part of the genome (8 whole-chromosome homeolog pairs, 22% of the genome in homeolog blocks), too little to read duplicated sets from. copies from the assembly (2 per number); 8 whole-chromosome homeolog pairs, 22% of the genome in homeolog blocks
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.13). This is also what a long-rediploidized autopolyploid looks like. 16 chromosome numbers assessed: 10 split into lineages, 0 tetrasomic-like, 4 one divergent copy, 2 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3535

Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 7 chromosome(s) carry blocks homeologous to two or more others (chr02:chr21+chr31, chr07:chr08+chr11, chr08:chr07+chr16+chr26, chr09:chr13+chr22, chr11:chr07+chr12, chr16:chr03+chr08, chr21:chr02+chr30): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.42), as when regions resolve at different times; homeologous-exchange candidates on 13 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 0/28 homeolog-paired chromosome numbers; pair-depth CV 0.13; homeology map: 22% of the genome in 63 homeolog blocks, 17 chromosome pairs, block-divergence CV 0.42; exchange: chr01(chr25:1/2);chr02(chr21:1/2;chr31:1/2);chr09(chr13:1/2);chr14(chr29:1/2);chr16(chr03:1/2);chr17(chr04:1/2);ch

Caveats: 1 chromosome-numbering mismatch(es) left ambiguous (not applied)

`fSalAlp3` *Salvelinus alpinus*

Known: salmonid whole-genome duplication (Ss4R) (Lien et al. 2016)

Assembly	HAP1 (40), HAP2 (40); 40 chromosome numbers, 2 copies each; 3159 sequences unplaced
Numbering fixes	none
Distances	same number 0.0048, different numbers 0.1160
Homeolog pairs	12 (median 0.105, CV 0.0386)
Homeology map	18% of the genome in 72 blocks; 18 chromosome pairs; block CV 0.161; two or more partners: chr04:chr18+chr27, chr05:chr07+chr13, chr07:chr05+chr16+chr38, chr10:chr32+chr35, chr11:chr09+chr25, chr13:chr05+chr23
States	resolved lineages 24, not assessable 16
Fusions	none
Residual tetrasomy	0 of 29 tested, 0.0 Mb
Exchange candidates	chr09 5.0 Mb (chr11:1/2)

Ploidy	medium	2 haplotype copies of each of 40 chromosome numbers; the numbers form sets of 2 (12 homeolog pairs covering 60% of numbers), so x ≈ 20 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 12 homeolog pairs covering 60% of numbers; homeolog distance 0.105 vs unrelated 0.116
Auto- or allo-like	high	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.04). This is also what a long-rediploidized autopolyploid looks like. 24 chromosome numbers assessed: 24 split into lineages, 0 tetrasomic-like, 0 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.2206
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 6 chromosome(s) carry blocks homeologous to two or more others (chr04:chr18+chr27, chr05:chr07+chr13, chr07:chr05+chr16+chr38, chr10:chr32+chr35, chr11:chr09+chr25, chr13:chr05+chr23): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.16), as when regions resolve at different times; 2 low-divergence homeolog blocks (8 Mb, 2% of the block span, mostly at chromosome ends; chr09xchr11): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication; homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 0/29 homeolog-paired chromosome numbers; pair-depth CV 0.04; homeology map: 18% of the genome in 72 homeolog blocks, 18 chromosome pairs, block-divergence CV 0.16; LORe-like blocks: 2, 8.5 Mb; exchange: chr09(chr11:1/2)

`fCorLav1` *Coregonus lavaretus*

Known: salmonid whole-genome duplication (Ss4R) (Lien et al. 2016)

Assembly	HAP1 (40), HAP2 (40); 40 chromosome numbers, 2 copies each; 15050 sequences unplaced
Numbering fixes	corrected 30, displaced 1
Distances	same number 0.0050, different numbers 0.1057
Homeolog pairs	14 (median 0.092, CV 0.1934)
Homeology map	30% of the genome in 84 blocks; 21 chromosome pairs; block CV 0.344; two or more partners: chr03:chr07+chr13, chr05:chr13+chr16+chr19, chr06:chr21+chr40, chr13:chr03+chr05, chr15:chr10+chr24, chr19:chr05+chr23

States	resolved lineages 26, one divergent copy 2, not assessable 12
Fusions	none
Residual tetrasomy	0 of 37 tested, 0.0 Mb
Exchange candidates	chr35 4.2 Mb (chr38:1/2)

Ploidy	medium	2 haplotype copies of each of 40 chromosome numbers; the numbers form sets of 2 (18 groups of mostly 2 chromosome numbers), so x ≈ 18 and up to 4x relative to x; the sets are nearly as diverged as unrelated chromosomes, so they may be an old (paleo) duplication. copies from the assembly (2 per number); 18 groups of mostly 2 chromosome numbers (genome partition z = 19.7); homeolog distance 0.092 vs unrelated 0.106
Auto- or allo-like	medium	Allo-like (disomic-like): the duplicated sets do not interchange, from homeolog pairs (pair-depth CV 0.19). This is also what a long-rediploidized autopolyploid looks like. 28 chromosome numbers assessed: 26 split into lineages, 0 tetrasomic-like, 2 one divergent copy, 0 candidate
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1072
Rediploidization	medium	Advanced: no stretch where homeologs are as close as alleles, and the karyotype has been rearranged since the duplication. Also: 6 chromosome(s) carry blocks homeologous to two or more others (chr03:chr07+chr13, chr05:chr13+chr16+chr19, chr06:chr21+chr40, chr13:chr03+chr05, chr15:chr10+chr24, chr19:chr05+chr23): fusions, fissions or translocations since the duplication; homeolog blocks diverged to different depths (CV 0.34), as when regions resolve at different times; 13 low-divergence homeolog blocks (243 Mb, 40% of the block span; chr02xchr28, chr03xchr13, chr04xchr37, chr05xchr13, chr06xchr40): regions that stopped recombining later than the rest (delayed, LORe-like rediploidization), or a younger duplication; homeologous-exchange candidates on 1 chromosome(s), stretches closer to the homeolog than to the own homolog (exchange, or a contig placed on the wrong chromosome). residual tetrasomy on 0/37 homeolog-paired chromosome numbers; pair-depth CV 0.19; homeology map: 30% of the genome in 84 homeolog blocks, 21 chromosome pairs, block-divergence CV 0.34; LORe-like blocks: 13, 243.0 Mb; exchange: chr35(chr38:1/2)

Caveats: HAP2's NCBI chromosome numbers differ from HAP1's on 30 of 40 chromosomes (cycles); renumbered by the matrix stage. HAP2 chr33 has no allele in HAP1.

13.5 Diploid-like and negative controls

`ddMalSylv1` *Malva sylvestris*

Known: dead diploid per user; origin: n/a (diploid) (user-provided)

Assembly	HAP1 (21), HAP2 (21); 21 chromosome numbers, 2 copies each; 1555 sequences unplaced
Numbering fixes	none
Distances	same number 0.0038, different numbers 0.1159
Homeolog pairs	0
Homeology map	1% of the genome in 2 blocks; 1 chromosome pairs; block CV –
States	not assessable 21
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 21 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 0 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1346
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 1% of the genome in 2 homeolog blocks, 1 chromosome pairs

`fLepOcu1` *Lepisosteus oculatus*

Known: no duplication since vertebrate 2R; outgroup to the teleost duplication (negative control) (Braasch et al. 2016)

Assembly	HAP1 (29), HAP2 (29); 29 chromosome numbers, 2 copies each; 1590 sequences unplaced
Numbering fixes	none
Distances	same number 0.0040, different numbers 0.2026
Homeolog pairs	1 (median 0.128, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 27
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 29 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1425
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

fAmiCal2 *Amia calva*

Known: no duplication since vertebrate 2R; outgroup to the teleost duplication (negative control) (Thompson et al. 2021)

Assembly	HAP1 (23), HAP2 (23); 23 chromosome numbers, 2 copies each; 641 sequences unplaced
Numbering fixes	none
Distances	same number 0.0009, different numbers 0.1736
Homeolog pairs	1 (median 0.142, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 21
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 23 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0784
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgCepNemo3 *Cepaea nemoralis*

Known: selected as a WGD candidate (land snail) (candidate list)

Assembly	HAP1 (22), HAP2 (22); 22 chromosome numbers, 2 copies each; 7601 sequences unplaced
Numbering fixes	corrected 16
Distances	same number 0.0137, different numbers 0.1028
Homeolog pairs	1 (median 0.098, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 20
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 22 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets

TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1439
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgHygCinc1 *Hygromia cinctella*

Known: selected as a WGD candidate (land snail) (candidate list)

Assembly	HAP1 (21), HAP2 (21); 21 chromosome numbers, 2 copies each; 6993 sequences unplaced
Numbering fixes	none
Distances	same number 0.0152, different numbers 0.1262
Homeolog pairs	1 (median 0.124, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 19
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 21 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1391
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgDauRufa1 *Daudebardia rufa*

Known: selected as a WGD candidate (land snail) (candidate list)

Assembly	HAP1 (28), HAP2 (28); 28 chromosome numbers, 2 copies each; 3136 sequences unplaced
Numbering fixes	none
Distances	same number 0.0200, different numbers 0.1088
Homeolog pairs	1 (median 0.103, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 26
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 28 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.6125
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgLitLitt3 *Littorina littorea*

Known: selected as a WGD candidate (periwinkle) (candidate list)

Assembly	HAP1 (17), HAP2 (17); 17 chromosome numbers, 2 copies each; 5851 sequences unplaced
Numbering fixes	corrected 17
Distances	same number 0.0118, different numbers 0.1163
Homeolog pairs	1 (median 0.100, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	resolved lineages 2, not assessable 15

Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 17 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 1 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1044
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgMonCant1 *Monacha cantiana*

Known: selected as a WGD candidate (land snail) (candidate list)

Assembly	HAP1 (23), HAP2 (24); 24 chromosome numbers, 2 copies each; 3271 sequences unplaced
Numbering fixes	corrected 23
Distances	same number 0.0082, different numbers 0.1230
Homeolog pairs	1 (median 0.071, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	fusion lineages 2, not assessable 22
Fusions	chr23+chr24 (HAP1)
Residual tetrasomy	0 of 1 tested, 0.0 Mb
Exchange candidates	chr23 28.0 Mb (chr24:1/2)

Ploidy	high	2 haplotype copies of each of 24 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare; the only homeolog-like pair comes from a fusion carried by one haplotype. Diploid-like as assembled. 1 fusion(s) on one haplotype only: chr23+chr24 (HAP1); no homeolog pairs besides the fused components
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1928
Rediploidization	medium	Not read as rediploidization: a fusion carried by one haplotype, with no duplicated sets otherwise, is a fusion polymorphism or a scaffolding join. 1 fusion(s) on one haplotype only: chr23+chr24 (HAP1); no homeolog pairs besides the fused components

Caveats: copy number varies between chromosomes: 1 copies x1, 2 copies x23; HAP1 carries chr23+chr24 fused, HAP2 has them apart: a fusion polymorphism or a scaffolding join; HAP2 was numbered automatically.

xgPomEleg1 *Pomatias elegans*

Known: selected as a WGD candidate (land snail) (candidate list)

Assembly	HAP1 (13), HAP2 (13); 13 chromosome numbers, 2 copies each; 1005 sequences unplaced
Numbering fixes	none
Distances	same number 0.0073, different numbers 0.1158
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	not assessable 13
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 13 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 0 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.1452

Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

xgStaPalu1 *Stagnicola palustris*

Known: selected as a WGD candidate (freshwater snail) (candidate list)

Assembly	HAP1 (18), HAP2 (18); 18 chromosome numbers, 2 copies each; 2932 sequences unplaced
Numbering fixes	corrected 2
Distances	same number 0.0125, different numbers 0.1441
Homeolog pairs	3 (median 0.137, CV 0.0605)
Homeology map	2% of the genome in 5 blocks; 1 chromosome pairs; block CV 0.167
States	resolved lineages 6, not assessable 12
Fusions	none
Residual tetrasomy	0 of 7 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 18 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 3 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.3202
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/7 homeolog-paired chromosome numbers; pair-depth CV 0.06; homeology map: 2% of the genome in 5 homeolog blocks, 1 chromosome pairs, block-divergence CV 0.17

lpTriTurg1_A *Triticum turgidum, A subgenome*

Known: 2n = 4x = 28 (AABB); A run alone (known)

Assembly	HAP1 (7), HAP2 (7); 7 chromosome numbers, 2 copies each; 1890 sequences unplaced
Numbering fixes	none
Distances	same number 0.0003, different numbers 0.0687
Homeolog pairs	0
Homeology map	0% of the genome in 5 blocks; 1 chromosome pairs; block CV 0.223
States	not assessable 7
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 7 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 0 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0122
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers; homeology map: 0% of the genome in 5 homeolog blocks, 1 chromosome pairs, block-divergence CV 0.22

lpTriTurg1_B *Triticum turgidum, B subgenome*

Known: 2n = 4x = 28 (AABB); B run alone (known)

Assembly	HAP1 (7), HAP2 (7); 7 chromosome numbers, 2 copies each; 1890 sequences unplaced
Numbering fixes	none
Distances	same number 0.0008, different numbers 0.0696
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	not assessable 7
Fusions	none
Residual tetrasomy	0 of 2 tested, 0.0 Mb
Exchange candidates	none

Ploidy	high	2 haplotype copies of each of 7 chromosome numbers; no older duplicated sets detected. copies from the assembly (2 per number)
Auto- or allo-like	not assessable	Not assessable: two haplotypes and no duplicated sets to compare. Diploid-like as assembled. 0 whole-chromosome homeolog pair(s); no duplicated sets
TE markers	not assessable	Only the absolute marker fraction is available (two copies per number); it does not separate diploid, auto- and allopolyploid, so no reading is made from it. te_marker_fraction mean 0.0152
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. residual tetrasomy on 0/2 homeolog-paired chromosome numbers (no unrelated-chromosome control; shared repeats not excluded); homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

13.6 One usable haplotype

daEupConf1 *Euphrasia confusa*

Known: unspecified; origin: allo (user annotation (meta.tsv))

Assembly	HAP1 (22); 22 chromosome numbers, 1 copies each; 1842 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	11 (median 0.058, CV 0.0353)
Homeology map	24% of the genome in 40 blocks; 11 chromosome pairs; block CV 0.066
States	not assessable 22
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (22 chromosome numbers). The numbers do form duplicated sets (11 groups of mostly 2 chromosome numbers (genome partition z = 18.9)). 11 groups of mostly 2 chromosome numbers (genome partition z = 18.9)
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable from this assembly. pair-depth CV 0.04; homeology map: 24% of the genome in 40 homeolog blocks, 11 chromosome pairs, block-divergence CV 0.07

Caveats: one usable haplotype: copy-level readings are not assessable

ddAraThal4 *Arabidopsis thaliana*

Known: diploid 2n=10; 2n=10 (x=5); origin: auto (true diploid, not a polyploid) (Arabidopsis Genome Initiative 2000)

Assembly	HAP1 (5); 5 chromosome numbers, 1 copies each; 38 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	0
States	not assessable 5
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (5 chromosome numbers). no duplicated sets of chromosome numbers detected
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from.

Caveats: one usable haplotype: copy-level readings are not assessable

ddPopNigr1

Assembly	HAP1 (19); 19 chromosome numbers, 1 copies each; 969 sequences unplaced
Numbering fixes	none

Distances	–
Homeolog pairs	0
Homeology map	23% of the genome in 21 blocks; 8 chromosome pairs; block CV 0.031; two or more partners: chr03:chr15+chr17, chr04:chr02+chr05
States	not assessable 19
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (19 chromosome numbers). no duplicated sets of chromosome numbers detected
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable from this assembly. homeology map: 23% of the genome in 21 homeolog blocks, 8 chromosome pairs, block-divergence CV 0.03

Caveats: one usable haplotype: copy-level readings are not assessable

ddSalCine1

Assembly	HAP1 (18); 18 chromosome numbers, 1 copies each; 745 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	1 (median 0.129, CV –)
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	not assessable 18
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (18 chromosome numbers). no duplicated sets of chromosome numbers detected
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: one usable haplotype: copy-level readings are not assessable

drIngLaur1

Assembly	HAP1 (13); 13 chromosome numbers, 1 copies each; 6363 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	not assessable 13
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (13 chromosome numbers). no duplicated sets of chromosome numbers detected
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: one usable haplotype: copy-level readings are not assessable

drTriDubi3

Assembly	HAP1 (15); 15 chromosome numbers, 1 copies each; 934 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	6 (median 0.076, CV 0.0943)

Homeology map	35% of the genome in 30 blocks; 9 chromosome pairs; block CV 0.051; two or more partners: chr02:chr12+chr15, chr03:chr13+chr15, chr15:chr02+chr03
States	not assessable 15
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (15 chromosome numbers). The numbers do form duplicated sets (6 homeolog pairs covering 80% of numbers). 6 homeolog pairs covering 80% of numbers
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable from this assembly. pair-depth CV 0.09; homeology map: 35% of the genome in 30 homeolog blocks, 9 chromosome pairs, block-divergence CV 0.05

Caveats: one usable haplotype: copy-level readings are not assessable

drUrtDioi1

Assembly	HAP1 (13); 13 chromosome numbers, 1 copies each; 4058 sequences unplaced
Numbering fixes	none
Distances	–
Homeolog pairs	0
Homeology map	0% of the genome in 0 blocks; 0 chromosome pairs; block CV –
States	not assessable 13
Fusions	none

Ploidy	not assessable	Not assessable from this assembly: only one usable haplotype (13 chromosome numbers). no duplicated sets of chromosome numbers detected
Auto- or allo-like	not assessable	Not assessable from this assembly. fewer than three copies and nothing to pool
TE markers	not assessable	TE markers not run for this species.
Rediploidization	not assessable	Not assessable: no duplicated sets to read rediploidization from. homeology map: 0% of the genome in 0 homeolog blocks, 0 chromosome pairs

Caveats: one usable haplotype: copy-level readings are not assessable

14 The literature: where ploidyspec stands

This chapter reads ploidyspec against the 86 works in Appendix C (reviewed 12 October 2026). It answers four questions: which methods already address parts of our problem, and what each needs (14.1); where ploidyspec is new and where it is not (14.2); whether our conclusions hold up against the published record (14.3, 14.4); and what the literature changed in the tool and still leaves open (14.5, 14.6). Section 14.7 indexes every work by theme.

The short version

The literature supports our framing (a continuum with two axes, read per chromosome) and several of our results quantitatively: the snow carp fusion waves, the *Euphrasia* subgenome divergence, the wheat subgenome partition. Our genuinely new contribution is reading both axes, per chromosome and along chromosomes, from one individual's phased assembly with no annotation, reference, reads or population. Subgenome partitioning by repeats is **not** new; we converge with TE-signature phasing. Three of our readings needed correcting: salmonids were missing delayed (LORe) rediploidization, masu rested on our own scaffolding, and "allo-like" for rose was worded too strongly. *Ranunculus*'s "no structure" was a fault of ours: its two subgenomes and eight homeolog pairs were hidden by the size adjustment and by subgenome-specific repeats, and are now found.

14.1 The methods landscape

Approach	Needs	Reads	Limit for our question
Read k-mer spectra: GenomeScope 2.0, Smudgeplot (Ranallo-Benavidez et al., 2020); Tetmer (Becher et al., 2020); review (Jenike et al., 2025)	short reads	genome-wide ploidy, heterozygosity, subgenome divergence (Khalaf et al., 2024)	one number per genome; no chromosome or regional resolution; auto-like and young allo can look alike
Ks distributions and WGD dating (Dunn & Sethuraman, 2025; Qiao et al., 2019; Ren et al., 2018)	annotated genes; often transcriptomes	timing of duplications from duplicate-gene divergence	Ks accumulates only between non-recombining copies (Gaynor et al., 2026); the peak can reflect parental divergence, migration or coalescence (Dunn & Sethuraman, 2025)
Fractionation balance: P-index (Wang et al., 2019)	annotation, synteny, an outgroup genome	auto vs allo from balanced vs biased gene loss (threshold 0.3)	biased loss also arises after autopolyploid rediploidization (Parey et al., 2022; Xie et al., 2026)
Gene trees and ancestral karyotypes (Parey et al., 2022; Redmond et al., 2023; Robertson et al., 2017)	many annotated genomes of related species	when each duplicated region stopped recombining (AORe vs LORe); auto vs allo origin	needs a clade of genomes; not per individual
Duplicated BUSCOs, Hox clusters (Farhat et al., 2023; McHale et al., 2025)	annotation	old WGDs beyond k-mer reach	presence of a WGD only; nothing on inheritance
Subgenome phasing by repeats or k-mers: PolyCRACKER (Gordon et al., 2019), TE signatures (Session & Rokhsar, 2023), SubPhaser (Jia et al.,	an assembly of a known allopolyploid; homeolog groups assigned first	which chromosomes belong to which progenitor	assumes allopolyploidy; fails on autopolyploids by design (Jia et al., 2022); says nothing about rediploidization

Approach	Needs	Reads	Limit for our question
2022); benchmark (Zhang et al., 2023); review (Session, 2024)			
Differential divergence against relatives: M3 (Lv et al., 2025)	the polyploid plus related diploid genomes	subgenome assignment, fast	needs references from relatives
Inheritance from crosses, families, populations (Bicknell et al., 2023; Bourke et al., 2017; Dalle Palle et al., 2022; Delaval et al., 2024; Dufresne et al., 2014; Roux & Pannell, 2015)	mapping populations, pedigrees or population samples; markers	polysomic vs disomic inheritance, per chromosome or locus; origin by model choice	the direct answer, but expensive and rare for wild species
Resequencing, allele composition, meiotic cytology (Gaynor et al., 2026; Grandont et al., 2013; Xie et al., 2026)	a phased assembly plus population data or cytology	tetrasomic vs disomic regions, mechanisms	population data needed
ploidyspec	**one individual's haplotype-resolved, chromosome-scale assembly**	**copies, duplicated sets, per-chromosome and per-window lineage structure, fusions, residual tetrasomy, exchange, switches, LORe-like blocks, karyotype change**	**structure, not inheritance or origin; blind beyond 8–10% divergence; the assembly is part of the measurement**

Table 4: Methods that address parts of the auto/allo and rediploidization question.

14.2 Where ploidyspec is new, and where it is not

New, as far as the reviewed literature shows:

1. **Both axes from one individual's assembly.** Every other route to rediploidization (gene trees across a clade, resequenced populations, crosses) needs data beyond one genome. ploidyspec reads origin-like structure and rediploidization per chromosome and along chromosomes from the phased assembly alone, with no annotation, reference, outgroup or reads. That is what the stream of phased reference genomes from the Darwin Tree of Life and Earth BioGenome projects provides (Exposito-Alonso et al., 2020; Howe et al., 2026).
2. **Fusion-driven rediploidization recovered blind.** Xie et al. (2026) needed haplotype-resolved genomes, resequencing, meiotic cytology and gene trees to show that fusions trigger rediploidization and that divergence spreads from the fusion site. ploidyspec finds every published snow carp fusion, places each junction at 94–100% agreement, and orders the waves by junction gradient (oldest 2.7–4.1, middle 2.1–2.6, youngest 0.6–0.7), from the assemblies alone (Section 10.3).
3. **Regional readouts with artefact guards.** Residual tetrasomy, homoeologous exchange, lineage switches and LORe-like blocks are each read per window and per pair, with guards against the failure modes the literature documents: centromeric satellites (Chen et al., 2024; Su et al., 2019), shared repeats, swapped contigs from reference-guided scaffolding (Alonge et al., 2019), chromosome size, and numbering errors between haplotype files.
4. **A homeology map without genes.** Genome-wide segmental homeology from k-mer windows, which shows karyotype change since the duplication (chromosomes with blocks on two partners) and regions of different age.
5. **Robust to real assemblies.** Odd ploidy and a second chromosome set inside one haplotype file (AAB), renumbering cycles, and size artefacts are handled automatically; published methods mostly assume clean input.
6. **One scale across the tree of life.** Plants, fish, snails and a weevil on the same distances and the same rules.

Not new, and we should say so:

- Partitioning subgenomes by repeat history. Our durum wheat partition and the TE-marker split use the principle of TE-signature phasing (Gordon et al., 2019; Jia et al., 2022; Session & Rokhsar, 2023); they are a by-product, not a claim.
- Ploidy from k-mers (Becher et al., 2020; Ranallo-Benavidez et al., 2020); we read copies from the assembly and use read spectra only as a check.
- The concepts themselves: the continuum (Twyford et al., 2025), divergence as a record of isolation (Gaynor et al., 2026), LORe (Robertson et al., 2017), fusion-triggered rediploidization (Xie et al., 2026).

14.3 Our conclusions checked against the literature

Our conclusion	Literature	Verdict	Consequence
Snow carp fusions in three waves; divergence peaks at the junction	Xie et al. (2026): unbalanced fusions trigger rediploidization, spreading from fusion sites in three waves	supported	an independent replication from assemblies alone
`SchCurv1` chr17 splits only towards its end: an inversion in HAP3	Xie et al.: a centromeric inversion made chr17′s short arm disomic	supported	coordinates still to compare (Section 12.5)
`daEupConf1`: duplicated sets 0.057 apart (one haplotype)	Tetmer: tetraploid *Euphrasia* is allopolyploid, subgenome divergence 5% (Becher et al., 2020)	supported	a quantitative match between read- and assembly-based estimates
Durum wheat: exact A/B partition and all seven A–B pairs, blind to the names	subgenome-specific repeats and satellites (Gordon et al., 2019; Jia et al., 2022; Session & Rokhsar, 2023; Su et al., 2019)	supported	present the partition as convergent with TE-signature phasing; the pairs show only against the cross-subgenome background, because each subgenome's own repeats make its chromosomes closer to each other than to their homeologs
Glechoma residual block is a centromeric satellite	centromeric satellites turn over and homogenize between and within subgenomes (Chen et al., 2024; Su et al., 2019; Yan et al., 2025)	supported	keep the complexity guard
Allo-like ≠ allopolyploid; a rediploidized autopolyploid looks the same	Twyford et al. (2025), Lv et al. (2024), Doyle & Sherman-Broyles (2017), Parey et al. (2022)	supported	wording already says so
Divergence is read as isolation of copies	Dunn & Sethuraman (2025); Gaynor et al. (2026)	supported	–
Mixed per-chromosome states are informative	mixed and partial inheritance in rose, sturgeon, banana, *Brachiaria* (Bourke et al., 2017; Dalle Palle et al., 2022; Jeridi et al., 2012; Worthington et al., 2016); theory (Le Comber et al., 2010)	supported	present mixtures as expected, not anomalous
Gar and bowfin: no duplication since 2R	the outgroups to the teleost WGD (Parey et al., 2022)	supported	negative controls
Pilosella: residual tetrasomy on 4 pairs (79 Mb), aligned at allelic identity	*P. piloselloides* is autopolyploid with tetrasomic inheritance (Bicknell et al., 2023)	consistent	an auto-like reading with residual tetrasomy fits the genus
Weevil HAP1 chr09: A to 40 Mb, B after (a lineage switch)	frequent exchange between A and B in AAB bananas (Xie et al., 2024); gene conversion and sporadic recombination in a clonal amphitriploid fish (Wang et al., 2022); clonal polyploidy common in weevils (Stenberg et al., 2003)	plausible	if *Strophosoma* is clonal, mitotic recombination or gene conversion, not meiotic exchange; or a phasing switch (Hi-C needed)

Our conclusion	Literature	Verdict	Consequence
Salmonids: “no residual tetrasomy” read as fully rediploidized	25% of each salmonid genome rediploidized late (LORe), >90% similar, telomere-including arms (Lien et al., 2016; Robertson et al., 2017); deviant SNPs concentrate there (Dallaire et al., 2023)	incomplete	LORe regions are not near-allelic, so the residual test misses them by design: LORe-like block class added (Section 6.6)
Masu: residual tetrasomy on one pair (chr17/chr29)	reference-guided scaffolding can misplace contigs (Alonge et al., 2019); a phased Hi-C assembly exists (Wu et al., 2026)	assembly-dependent	re-run on Wu et al.‘s assembly (Section 9.4)
Rosa spinosissima: allo-like (high), 7 pairs	rose pairing is partially preferential, varies along arms and between genotypes (Bourke et al., 2017); segmental allopolyploidy (Mason & Wendel, 2020; Stebbins, 1947)	overstated	read “copies do not mix today on any of the 7 pairs”; segmental allopolyploidy is as likely as strict allopolyploidy
Cerastium alpinum: 18 large–small homeolog pairs	2n = 72 (Boşcaiu et al., 1999), octoploid throughout its range on x = 9 (Brysting, 2000); an allopolyploid can look diploid-like in k-mers (Nieuwenhuis et al., 2024)	partial	the pairs are one level of an 8x genome; the deeper level (sets of four) is beyond k-mer reach
Ranunculus repens: two subgenomes of 8 and eight homeolog pairs (first reported as no structure)	diploid relatives have x = 8 (Karbstein et al., 2025), so 2n = 32 is 4x; subgenome-specific repeats are the TE-phasing signal (Session & Rokhsar, 2023)	supported (after a fix)	the earlier “no structure” was our size adjustment removing a larger and a smaller subgenome; read as allo-like with asymmetric subgenomes (A 1.23× B)
Eleocharis: auto-like origin, residual tetrasomy on 10 of 19 numbers	holocentric chromosomes: frequent fission and fusion, restricted chiasmata, inverted meiosis (Bureš et al., 2013; Hipp et al., 2009; Mandrioli & Manicardi, 2020)	caution	chromosome numbers need not track ancestral units; treat arm- and number-based readings as provisional
Hesperis: AAAB-like on 6 numbers	Brassicaceae chromosome numbers change by descending dysploidy after WGD; number is no proxy for history (Mandáková et al., 2016)	consistent	do not infer x from the count
Residual tetrasomy ≥ 90% terminal is low confidence	real delayed regions include telomeres (Lien et al., 2016); exchanges are subtelomeric (Zhang et al., 2020)	partly contradicted	keep the flag (repeats are also terminal), but terminal signal is expected biology, not presumed artefact

Table 5: Each conclusion of the panel that the literature bears on.

14.4 The panel in the light of the literature

Species	What the literature says	What ploidyspec reads
`SchCurv1`, `SchYoun1` snow carps	single ancestral autotetraploidy; rediploidization started by fusions, incomplete at speciation (Xie et al., 2026; Xie et al., 2026)	4 × 25; mixed states; every fusion and the wave order; chr17 inversion
`fSalAlp3`, `fCorLav1`, `OncMaso1` salmonids	Ss4R autotetraploidy 80–95 Mya; LORe over 25% (Lien et al., 2016; Robertson et al., 2017); charr and whitefish deviant SNPs in recently rediploidized regions (Dallaire et al., 2023); phased masu (Wu et al., 2026)	map reaches 16–30% of the genome; karyotype reshuffled; LORe-like blocks: whitefish 243 Mb (40% of the block span, five pairs), charr 8.5 Mb (one pair, at chromosome ends)
`fLepOcu1`, `fAmiCal2` gar, bowfin	non-duplicated outgroups of the teleost WGD (Parey et al., 2022)	one pair each, 0% in blocks: not assessable, as expected

Species	What the literature says	What ploidyspec reads
`lpTriTurg1` durum wheat	AABB; subgenome-specific centromeric satellites (Su et al., 2019); TE-based phasing separates wheat subgenomes (Gordon et al., 2019; Jia et al., 2022); HE in nascent AADD (Zhang et al., 2020)	partition exactly A vs B (z = 179) and all seven A–B pairs; none in the window map
`icStrMela3` weevil	AAB precedent with frequent exchange (Xie et al., 2024); clonal polyploids keep gene conversion (Wang et al., 2022); weevil clonality and polyploidy (Stenberg et al., 2003)	3 × 10, AAB-like (high); chr09 lineage switch
`daEupConf1` *Euphrasia*	tetraploid *Euphrasia* allopolyploid, 5% subgenome divergence (Becher et al., 2020)	one usable haplotype; 11 groups of 2 numbers, 0.057 apart
`daLatClan1`, `daLatSqua1` *Lathraea*	parasitic Orobanchaceae: genome inflation and gene loss (Lyko & Wicke, 2021); parasitic dodders are allopolyploid with homoeologous recombination (Segawa et al., 2025)	allo-like with rearranged karyotypes; *L. clandestina*'s near-allelic block is repeats; one short terminal LORe-like pair (chr05/chr07, 5 Mb), possibly subtelomeric repeats
`daPilAura1` *Pilosella*	tetrasomic autopolyploidy in *P. piloselloides*; apomixis loci (Bicknell et al., 2023)	residual tetrasomy on 4 pairs; one copy apart on most numbers; LORe-like blocks on two pairs (115 Mb)
`dcCerAlpi1` *Cerastium*	2n = 72, octoploid on x = 9 (Boşcaiu et al., 1999; Brysting, 2000)	18 large–small pairs (x ≈ 18): one level of the duplication
`dmRanRepe1` *Ranunculus*	*Ranunculus*: polyploidy, apomixis and reticulation; diploid x = 8, 2.69 Gb (Karbstein et al., 2025)	2 × 16: two subgenomes of 8 (A 1.23× B) and eight homeolog pairs (Section 14.3)
`drRosSpin1` *Rosa*	partial preferential pairing in tetraploid rose (Bourke et al., 2017)	7 pairs, synchronous (CV 0.03), no residual tetrasomy
`lpElePalu1` *Eleocharis*	holocentric Cyperaceae; fission and fusion dominate number change (Bureš et al., 2013; Hipp et al., 2009)	residual tetrasomy on 10 of 19 numbers, 310 Mb; 13 near-allelic map blocks, no LORe-like ones
`ddHesMatr1` *Hesperis*	Brassicaceae dysploidy and pseudo-ploidy (Mandáková et al., 2016)	4 × 6, AAAB-like; one lineage switch
`xgCepNemo3` and the land snails	stylommatophoran WGD shown by duplicated BUSCOs and Hox clusters (McHale et al., 2025); WGD in neogastropods (Farhat et al., 2023)	little or no k-mer-detectable homeology: the duplication is beyond k-mer reach

14.5 What the literature changed in ploidyspec

Change	Why
LORe-like block class in the homeology map, with a near-allelic lower bound and a two-way rule (Section 6.6)	salmonid delayed rediploidization (Lien et al., 2016; Robertson et al., 2017) was visible in the map but unreported
Masu re-run on the published phased assembly	(Wu et al., 2026); our reference-scaffolded HAP2 produced artefacts (Section 9.4)
Wording: positions on two axes, "copies do not mix today", never a species label	(Doyle & Sherman-Broyles, 2017; Gaynor et al., 2026; Lv et al., 2024; Twyford et al., 2025)
Limits: holocentric genomes, partial inheritance, repeat-based signals (Section 12.3)	(Bourke et al., 2017; Hipp et al., 2009; Session & Rokhsar, 2023)
Terminal residual signal flagged, not presumed artefact	(Lien et al., 2016; Zhang et al., 2020)
Reading of "no pairs" in a known polyploid (Section 12.2)	(Gaynor et al., 2026)

Not yet in the code: a holocentric flag in the summary, the segmental alternative in the "allo-like" sentence, and a confidence rule for terminal residual tetrasomy that accounts for real terminal biology.

14.6 Open questions the literature raises

- **Validate the LORe-like class.** Map charr and whitefish LORe-like blocks onto Atlantic salmon's seven delayed arm pairs (Lien et al., 2016) through synteny.
- **Whitefish versus charr.** Whitefish shows far more LORe-like sequence than charr; deviant SNPs in both (Dallaire et al., 2023) could test whether this is biology or assembly.
- **Masu on a clean assembly.** Does residual tetrasomy on chr17/chr29 survive on (Wu et al., 2026)?
- **Weevil reproduction.** Whether *Strophosoma melanogrammum* is clonal decides whether its chr09 switch can be meiotic exchange (Stenberg et al., 2003; Wang et al., 2022).
- **Rose ploidy and pairing.** Our read spectrum says tetraploid; a 2026 genome report says diploid; pairing in tetraploid rose is genotype-dependent (Bourke et al., 2017).
- **The deeper Cerastium level** (8x on x = 9) and a primary count for this *Ranunculus repens* individual (Brysting, 2000; Karbstein et al., 2025).
- **Head-to-head comparisons.** SubPhaser and M3 on wheat and Glechoma (Jia et al., 2022; Lv et al., 2025); GenomeScope, Smudgeplot and Tetmer on the panel's reads (Becher et al., 2020; Ranallo-Benavidez et al., 2020); Ks against our distances on the snow carps (Dunn & Sethuraman, 2025; Gaynor et al., 2026).
- **Beyond k-mer reach.** Combine with duplicated-BUSCO or synteny methods for the snails and the deepest salmonid homeology (Farhat et al., 2023; McHale et al., 2025).

14.7 The literature by theme

Theme	Works
Definitions and the continuum	(Doyle & Sherman-Broyles, 2017; Leal et al., 2024; Lv et al., 2024; Soltis et al., 2010; Stebbins, 1947; Thomas, 2026; Twyford et al., 2025)
WGD prevalence and consequences	(Crombez et al., 2025; Li et al., 2021; Peer et al., 2009; Qiao et al., 2019; Ren et al., 2018; Yu et al., 2023)
Meiosis and inheritance	(Bicknell et al., 2023; Bourke et al., 2017; Dalle Palle et al., 2022; Delaval et al., 2024; Dufresne et al., 2014; Grandont et al., 2013; Jeridi et al., 2012; Le Comber et al., 2010; Roux & Pannell, 2015; Soares et al., 2021; Worthington et al., 2016)
Rediploidization and LORe	(Dallaire et al., 2023; Feng et al., 2024; Lien et al., 2016; Mandáková et al., 2016; Parey et al., 2022; Redmond et al., 2023; Robertson et al., 2017; Xie et al., 2026; Xie et al., 2026)
Homoeologous exchange and segmental allopolyploidy	(Burns et al., 2021; Burns et al., 2024; Deb et al., 2023; Deng et al., 2026; Gaeta & Chris Pires, 2010; Hurgobin et al., 2018; Leal-Bertioli et al., 2018; Mason & Wendel, 2020; Mu et al., 2023; Wöhner et al., 2023; Zhang et al., 2020)
Repeats, TEs and centromeres	(Chen et al., 2024; Gordon et al., 2019; Session & Rokhsar, 2023; Shi et al., 2026; Su et al., 2019; Yan et al., 2025)
Subgenome phasing	(Jia et al., 2022; Lv et al., 2025; Robbins et al., 2026; Session, 2024; Song et al., 2024; Zhang et al., 2023)
Divergence and Ks	(Dunn & Sethuraman, 2025; Gaynor et al., 2026; Wang et al., 2019)
k-mer methods	(Becher et al., 2020; Jenike et al., 2025; Khalaf et al., 2024; Ondov et al., 2016; Ranallo-Benavidez et al., 2020)
Assembly of polyploids and standards	(Alonge et al., 2019; Exposito-Alonso et al., 2020; Howe et al., 2026; Kyriakidou et al., 2018; Sun et al., 2022; Yang et al., 2017; Zhao & Shi, 2025)
Phased polyploid genomes	(Nieuwenhuis et al., 2024; Segawa et al., 2025; Wang et al., 2022; Wu et al., 2026; Xie et al., 2024)
Holocentric chromosomes	(Bureš et al., 2013; Hipp et al., 2009; Mandrioli & Manicardi, 2020)
Taxa in the panel	(Boşcaiu et al., 1999; Brysting, 2000; Farhat et al., 2023; Karbstein et al., 2025; Lyko & Wicke, 2021; McHale et al., 2025; Stenberg et al., 2003)

15 Appendix A: parameters

Parameter (flag)	Default	Why this value
`--k`	15,23	largest k above the chance floor is used; k = 23 on every panel pair
`--min-len`	1 Mb	chromosome-scale sequences only
rename ratio / swap	3× / 2×	real mismatches were 4.6–96×; `ddHesMatrl`'s swap 3.1× and 2.3×
extra-set ratio, min units	1.3×, 3	weevil B set 1.5–2.4×; a fusion leftover is one unit
`--fdr-alpha`, `--min-effect`	0.05, 2%	a 1.25% difference reached z = −4 on simulated diploids
`--window`, `--window-k`	250 kb, 15	allelic resolution along chromosomes
map k, z, min block	23, 3, 6 windows + null	chance hits negligible at k = 23
map pair minimum	2 Mb reciprocal	one short block is not a pair
`--dist-split`, `--te-split`, `--window-split`	1.25, 2.0, 1.25	tuned on snow carps
`--min-segment-bp`	2 Mb	shortest segment or block
`--residual-ratio`	3	near-allelic = within 3× the allelic level
`--residual-homeolog-frac`	0.25	masu residual windows 0.1× homeolog; false ones 0.7–0.9×
`--residual-min-run`	8 windows	near-homozygous genomes had nulls of 3
allelic cap	1.5× genome median	a large exchange hid itself at 5×
exchange ratio	2×	homeolog clearly closer than own homolog
fusion: long ratio, z, sibling excess	1.4×, 10, 2.5×	SchCurv1 fusion 3.4×, homeology 0.9×
partition z for pooling	10	daInuConz1 10.6
subgenome partition z, smaller set, permutations	10, 25%, 200	subgenome species 109–179; no other species above 5 (*Malva* 2.6)
summary: residual pairs, duplication pairs, map fraction	2, 3, 30%	one pair is not an origin; gar and bowfin pass one pair
LORe-like band, min blocks	0.25–0.6× median block distance, 6	below 0.25× is the residual test's (near-allelic); whitefish LORe-like blocks 0.011–0.026 against a median of 0.045
LORe-like reading	≥ 1 two-way pair, ≥ 4 Mb	one short low block is noise

16 Appendix B: glossary

Allelic distance distance between copies of the same chromosome number.
Block a run of windows whose closest other chromosome is consistently the same.
Containment fraction of one sequence's k-mers present in another.
Disomic / tetrasomic pairing with a fixed partner / at random among four.
Extra set a second chromosome set held in one haplotype file under other numbers.
Homeolog a chromosome derived from the same ancestral chromosome through polyploidy.
Homeologous exchange a stretch of one homeolog replaced by the other's sequence.
LORe lineage-specific ohnologue resolution; regions that rediploidized late, after speciation (salmonids).
Mixed inheritance some chromosomes or loci segregate disomically, others polysomically, in one species.
Pooling reading two homeologous numbers' copies together.
Reciprocal block a block whose partner has a block back on it.
Residual tetrasomy regions of an otherwise rediploidized genome where homeologs still behave as one pool.
Rediploidization the return of a polyploid to disomic behaviour, region by region.
Segmental allopolyploid a polyploid whose subgenomes are only partly diverged, so pairing is partly preferential (Stebbins 1947).

17 Appendix C: references

All works cited in this guide and the further literature reviewed for Chapter 14 (86 works).

Alonge, M., Soyk, S., Ramakrishnan, S., Wang, X., Goodwin, S., Sedlazeck, F. J., Lippman, Z. B., & Schatz, M. C. (2019). RaGOO: fast and accurate reference-guided scaffolding of draft genomes. *Genome Biol.*, *20*(1), 224.

Becher, H., Brown, M. R., Powell, G., Metherell, C., Riddiford, N. J., & Twyford, A. D. (2020). Maintenance of species differences in closely related tetraploid parasitic Euphrasia (Orobanchaceae) on an isolated island. *Plant Commun.*, *1*(6), 100105.

Bicknell, R., Gaillard, M., Catanach, A., McGee, R., Erasmuson, S., Fulton, B., & Winefield, C. (2023). Genetic mapping of the LOSS OF PARTHENOGENESIS locus in Pilosella piloselloides and the evolution of apomixis in the Lactuceae. *Front. Plant Sci.*, *14*, 1239191.

Boşcaiu, M., Vicente, O., & Ehrendorfer, F. (1999). Chromosome numbers, karyotypes and nuclear DNA contents from perennial polyploid groups of Cerastium (Caryophyllaceae). *Plant Syst. Evol.*, *218*(1–2), 13–21.

Bourke, P. M., Arens, P., Voorrips, R. E., Esselink, G. D., Koning-Boucoiran, C. F. S., Van't Westende, W. P. C., Santos Leonardo, T., Wissink, P., Zheng, C., Geest, G. van, Visser, R. G. F., Krens, F. A., Smulders, M. J. M., & Maliepaard, C. (2017). Partial preferential chromosome pairing is genotype dependent in tetraploid rose. *Plant J.*, *90*(2), 330–343.

Brysting, A. K. (2000). Chromosome number variation in the polyploid Cerastium alpinum-C. arcticum complex (Caryophyllaceae). *Nord. J. Bot.*, *20*(2), 149–156.

Bureš, P., Zedek, F., & Marková, M. (2013). Holocentric Chromosomes. In *Plant Genome Diversity Volume 2: Plant Genome Diversity Volume 2* (pp. 187–208). Springer Vienna.

Burns, R., Glushkevich, A., Kulkarni, A., Kolesnikova, U., Kolář, F., Scott, A. D., & Novikova, P. Y. (2024). Diploid origins and early genome stabilization in the allotetraploid Arabidopsis suecica. *Biorxiv*.

Burns, R., Mandáková, T., Gunis, J., Soto-Jiménez, L. M., Liu, C., Lysak, M. A., Novikova, P. Y., & Nordborg, M. (2021). Gradual evolution of allopolyploidy in Arabidopsis suecica. *Nat. Ecol. Evol.*, *5*(10), 1367–1381.

Chen, C., Wu, S., Sun, Y., Zhou, J., Chen, Y., Zhang, J., Birchler, J. A., Han, F., Yang, N., & Su, H. (2024). Three near-complete genome assemblies reveal substantial centromere dynamics from diploid to tetraploid in Brachypodium genus. *Genome Biol.*, *25*(1), 63.

Crombez, E., Peer, Y. Van de, & Li, Z. (2025). The subordinate role of pseudogenization to recombinative deletion following polyploidization in angiosperms. *Nat. Commun.*, *16*(1), 6335.

Dallaire, X., Bouchard, R., Hénault, P., Ulmo-Diaz, G., Normandeau, E., Mérot, C., Bernatchez, L., & Moore, J.-S. (2023). Widespread deviant patterns of heterozygosity in whole-genome sequencing due to autopolyploidy, repeated elements, and duplication. *Genome Biol. Evol.*, *15*(12), evad229.

Dalle Palle, S., Boscari, E., Bordignon, S. G., Muñoz-Mora, V. H., Bertorelle, G., & Congiu, L. (2022). Different chromosome segregation patterns coexist in the tetraploid Adriatic sturgeon Acipenser naccarii. *Diversity (Basel)*, *14*(9), 745.

Deb, S. K., Edger, P. P., Pires, J. C., & McKain, M. R. (2023). Patterns, mechanisms, and consequences of homoeologous exchange in allopolyploid angiosperms: a genomic and epigenomic perspective. *New Phytol.*, *238*(6), 2284–2304.

Delaval, A., Glover, K. A., Solberg, M. F., Taggart, J. B., Besnier, F., Sørvik, A. G. E., Øyro, J., Garnes-Gutvik, S. N., Fjelldal, P. G., Hansen, T., & Harvey, A. (2024). A genetic method to infer ploidy and aberrant inheritance in triploid organisms. *Mol. Ecol. Resour.*, *24*(7), e14004.

Deng, P., Xie, J., Li, X., Chen, X., Dai, H., Xia, Z., Fan, J., & others. (2026). The homoeolog-resolved assembly of the 40-Gb decaploid Thinopyrum ponticum genome reveals extensive homoeologous exchange and recombination, segmental …. *Mol. Plant*.

Doyle, J. J., & Sherman-Broyles, S. (2017). Double trouble: taxonomy and definitions of polyploidy. *New Phytol.*, *213*(2), 487–493.

Dufresne, F., Stift, M., Vergilino, R., & Mable, B. K. (2014). Recent progress and challenges in population genetics of polyploid organisms: an overview of current state-of-the-art molecular and statistical tools. *Mol. Ecol.*, *23*(1), 40–69.

Dunn, T., & Sethuraman, A. (2025). Hidden in Plain Sight. How Ks histogram dynamics can reveal and obscure ancient whole genome duplications. *Biorxiv*.

Exposito-Alonso, M., Drost, H.-G., Burbano, H. A., & Weigel, D. (2020). The Earth BioGenome project: opportunities and challenges for plant genomics and conservation. *Plant J.*, *102*(2), 222–229.

Farhat, S., Modica, M. V., & Puillandre, N. (2023). Whole genome duplication and gene evolution in the hyperdiverse venomous gastropods. *Mol. Biol. Evol.*, *40*(8), msad171.

Feng, X., Chen, Q., Wu, W., Wang, J., Li, G., Xu, S., Shao, S., Liu, M., Zhong, C., Wu, C.-I., Shi, S., & He, Z. (2024). Genomic evidence for rediploidization and adaptive evolution following the whole-genome triplication. *Nat. Commun.*, *15*(1), 1635.

Gaeta, R. T., & Chris Pires, J. (2010). Homoeologous recombination in allopolyploids: the polyploid ratchet: Research review. *New Phytol.*, *186*(1), 18–28.

Gaynor, M. L., Feng, K., Soltis, D. E., Soltis, P. S., & Smith, S. A. (2026). All detectable ancient whole-genome duplications involve hybridization. *Biorxiv*, 2026.08.25.747000.

Gordon, S. P., Levy, J. J., & Vogel, J. P. (2019). PolyCRACKER, a robust method for the unsupervised partitioning of polyploid subgenomes by signatures of repetitive DNA evolution. *BMC Genomics*, *20*(1), 580.

Grandont, L., Jenczewski, E., & Lloyd, A. (2013). Meiosis and its deviations in polyploid plants. *Cytogenet. Genome Res.*, *140*(2–4), 171–184.

Hipp, A. L., Rothrock, P. E., & Roalson, E. H. (2009). The evolution of chromosome arrangements in Carex (Cyperaceae). *Bot. Rev.*, *75*(1), 96–109.

Howe, K., Agius, D. R., Alioto, T., Aury, J.-M., Biello, R., Blaxter, M., Brown, M. W., Brown, T., Burriel-Carranza, B., Cheng, H., Copetti, D., Correard, S., Cruz, F., Davis, P., Denisova, A., Formenti, G., Galante, P. A. F., Howard, C., Jaron, K. S., … Durbin, R. (2026). Generating eukaryotic reference genome assemblies: Earth BioGenome Project quality standards and recommendations. *Arxiv [Q-Bio.gn]*.

Hurgobin, B., Golicz, A. A., Bayer, P. E., Chan, C.-K. K., Tirnaz, S., Dolatabadian, A., Schiessl, S. V., Samans, B., Montenegro, J. D., Parkin, I. A. P., Pires, J. C., Chalhoub, B., King, G. J., Snowdon, R., Batley, J., & Edwards, D. (2018). Homoeologous exchange is a major cause of gene presence/absence variation in the amphidiploid Brassica napus. *Plant Biotechnol. J.*, *16*(7), 1265–1274.

Jenike, K. M., Campos-Domínguez, L., Boddé, M., Cerca, J., Hodson, C. N., Schatz, M. C., & Jaron, K. S. (2025). K-mer approaches for biodiversity genomics. *Genome Research*, *35*(2), 219–230.

Jeridi, M., Perrier, X., Rodier-Goud, M., Ferchichi, A., D#39;Hont, A., & Bakry, F. (2012). Cytogenetic evidence of mixed disomic and polysomic inheritance in an allotetraploid (AABB) Musa genotype. *Annals of Botany*, *110*(8), 1593–1606.

Jia, K.-H., Wang, Z.-X., Wang, L., Li, G.-Y., Zhang, W., Wang, X.-L., Xu, F.-J., Jiao, S.-Q., Zhou, S.-S., Liu, H., Ma, Y., Bi, G., Zhao, W., El-Kassaby, Y. A., Porth, I., Li, G., Zhang, R.-G., & Mao, J.-F. (2022). SubPhaser: a robust allopolyploid subgenome phasing method based on subgenome-specific k-mers. *New Phytol.*, *235*(2), 801–809.

Karbstein, K., Choudhary, N., Xie, T., Tomasello, S., Wagner, N. D., Barke, B. H., Paetzold, C., Bradican, J. P., Preick, M., Himmelbach, A., Stein, N., Papantonis, A., Irisarri, I., Vries, J. de, Pucker, B., & Hörandl, E. (2025). Assembling genomes of non-model plants: A case study with evolutionary insights from Ranunculus (Ranunculaceae). *Plant J.*, *123*(6), e70390.

Khalaf, A., Lawniczak, M. K. N., Blaxter, M. L., & Jaron, K. S. (2024). Polyploidy is widespread in Microsporidia. *Microbiol. Spectr.*, *12*(2), e0366923.

Kyriakidou, M., Tai, H. H., Anglin, N. L., Ellis, D., & Strömvik, M. V. (2018). Current strategies of polyploid plant genome sequence assembly. *Front. Plant Sci.*, *9*, 1660.

Le Comber, S. C., Ainouche, M. L., Kovarik, A., & Leitch, A. R. (2010). Making a functional diploid: from polysomic to disomic inheritance. *New Phytol.*, *186*(1), 113–122.

Leal-Bertioli, S. C. M., Godoy, I. J., Santos, J. F., Doyle, J. J., Guimarães, P. M., Abernathy, B. L., Jackson, S. A., Moretzsohn, M. C., & Bertioli, D. J. (2018). Segmental allopolyploidy in action: Increasing diversity through polyploid hybridization and homoeologous recombination. *Am. J. Bot.*, *105*(6), 1053–1066.

Leal, J. L., Milesi, P., Hodková, E., Zhou, Q., James, J., Eklund, D. M., & Lascoux. (2024). Complex polyploids: origins, genomic composition, and role of introgressed alleles. *Systematic Biology*, *73*(2), 392–418.

Li, Z., McKibben, M. T. W., Finch, G. S., Blischak, P. D., Sutherland, B. L., & Barker, M. S. (2021). Patterns and processes of diploidization in land plants. *Annu. Rev. Plant Biol.*, *72*, 387–410.

Lien, S., Koop, B. F., Sandve, S. R., Miller, J. R., Kent, M. P., Nome, T., Hvidsten, T. R., Leong, J. S., Minkley, D. R., Zimin, A., Grammes, F., Grove, H., Gjuvsland, A., Walenz, B., Hermansen, R. A., Schalburg, K. von, Rondeau, E. B., Di Genova, A., Samy, J. K. A., … Davidson, W. S. (2016). The Atlantic salmon genome provides insights into rediploidization. *Nature*, *533*(7602), 200–205.

Lv, Y., Li, J., Li, Y., Huang, Y., Lai, Q., Wen, Z., Wang, J., He, Y., Shi, J., Huang, Z., Jiang, Y., Peer, Y. Van de, Shi, Q., Xie, B., & Wang, Y. (2025). Subgenome partitioning and polyploid genome evolution in the loach family botiidae (order Cypriniformes). *Adv. Sci. (Weinh.)*, *12*(36), e05411.

Lv, Z., Addo Nyarko, C., Ramtekey, V., Behn, H., & Mason, A. S. (2024). Defining autopolyploidy: Cytology, genetics, and taxonomy. *Am. J. Bot.*, *111*(8), e16292.

Lyko, P., & Wicke, S. (2021). Genomic reconfiguration in parasitic plants involves considerable gene losses alongside global genome size inflation and gene births. *Plant Physiol.*, *186*(3), 1412–1423.

Mandáková, T., Gloss, A. D., Whiteman, N. K., & Lysak, M. A. (2016). How diploidization turned a tetraploid into a pseudotriploid. *Am. J. Bot.*, *103*(7), 1187–1196.

Mandrioli, M., & Manicardi, G. C. (2020). Holocentric chromosomes. *Plos Genet.*, *16*(7), e1008918. https://doi.org/10.1371/journal.pgen.1008918

Mason, A. S., & Wendel, J. F. (2020). Homoeologous exchanges, segmental allopolyploidy, and polyploid genome evolution. *Front. Genet.*, *11*, 1014.

McHale, F., Mulhair, P. O., & Holland, P. W. H. (2025). Evolution of duplicated Hox gene clusters in land snails and slugs. *J. Exp. Zool. B Mol. Dev. Evol.*, *344*(6), 363–368.

Mu, W., Li, K., Yang, Y., Breiman, A., Yang, J., Wu, Y., Zhu, M., Wang, S., Catalan, P., Nevo, E., & Liu, J. (2023). Subgenomic stability of progenitor genomes during repeated allotetraploid origins of the same grass Brachypodium hybridum. *Mol. Biol. Evol.*, *40*(12), msad259.

Nieuwenhuis, R., Hesselink, T., Broeck, H. C. van den, Cordewener, J., Schijlen, E., Bakker, L., Diaz Trivino, S., Struss, D., Hoop, S.-J. de, Jong, H. de, & Peters, S. A. (2024). Genome architecture and genetic diversity of allopolyploid okra (Abelmoschus esculentus). *Plant J.*, *118*(1), 225–241.

Ondov, B. D., Treangen, T. J., Melsted, P., Mallonee, A. B., Bergman, N. H., Koren, S., & Phillippy, A. M. (2016). Mash: fast genome and metagenome distance estimation using MinHash. *Genome Biol.*, *17*(1), 132.

Parey, E., Louis, A., Montfort, J., Guiguen, Y., Crollius, H. R., & Berthelot, C. (2022). An atlas of fish genome evolution reveals delayed rediploidization following the teleost whole-genome duplication. *Genome Res.*, *32*(9), 1685–1697.

Peer, Y. Van de, Maere, S., & Meyer, A. (2009). The evolutionary significance of ancient genome duplications. *Nat. Rev. Genet.*, *10*(10), 725–732.

Qiao, X., Li, Q., Yin, H., Qi, K., Li, L., Wang, R., Zhang, S., & Paterson, A. H. (2019). Gene duplication and evolution in recurring polyploidization-diploidization cycles in plants. *Genome Biol.*, *20*(1), 38.

Ranallo-Benavidez, T. R., Jaron, K. S., & Schatz, M. C. (2020). GenomeScope 2.0 and Smudgeplot for reference-free profiling of polyploid genomes. *Nat. Commun.*, *11*(1), 1432.

Redmond, A. K., Casey, D., Gundappa, M. K., Macqueen, D. J., & McLysaght, A. (2023). Independent rediploidization masks shared whole genome duplication in the sturgeon-paddlefish ancestor. *Nat. Commun.*, *14*(1), 2879.

Ren, R., Wang, H., Guo, C., Zhang, N., Zeng, L., Chen, Y., Ma, H., & Qi, J. (2018). Widespread whole genome duplications contribute to genome complexity and species diversity in angiosperms. *Mol. Plant*, *11*(3), 414–428.

Robbins, M. D., Park, S., Bushman, B. S., Warnke, S. E., & Barnaby, J. Y. (2026). Haplotype-resolved chromosome-level genome assembly of creeping bentgrass, Agrostis stolonifera. *Sci. Data*, *13*(1), 241.

Robertson, F. M., Gundappa, M. K., Grammes, F., Hvidsten, T. R., Redmond, A. K., Lien, S., Martin, S. A. M., Holland, P. W. H., Sandve, S. R., & Macqueen, D. J. (2017). Lineage-specific rediploidization is a mechanism to explain time-lags between genome duplication and evolutionary diversification. *Genome Biol.*, *18*(1), 111.

Roux, C., & Pannell, J. R. (2015). Inferring the mode of origin of polyploid species from next-generation sequence data. *Mol. Ecol.*, *24*(5), 1047–1059.

Segawa, T., Yoshizumi, S., Toyonaga, H., Shiraishi, A., Sato, K., Yamabe, T., Takagi, M., Takagawa, M., Yokoyama, R., Itoh, T., & Ono, E. (2025). Chromosome-scale genome assemblies of two allopolyploid Cuscuta species uncover genomic signatures of parasitic lifestyle and polyploid evolution. *Biorxiv*.

Session, A. M. (2024). Allopolyploid subgenome identification and implications for evolutionary analysis. *Trends Genet.*, *40*(7), 621–631.

Session, A. M., & Rokhsar, D. S. (2023). Transposon signatures of allopolyploid genome evolution. *Nat. Commun.*, *14*(1), 3180.

Shi, G., Zhu, Y., Li, Y., Zheng, Q., Wang, W., Wu, X., Feng, X., Luo, W., Wang, F., Xie, Q., Tang, S., Liu, H., Li, H., & Meng, Z. (2026). Transposable element landscapes provide insights into TE's evolutionary convergence driven by asymmetric subgenome dynamics following allopolyploidization in cotton. *Ind. Crops Prod.*, *246*(123434), 123434.

Soares, N. R., Mollinari, M., Oliveira, G. K., Pereira, G. S., & Vieira, M. L. C. (2021). Meiosis in polyploids and implications for genetic mapping: A review. *Genes (Basel)*, *12*(10), 1517.

Soltis, D. E., Buggs, R. J. A., Doyle, J. J., & Soltis, P. S. (2010). What we still don't know about polyploidy. *Taxon*, *59*(5), 1387–1403.

Song, Y., Peng, Y., Liu, L., Li, G., Zhao, X., Wang, X., Cao, S., Muyle, A., Zhou, Y., & Zhou, H. (2024). Phased gap-free genome assembly of octoploid cultivated strawberry illustrates the genetic and epigenetic divergence among subgenomes. *Hortic. Res.*, *11*(1), uhad252.

Stebbins, G. L., Jr. (1947). Types of polyploids; their classification and significance. *Adv. Genet., Advances in Genetics*, *1*, 403–429.

Stenberg, P., Lundmark, M., Knutelski, S., & Saura, A. (2003). Evolution of clonality and polyploidy in a weevil system. *Molecular Biology and Evolution*, *20*(10), 1626–1632.

Su, H., Liu, Y., Liu, C., Shi, Q., Huang, Y., & Han, F. (2019). Centromere satellite repeats have undergone rapid changes in polyploid wheat subgenomes. *Plant Cell*, *31*(9), 2035–2051.

Sun, H., Jiao, W.-B., Krause, K., Campoy, J. A., Goel, M., Folz-Donahue, K., Kukat, C., Huettel, B., & Schneeberger, K. (2022). Chromosome-scale and haplotype-resolved genome assembly of a tetraploid potato cultivar. *Nat. Genet.*, *54*(3), 342–348.

Thomas, W. J. W. (2026). Heard it through the grapevine: a phased reference genome for the polyploid Kyoho grape. *Plant Physiol.*, *201*(4), kiag378.

Twyford, A. D., Conover, J. L., Doyle, J. J., Mason, A. S., Soltis, D. E., Soltis, P. S., & Wendel, J. F. (2025). The polyploid continuum and the landscape of polyploid genomic variation. *Am. J. Bot.*, *112*(11), e70121.

Wang, J., Qin, J., Sun, P., Ma, X., Yu, J., Li, Y., Sun, S., Lei, T., Meng, F., Wei, C., Li, X., Guo, H., Liu, X., Xia, R., Wang, L., Ge, W., Song, X., Zhang, L., Guo, D., … Wang, X. (2019). Polyploidy index and its implications for the evolution of polyploids. *Front. Genet.*, *10*, 807.

Wang, Y., Li, X.-Y., Xu, W.-J., Wang, K., Wu, B., Xu, M., Chen, Y., Miao, L.-J., Wang, Z.-W., Li, Z., Zhang, X.-J., Yin, Z., Zhou, B.-T., Yang, Y.-L., Zhu, C.-L., Hu, M.-L., Zheng, J.-M., Feng, C.-G., Qiu, Q., … Gui, J.-F. (2022). Comparative genome anatomy reveals evolutionary insights into a unique amphitriploid fish. *Nat. Ecol. Evol.*, *6*(9), 1354–1366.

Wöhner, T. W., Emeriewen, O. F., Wittenberg, A. H. J., Nijbroek, K., Wang, R. P., Blom, E.-J., Schneiders, H., Keilwagen, J., Berner, T., Hoff, K. J., Gabriel, L., Thierfeldt, H., Almolla, O., Barchi, L., Schuster, M., Lempe, J., Peil, A., & Flachowsky, H. (2023). The structure of the tetraploid sour cherry 'Schattenmorelle' (Prunus cerasus L.) genome reveals insights into its segmental allopolyploid nature. *Front. Plant Sci.*, *14*, 1284478.

Worthington, M., Heffelfinger, C., Bernal, D., Quintero, C., Zapata, Y. P., Perez, J. G., De Vega, J., Miles, J., Dellaporta, S., & Tohme, J. (2016). A parthenogenesis gene candidate and evidence for segmental allopolyploidy in apomictic Brachiaria decumbens. *Genetics*, *203*(3), 1117–1132.

Wu, B., Yu, Y., Zhang, X., Chen, H., Qin, C., Xie, J., Li, L., Wu, W., & Chen, J. (2026). Chromosome-level genome assembly of masu salmon (Oncorhynchus masou masou). *Sci. Data*, *13*(1), 534.

Xie et al. (2026). Snow carps as an emerging model for polyploid genome evolution. *Zool. Res.*

Xie et al. (2026). Chromosomal fusions trigger rediploidization of autopolyploid genomes. *Nature*, *654*(8119), 706–713.

Xie, W.-Z., Zheng, Y.-Y., He, W., Bi, F., Li, Y., Dou, T., Zhou, R., Guo, Y.-X., Deng, G., Zhang, W., Yuan, M.-H., Sanz-Jimenez, P., Zhu, X.-T., Xu, X.-D., Zhou, Z.-W., Zhou, Z.-W., Feng, J.-W., Liu, S., Li, C., … Chen, L.-L. (2024). Two haplotype-resolved genome assemblies for AAB allotriploid bananas provide insights into banana subgenome asymmetric evolution and Fusarium wilt control. *Plant Commun.*, *5*(2), 100766.

Yan, H., Han, J., Jin, S., Han, Z., Si, Z., Yan, S., Xuan, L., Yu, G., Guan, X., Fang, L., Wang, K., & Zhang, T. (2025). Post-polyploidization centromere evolution in cotton. *Nat. Genet.*, *57*(4), 1021–1030.

Yang, J., Moeinzadeh, M.-H., Kuhl, H., Helmuth, J., Xiao, P., Haas, S., Liu, G., Zheng, J., Sun, Z., Fan, W., Deng, G., Wang, H., Hu, F., Zhao, S., Fernie, A. R., Boerno, S., Timmermann, B., Zhang, P., & Vingron, M. (2017). Haplotype-resolved sweet potato genome traces back its hexaploidization history. *Nat. Plants*, *3*(9), 696–703.

Yu, R.-M., Zhang, N., Zhang, B.-W., Liang, Y., Pang, X.-X., Cao, L., Chen, Y.-D., Zhang, W.-P., Yang, Y., Zhang, D.-Y., Pang, E.-L., & Bai, W.-N. (2023). Genomic insights into biased allele loss and increased gene numbers after genome duplication in autotetraploid Cyclocarya paliurus. *BMC Biol.*, *21*(1), 168.

Zhang, R.-G., Shang, H.-Y., Jia, K.-H., & Ma, Y.-P. (2023). Subgenome phasing for complex allopolyploidy: case-based benchmarking and recommendations. *Brief. Bioinform.*, *25*(1), bbad513.

Zhang, Z., Gou, X., Xun, H., Bian, Y., Ma, X., Li, J., Li, N., Gong, L., Feldman, M., Liu, B., & Levy, A. A. (2020). Homoeologous exchanges occur through intragenic recombination generating novel transcripts and proteins in wheat and other polyploids. *Proc. Natl. Acad. Sci. U. S. A.*, *117*(25), 14561–14571.

Zhao, Z., & Shi, T. (2025). Haplotype-resolved assembly in polyploid plants: Methods, challenges, and implications for evolutionary and breeding research. *Genes (Basel)*, *16*(6), 636.